OSDP 3.0 Proposed PIV Enhancements

Security Industry Association

Redline: ce0c2e1c8433 to ce0c2e1c8433 working tree
October 14, 2026

Contents

Changelog **5**

1 Introduction **10**

2 Changes to OSDP 2.2 **11**

2.1 Updated Command Definitions . . . 11

2.1.1 Authentication Challenge (osdp_CRAUTH) . . . 11

2.1.2 Response to Challenge (osdp_CRAUTHR) . . . 13

2.1.3 Get PIV Data (osdp_PIVGETDATA) . . . 14

2.1.4 PIV Data Reply (osdp_PIVGETDATAR) . . . 18

2.2 PIV Error Codes . . . 19

2.2.1 PIV Error Reply (osdp_PIVERROR) . . . 20

2.3 Updated Function Codes . . . 20

2.3.1 Function Code 12 – Smart Card Support . . . 20

2.3.2 Function Code 13 – Reader Interfaces . . . 21

2.3.3 Function Code 14 – Biometric Support . . . 21

3 New or Enhanced PIV Commands **23**

3.1 Credential preparation and Secure Messaging . . . 23

3.2 PIV Put Data (osdp_PIVPUTDATA) . . . 31

DRAFT

3.3 PIV Application Select (osdp_PIVSELECT) . . . 32
3.4 Card Status (osdp_CARDSTATUS) . . . 34
3.5 Card Status Reply (osdp_CARDSTATUSR) . . . 34
3.6 PIV Secure PIN Entry (osdp_PIVSPE) . . . 38
3.7 PIV Secure PIN Entry Reply (osdp_PIVSPER) . . . 43
3.8 PIV Biometric Match (osdp_PIVBIOMATCH) . . . 45
3.9 PIV Biometric Match Reply (osdp_PIVBIOMATCHR) . . . 48
3.10 TWIC Privacy Key Load (osdp_PIVLOADTPK) . . . 49
3.11 PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING) . . . 50
3.12 Set PIV Mode (osdp_PIVMODE) . . . 51
3.13 PIV Auto Status Reply (osdp_PIVSTATUS) . . . 58
3.14 VCI Trust Anchor Load (osdp_PIVVCILOADTA) . . . 62
3.15 VCI Trust Anchor Status (osdp_PIVVCITASTATUS) . . . 67

A References **69**

A.1 Normative References . . . 69
A.2 Informative References . . . 70

B PIVGETDATA Worked Examples (Informative) **71**

B.1 Reading the exchanges . . . 71
B.2 Source fixtures . . . 72
B.3 Complete Discovery Object . . . 72
B.4 Discovery PIN Usage Policy (0x5F2F) . . . 73
B.5 Read the FASC-N element (0x30) . . . 74
B.6 Read the FASC-N value . . . 75
B.7 Read the UUID element (0x34) . . . 76
B.8 Read the UUID value . . . 77
B.9 Bounded CHUID prefix . . . 78
B.10 Complete certificate object . . . 79

B.11 Read the certificate element (0x70) . . . 80
B.12 Read a compressed certificate element . . . 81
B.13 Read a range within the certificate element . . . 83
B.14 Complete Key History object . . . 84
B.15 BIT Group with no biometric templates . . . 85
B.16 Empty 0x53 container . . . 86
B.17 Zero-byte successful response . . . 87
B.18 Existing child with an empty value . . . 88
B.19 Missing direct child . . . 89
B.20 Element request against an empty response . . . 90
B.21 Ambiguous direct-child selection . . . 91

C High-Speed Autonomous PIV Data Templates (Informative) 92

C.1 Vector Sources . . . 92
C.2 Key Selection and Certificate Discovery . . . 92
C.3 Illustrative PIV Authentication Resume . . . 94
C.4 Illustrative TWIC Auto Key Proof . . . 95
C.5 Supported Profiles . . . 95
C.6 Fixed-Width KMAC Challenge Template . . . 96
C.7 Dynamic Authentication Templates . . . 97
C.7.1 RSA Key Challenge Input . . . 97
C.7.2 ECC Signature Input . . . 98
C.7.3 Worked Dynamic Authentication Examples . . . 98
C.8 APDUs . . . 107

D Credential SM and VCI Worked Examples (Informative) 110

D.1 Credential SM over contact . . . 111
D.2 PIN before credential presentation . . . 112
D.3 PIV Auto needs commanded pairing . . . 113

D.4 SM failure diagnostics . . . 114
D.5 PIN cache expires while idle . . . 115
D.6 Expiration while waiting for a credential . . . 116
D.7 Expiration before VERIFY submission . . . 117
D.8 Expiration after VERIFY submission . . . 118
D.9 PIN reuse preserves the original deadline . . . 119
D.10 Credential failure status across two fragments . . . 120
D.11 Incorrect PIN preserves SM and VCI . . . 121
D.12 Recovering trust for contactless PIV Auto . . . 122

E CRAUTH Worked Examples (Informative) 124

F Deprecated `osdp_GENAUTH` 126

DRAFT

Changelog

2026-10-14 Added a key-reference byte to each PIV Auto profile. The PD reads the card's certificate to choose the algorithm, so the same profile can handle RSA and ECC cards. Added a resume option after pairing or PIN entry, defined counter handling and the challenge format, and updated the appendix with PIV and Legacy/NextGen TWIC examples.

2026-10-07 Added "Invalid TPK" for missing or incorrect TWIC Privacy Keys and corrected the TPK field to 16 bytes. Clarified PIV/TWIC application selection and added a Data Model byte to PIVSELECT. Updated the Legacy/NextGen fingerprint and facial workflows and the BIOMATCH hash description. Clarified CRAUTH cleanup so each new request starts a fresh transaction while the previous reply remains available for retransmission.

2026-09-16 Changes from OSDP 2.2:

- Renamed `osdp_PIVDATA`/`osdp_PIVDATAR` to `osdp_PIVGETDATA`/`osdp_PIVGETDATAR`, retaining command code `0xA3`. Added identifier and tag lengths, two-byte offsets and requested lengths, and whole-object or unique direct-child selection with the original tag and length bytes preserved. Defined OSDP 3.0 multi-part delivery and an all-zero first reply for empty success. Command data length distinguishes the new request from the fixed-length OSDP 2.2 form.
- Added Format Marker `0xFF` to the reassembled `osdp_CRAUTH` request, retaining command code `0xA5`. Multi-part lengths and offsets cover the marker, algorithm, key reference, and challenge. OSDP 3.0 ACUs use the new CRAUTH and PIVGETDATA forms with supporting PDs; PDs may additionally support the OSDP 2.2 forms, with format-specific response handling.
- Renamed `osdp_PDERROR` to `osdp_PIVERROR` and added GET DATA selection and transfer errors. The new CRAUTH and PIVGETDATA forms return one terminal PIVERROR on failure, replacing CRAUTH's empty failure reply and optional follow-up error. Renumbered tentative PIV-specific errors consecutively from `0x1020` through `0x1029`; credential removal is now `0x1026`.
- Separated credential SM setup, VCI pairing, and PIN verification. Clarified automatic preparation over contact and contactless, PIV Auto failure reporting, and ACU-directed recovery. Authenticated PIN and pairing rejections preserve SM; SM processing and integrity failures end the session.
- Added `osdp_CARDSTATUS` Type `0x02` diagnostics containing the SM outcome and credential CVC hash. Every CARDSTATUSR now includes Status Type after reply code `0x85`: `0x01` for standard status and pairing completion, `0x02` before the multi-part fields in each diagnostic fragment. Extended `osdp_PIVSTATUS` with SW1/SW2 when a verified credential failure status is available; the status word follows the signed response in the reassembled payload.
- Defined cached-PIN reuse without extending the original deadline, silent idle expiration, and completion of requests affected by expiration. Bounded PD timeout defaults to 1–255 seconds. Reassigned SPE flags `0x01` for retention across removal and `0x02` for Auto Execute; added result `0x0B` for credential communication failure or timeout, requiring preparation to restart after an unconfirmed PIN VERIFY. Applied the same restart rule to pairing VERIFY timeouts, reported through existing PIVERROR `0x0002` (smart-card timeout).

- Added workflow diagrams and worked packet examples for selection, SM/VCI recovery, PIN-cache timing, and multi-part replies. Standardized table sizes and protocol values on hexadecimal notation.

2026-09-02 Made proposal builds self-contained by vendoring the SIL-licensed Carlito typeface, replacing system Calibri and Arial dependencies with real regular, bold, italic, and bold-italic faces, and documenting the XeLaTeX and `latexmk` requirements. Completed the Enhanced PIV message-name cleanup by applying the compact OSDP naming convention consistently to `osdp_PIVLOADTPK`, `osdp_PIVXMITPAIRING`, `osdp_PIVVCILOADTA`, `osdp_PIVVCITASTATUS`, and `osdp_PIVVCITASTATUSR`; command and reply codes, payloads, and behavior are unchanged. Standardized multi-part messages on the generic OSDP *MpSizeTotal*, *MpOffset*, and *MpFragmentSize* field names and moved the multi-part fields ahead of the reassembled `osdp_PIVMODE` and `osdp_PIVVCILOADTA` data. The current `osdp_PIVVCILOADTA` definition supersedes the anchor-only multi-part sizing described in the 2026-05-30 entry: its total now includes the 12-byte metadata block. Updated the PIV Auto simulator, generated fixtures, proposal PDF, and redline PDF to exercise and publish the same framing and naming.

2026-08-12 Proposed Secure PIN Entry changes and consistency cleanup:

- Proposed clear, mode-specific SPE behavior for status, PIN acquisition and verification, cached-PIN verification, cache/abort, and attempts-remaining queries. The proposal also defines finite default timeouts, cache and PIN-verified lifecycles, and ACK/poll completion behavior.
- Proposed a one-byte `osdp_PIVSPER` reply whose value is interpreted by the requested mode. The OSDP 2.2 `osdp_PR01SPER` reply remains unchanged.
- Standardized Enhanced PIV names, including renaming the autonomous PIV Auto reply to `osdp_PIVSTATUS`. Corrected tentative code conflicts by assigning `osdp_PIVSELECT` to `0xB0` and `osdp_PDERROR` to `0x8A`.

2026-06-18 PIVSELECT, PDERROR, and VCI consistency updates:

- Added `osdp_PIVSELECT` as an OSDP-native direct AID selection command using tentative command code `0xA7`; clarified its ISO/IEC 7816 `SELECT by DF name` mapping, selected-AID status update, and application-state clearing rules.
- Removed stale `osdp_CRAUTHR` `osdp_PDERROR` wording so codes `0x0006`/`0x0007` retain their common meanings for secure-channel and multi-part errors.
- Clarified that VCI pairing codes are transmitted from the ACU to the PD only over OSDP Secure Channel 2.0 and are used by the PD to establish VCI secure messaging with the credential.
- Tightened and clarified verbiage by replacing imprecise “legacy” terminology throughout normative text with neutral wording such as unchanged, OSDP 2.2-format, and poll-based; retained “legacy” only for withdrawn algorithms such as RSA-1024 test material.

2026-05-30 Added source-backed VCI and PIV Auto examples, clarified PIV Auto interoperability requirements, and improved reviewability:

- Added NIST-derived VCI and PIV APDU fixture coverage, GSA-backed demonstration keys, and reproducible worked examples for PIV Auto and VCI validation. RSA-1024 material is retained only as informative legacy/non-current test-vector material.
- Clarified PIV Auto cryptographic inputs and response handling, including KMAC-derived challenge construction, RSA and ECC `GENERAL AUTHENTICATE` templates, nonzero UUID requirements, and complete returned `0x7C` Dynamic Authentication Templates in `osdp_PIVSTATUS`.
- Aligned related command semantics for capability advertising, `osdp_CARDSTATUS` errors, selective `osdp_PIVDATA` replies, `osdp_PIVSPE` status/abort modes, and VCI trust-anchor inventory.
- Reworked Appendix C and the references section for standards review: examples are explicitly informative, vector provenance is documented, source fixtures are framed as public test material, and normative references now include the needed APDU, KMAC, X.509, and requirement-language specifications.
- Added supporting scripts and tests for deterministic simulation, fixture replay, VCI import/validation, Appendix C generation, YubiKey material loading, live APDU capture, and reproducible PDF/table builds.

2026-05-27 Version renamed from OSDP 2.3 to OSDP 3.0 throughout. Consolidated PIV Auto, biometric-result, VCI trust-anchor, and SPE updates:

- Removed `osdp_PIVSPE` flag `0x01` (retry continuation); each `osdp_PIVSPE` command now initiates exactly one PIN-entry session and `osdp_PIVSPER` terminates the session unconditionally, including on PIN mismatch. ACUs wishing to allow additional attempts **MAY** re-issue `osdp_PIVSPE`, using the *Attempts Remaining* count from the preceding `osdp_PIVSPER` to guide retry policy. Bit `0x01` is now reserved.
- Aligned `osdp_PIVSPER` and `osdp_PIVBIOMATCHR` result codes for credential removal, changed `osdp_PIVBIOMATCHR 0x03` to PIN verification required, reserved `osdp_PIVBIOMATCHR 0x07`, clarified biometric assurance limits, and specified that only off-card biometric match success returns the 32-byte SHA-256 hash of tag `0xBC`.
- Clarified the Function Code 14 biometric-support reference for `osdp_PIVLOADTPK` and added CAC as PIV mode data model `0x03`.
- Added PIV Auto KMAC challenge derivation, required series and sequence counters, fixed-width supplemental entropy, and the `osdp_PIVSTATUS` poll reply for reporting credential identity and signed challenge responses. PIV Auto is configured through `osdp_PIVMODE` and is restricted to Secure Channel 2.0 sessions that are not using factory keys.
- Replaced the SP 800-73 Intermediate CVC anchor model with an OSDP-defined VCI trust anchor record (tag `0x7F50`) carrying a Profile Identifier (`0x80` = `0x01`), an eight-byte Issuer Identifier Number (`0x42`), and a DER-encoded SubjectPublicKeyInfo (`0x7F49`). Maximum conforming record size is 576 bytes (`0x0240`), derived from the RSA-4096 key case.
- Added normative IIN derivation rule: the IIN in `osdp_PIVVCILOADTA` and in the trust anchor record **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate.

- Tightened osdp_PIVVCILOADTA multi-part encoding: Anchor Length and MpSizeTotal ranges corrected to 0x0000-0x0240; added normative constraint that MpOffset + MpFragmentSize **SHALL NOT** exceed Anchor Length; PD **SHALL** verify the reassembled byte string is exactly *Anchor Length* bytes and a single 0x7F50 record before committing.
- Documented variable-length PD storage accounting: PDs **SHALL** account for anchors by their actual *Anchor Length*, not a fixed maximum; replacement is atomic (prior anchor remains if replacement fails); delete releases actual stored bytes.
- Specified the two-tier PD verification path during VCI establishment: direct (trust anchor IIN matches secure messaging CVC IIN, anchor SPKI verifies CVC signature) and intermediate (no direct match triggers retrieval of PIV data object 0x1017/5FC122; Intermediate CVC Subject Identifier must equal CVC IIN, Role 0x5F4C must be 0x12, anchor IIN must match Intermediate CVC IIN, anchor SPKI verifies Intermediate CVC, Intermediate CVC verifies secure messaging CVC). PDs **SHALL NOT** perform X.509 path validation or revocation for VCI trust anchors.
- Added normative ACU post-VCI validation requirements: after osdp_CARDSTATUSR reports VCI established, the ACU **SHALL** obtain the content-signing certificate (from cache or via osdp_PIVDATA for 0x1017/inner tag 0x70), verify SKI matches *VCI Anchor IIN*, verify SPKI matches the stored anchor record, verify PIV or PIV-I content-signing EKU OID, verify validity period (expired certificate **SHALL** cause session rejection and anchor unload), and perform X.509 path validation and revocation per deployment policy.
- Restored *VCI Trust Anchor ID* to osdp_CARDSTATUSR with normative language; added *VCI CVC IIN* and *VCI Anchor IIN* fields (each 8 bytes) so the ACU receives the IIN pair needed to distinguish direct and intermediate verification paths.
- Updated osdp_PIVVCITASTATUSR: clarified that *Available Bytes* reflects variable-length accounting and that the per-record IIN is the anchor selector from the stored trust anchor record, not necessarily the secure messaging CVC IIN reported in osdp_CARDSTATUSR.
- Added VCI security preconditions: all VCI-related commands **SHALL** be rejected unless OSDP Secure Channel 2.0 is active and not using factory/default keys; failures reported via osdp_PDERROR 0x0006.

2026-01-21 Error-handling cleanup and code assignments: **(1)** osdp_PIVPUTDATA now **SHALL** report failures via osdp_PDERROR (status word optional) so writes can't silently succeed. **(2)** osdp_PIVSPE adds result code 0x09 in osdp_PIVSPER for "VCI not established." **(3)** PDERROR is credential/global only; PD capability gaps stay osdp_NAK. **(4)** OSDP 2.2-format replies (e.g., osdp_CRAUTHR, osdp_PIVDATAR) remain unchanged, with optional follow-up osdp_PDERROR detail if the PD has it. **(5)** Documented tentative command/reply codes for the new PIV set: commands 0xA6 (VCI trust-anchor status) through 0xAF (VCI trust-anchor load), and replies 0x85-0x89 for the new responses.

2026-01-07 Added a PIV Mode interface-mask table aligned to Function Code 13 bits, clarified osdp_CRAUTHR reply structure (full GENERAL AUTHENTICATE response without SW1|SW2; status words reported via osdp_PDERROR only), tightened osdp_CARDSTATUS/osdp_CARDSTATUSR behavior (solicited-only after PD preparation; error on credential removal), remapped the PIV error codes to the 0x10xx block per subcommittee guidance, added PIV APDU references,

aligned osdp_PIVDATA tag selection to allow Discovery (0x7E)/BIT (0x7F61) subtags in line with SP 800-73-4, clarified that osdp_PDERROR conveys a single error per reply, and clarified osdp_PIVXMITPAIRING handling so pairing codes can be provided before VCI is established when the credential requires them.

2025-12-10 Consolidated application-level errors on osdp_PDERROR (two-byte codes, optional SW1|SW2 for APDU failures, multi-error support); moved the PIV error set to 0x0100-0x01FF and updated PIV commands to use osdp_PDERROR (transport faults remain osdp_NAK); made the Global Interface Mask in osdp_PIVMODE a required byte (no TLV tag) with an updated field layout; clarified that osdp_CARDSTATUSR is solicited-only after PD preparation with poll-based osdp_RAW as the aggregated card-present indication; and noted the deprecation of osdp_GENAUTH with a recommendation to migrate to osdp_CRAUTH.

2025-12-04 Clarified that osdp_CRAUTHR returns the full GENERAL AUTHENTICATE data including the Dynamic Authentication Template (excluding only the ISO/IEC 7816 status word).

2025-11-12 Added PIV Biometric Match request/reply and tables (one-bit type values only); moved Enhanced PIV NAKs to a 0x20+ base (added 0x27 Credential Not Present/Removed, 0x28 Biometric Type Not Supported; clarified 0x23 Credential Security Status Not Satisfied); aligned CRAUTHR error handling to reference the enhanced NAK set; relocated Enhanced PIV (0x04) and PIV Auto (0x08) indicators to Function Code 12 and cleaned Function Code 13 to interface-only bits (removed non-interface flags; removed PIV Auto from Function Code 14); updated CARDSTATUS gating to Function Code 12 bit 0x04; added a PIV Mode Flags table and clarified PIV Mode AID encoding; enabled PDF TOC hyperlinks and improved table/caption pagination.

2025-10-30 Clarified CRAUTH command/reply behaviour (including RSA, AES, and ECC support), overhauled PIV Mode configuration with TLV-based application and interface controls (including PIV Auto), added card status request/reply messages, expanded PIV data retrieval semantics, introduced PIV Put Data, PIV Secure PIN Entry, TWIC privacy-key load, VCI pairing transmit, and VCI trust-anchor load/status commands, and documented the enhanced error-code set (including VCI/PIV Auto conditions).

2025-09-17 Major PIV Mode and SPE revisions.

2025-06-25 Added multi-AID usage and VCI support notes.

2025-05-27 New function code definitions for 3.0 biometric handling.

1 Introduction

This document captures proposed enhancements to the SIA Open Supervised Device Protocol (OSDP) needed to carry FICAM-compliant Personal Identity Verification (PIV) workflows. The material consolidates draft content, reviewer comments, and interoperability feedback collected during the 2025 working sessions and re-expresses it as normative guidance for command authors and device implementers.

The chapters that follow highlight changes required for existing OSDP 2.2 commands, introduce new transactions where no equivalent existed, and provide supplemental examples that reflect current federal credentialing practices. Wherever the proposal diverges from the baseline specification, the text records both the behavioral requirements and the error handling expectations so that access control units (ACUs) and peripheral devices (PDs) can adopt the profile without relying on vendor extensions.

The key words **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are interpreted as described in RFC 2119 and RFC 8174 when they appear in capitals.

Tables express protocol sizes, lengths, offsets, and numeric values in hexadecimal with the prefix `0x`. Packet and APDU dumps use hexadecimal byte pairs without repeated prefixes. Standard names, bit positions, and elapsed-time annotations retain their usual notation.

OSDP 3.0 introduces a new PIVGETDATA request format at command code `0xA3`. PDs support the new format and may also accept the OSDP 2.2 format; the command data length distinguishes the two. OSDP 3.0 ACUs use the new format with PDs that support it. The revised CRAUTH and PIVGETDATA transactions return data on success and one terminal PIVERROR on failure.

2 Changes to OSDP 2.2

This section documents the normative deltas that apply when an OSDP 2.2 implementation adopts the Enhanced PIV profile. Each subsection references the original command definition and records the precise behavioral updates, encoding clarifications, and new error handling rules introduced by this proposal.

2.1 Updated Command Definitions

2.1.1 Authentication Challenge (osdp_CRAUTH)

The `osdp_CRAUTH` command, code `0xA5`, transports a `GENERALAUTHENTICATE` operation to the credential. An OSDP 3.0 ACU **SHALL** use the new format with a PD that supports it. An OSDP 3.0 PD **SHALL** support the new format and **MAY** also support the OSDP 2.2 format.

Request format and recognition The new format uses the OSDP 3.0 multi-part fields immediately after the command code. The reassembled data consists of Format Marker `0xFF`, Algorithm, Reference Identifier, and Challenge, in that order. The three leading fields are each one byte. *MpSizeTotal* counts all of these bytes, including the marker, algorithm, and key reference.

The first data fragment **SHALL** start at offset `0x0000` and contain at least one data byte. The PD **SHALL** select the new format when that first byte is `0xFF`. Otherwise, the PD **SHALL** select the OSDP 2.2 format if supported, or reject the request with `osdp_NAK` `0x09` (unable to process command). This proposal reserves `0xFF` in that position as an OSDP format marker. A dual-format PD **SHALL NOT** interpret that byte as a legacy algorithm identifier. The marker is separate from the Algorithm field and is removed before the PD constructs the credential command.

The PD **SHALL** retain the selected format until completion or cancellation. Later fragments carry the next bytes of the same data record; the marker and other fields are not repeated. Fragment boundaries may split the leading fields or the challenge. The PD **SHALL** validate the complete record before sending the credential command. The minimum record length is `0x0003`; the selected algorithm determines the required challenge length. A malformed new-format record **SHALL NOT** be retried as an OSDP 2.2 record. Multi-part framing errors use the applicable OSDP NAK rules. Invalid reassembled parameters return PIVERROR `0x0005` (invalid parameter) after acceptance.

A PD accepting the OSDP 2.2 form **SHALL** use that form's request and response rules for the transaction. The rules below apply to the new form. The worked example in Appendix E shows the marker, algorithm, and key reference divided between command fragments.

- Codes `0x05`, `0x06`, and `0x07` identify RSA private-key operations. Codes `0x08`, `0x0A`, and `0x0C` identify AES encryption with 128-, 192-, and 256-bit keys respectively. Codes `0x11` and `0x14` identify ECC P-256 and P-384 mechanisms; the selected key reference determines whether the operation is ECDSA signature generation or ECDH key establishment. Algorithm selection remains subject to the applicable NIST requirements.
- The *Key* field retains its role as the card's reference identifier. PDs **SHALL NOT** reinterpret or remap the identifier.

- For RSA algorithms (0x05, 0x06, 0x07), the *Challenge* field carries the encoded datum for the RSA private-key operation. The PD forwards the value unmodified; padding and digest encoding are the responsibility of the ACU.
- For AES algorithms (0x08, 0x0A, 0x0C), the *Challenge* field contains the plaintext block(s) to be encrypted. PDs **SHALL** send the data to the credential without padding; ACUs **SHALL** provide data aligned to the block size dictated by the selected key length and card profile.
- For ECC signature operations using key references 0x9A, 0x9C, or 0x9E with algorithms 0x11 or 0x14, the *Challenge* field carries the pre-hashed input to be signed. The PD **SHALL** issue GENERALAUTHENTICATE with command data encoded as 7C Len 82 00 81 Len Challenge; the credential returns the ECDSA signature in tag 0x82. Exponentiation tag 0x85 **SHALL NOT** be used with 0x9A, 0x9C, or 0x9E.
- For ECC key-establishment operations using key reference 0x9D or retired key-management references, algorithms 0x11 and 0x14 use the *Challenge* field to convey the peer public key. The PD **SHALL** issue GENERALAUTHENTICATE with command data encoded as 7C Len 82 00 85 Len (04 || X || Y), where X and Y are the uncompressed coordinates for the selected curve and are left-padded with 0x00 as required. The credential performs elliptic-curve Diffie-Hellman in accordance with SP 800-73.

Table 1: Authentication Challenge (osdp_CRAUTH)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA5
0x02	MpSizeTotal	Length of the complete record, including Format Marker, Algorithm, Reference Identifier, and Challenge; least-significant byte first.	0x0003–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete message, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0x01	Format Marker	First byte of the reassembled record. Identifies the OSDP 3.0 format; removed before constructing the credential command.	0xFF

Size (bytes)	Name	Meaning	Value
0x01	Algorithm	Cryptographic mechanism. Supported values include 0x05/0x06/0x07 (RSA private-key operation), 0x08/0x0A/0x0C (AES encryption with 128/192/256-bit keys), and 0x11/0x14 (ECC P-256/P-384; operation depends on the key reference).	0x00–0xFF
0x01	Reference Identifier	Key reference passed to GENERALAUTHENTICATE.	0x00–0xFF
0x00-n	Challenge	For RSA algorithms, the encoded datum for the private-key operation. For AES algorithms, the plaintext block(s) to be encrypted, pre-padded by the ACU as required. For ECC 0x9A/0x9C/0x9E, the pre-hashed signature input in tag 0x81; for ECC 0x9D/retired key-management references, the peer public key in tag 0x85.	0x00–0xFF

When credential SM is established, the PD **SHALL** protect the resulting credential command using that context. The credential enforces the selected key's access conditions. An available card-authentication key (0x9E) may be used without SM or PIN verification when the credential permits it. Shared card-transport rules appear in Subsection 3.1.

Deprecated osdp_GENAUTH The osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead.

2.1.2 Response to Challenge (osdp_CRAUTHR)

OSDP 3.0 uses the same completion rule for CRAUTH and PIVGETDATA: data replies carry successful results; one terminal PIVERROR reports a failed accepted operation. This replaces CRAUTH's OSDP 2.2 failure behavior.

After accepting the complete CRAUTH request, the PD **SHALL** return the result on subsequent osdp_POLL commands. On success, the PD **SHALL** return osdp_CRAUTHR using the OSDP 3.0 multi-part format. On failure, the PD **SHALL** return one osdp_PIVERROR with the applicable error code from Subsection 2.2. That reply ends the operation. The PD **SHALL NOT** precede the error with an empty CRAUTHR. If CRAUTHR fragments have already been returned, the ACU **SHALL** discard the partial response when PIVERROR arrives.

OSDP retransmissions repeat the same reply without repeating the credential operation. Command rejection and transport faults continue to use the applicable OSDP NAK rules.

After returning the final osdp_CRAUTHR fragment or a terminal osdp_PIVERROR, the PD **SHALL** end the operation and clear its assembled request, request length, response offset, pending error, and completion state. The PD **SHALL** then accept a new osdp_CRAUTH beginning at offset zero. Clearing the completed operation **SHALL NOT** invalidate the cached OSDP reply required for link-layer retransmission. A retransmission replays that reply without advancing the response offset or repeating the credential operation.

Reply structure

osdp_CRAUTHR **SHALL** return the credential's full ISO/IEC 7816 GENERALAUTHENTICATE response payload, including the Dynamic Authentication Template, framed by the standard OSDP multi-part fields. PDs **SHALL NOT** include SW1|SW2 in osdp_CRAUTHR; any status words from the credential **SHALL** be carried only in osdp_PIVERROR under the status-word rules in Subsection 2.2.1. PDs **SHALL NOT** strip outer BER-TLV tags or lengths from the returned payload.

Table 2: Response to Challenge (osdp_CRAUTHR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x82
0x02	MpSizeTotal	Length of the complete response, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete response, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0x00-n	Response Data	Signature, encrypted block, or shared-secret material returned by the credential. The credential's BER-TLV structure is preserved exactly as received (status word excluded).	0x00–0xFF

2.1.3 Get PIV Data (osdp_PIVGETDATA)

OSDP 3.0 retains command code 0xA3 and introduces the PIVGETDATA request format defined here. An OSDP 3.0 ACU **SHALL** send the new format to a PD that supports it. An OSDP 3.0 PD **SHALL** support the new format and **MAY** also support the OSDP 2.2 format. The ACU **SHALL** confirm Enhanced PIV support (Function Code 12 bit 0x04) before using the new format. The reply uses the OSDP 3.0 multi-part layout. Worked exchanges appear in Appendix B.

Recognizing the request format The PD **SHALL** determine the format from the command data length, excluding the command code, OSDP header, security block, and trailer. The OSDP 2.2 format contains exactly 0x05 data bytes: a three-byte object identifier, a one-byte element identifier, and a one-byte offset. The new format contains 0x06 plus Identifier Length plus Tag Length data bytes, giving a range of 0x07 through 0x0C bytes. These lengths do not overlap.

A PD that supports both formats **SHALL** interpret exactly 0x05 data bytes using the OSDP 2.2 definition, including its selection and response rules. A new-format-only PD **SHALL** reject that request with PIVERROR 0x0005 (invalid parameter) on a later poll after ACK. For a new-format request, the PD **SHALL** validate both length fields and the exact total length before accessing the credential. Any other data length or invalid new-format field length produces the same error. The PD **SHALL NOT** reinterpret a malformed new-format request as an OSDP 2.2 request.

Table 3: Get PIV Data (osdp_PIVGETDATA)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA3
0x01	Identifier Length	Number of bytes in Object Identifier.	0x01–0x03
0x01-0x03	Object Identifier	Complete BER tag identifying the credential data object. Native byte order; no leading padding.	BER tag
0x01	Tag Length	Number of bytes in Element Tag. Zero selects the complete object and omits Element Tag.	0x00–0x03
0x00-0x03	Element Tag	Complete BER tag of a unique direct child of the object's outer container.	BER tag
0x02	Offset	Starting byte within the selected object or element. Zero identifies its first tag byte. Least-significant byte first.	0x0000–0xFFFF
0x02	Requested Length	Maximum bytes to return, least-significant byte first. Zero requests all remaining bytes.	0x0000–0xFFFF

Request encoding The ACU **SHALL** provide valid Object Identifier and Element Tag encodings. The PD **SHALL** pass Object Identifier to the card unmodified and use Element Tag unmodified to select a matching element from the response. Each length field **SHALL** equal the number of tag bytes that follow it. Identifier Length is one to three; Tag Length is zero to three. Tag Length zero omits Element Tag. The PD **SHALL** reject invalid field lengths, truncated requests, and trailing request bytes with osdp_PIVERROR 0x0005 (invalid parameter).

The PD **SHALL** issue the card's GET DATA operation for the requested object on the active credential interface. Each request reads the credential; the PD **SHALL NOT** substitute object data retained from an earlier request. If the credential denies access through that interface, the PD **SHALL** report the failure using

osdp_PIVERROR. When credential Secure Messaging (SM) is established, the PD **SHALL** use that SM context. The credential enforces VCI availability and the requested object's access conditions. The PD **SHALL** complete card-side response chaining and remove status words and secure-messaging envelopes before selecting response data. A requested OSDP length does not limit the card-side transfer needed to complete that operation.

Selection When Tag Length is zero, the PD **SHALL** select the complete object, including its outer tag, length, and value. The PD **SHALL** preserve the original bytes returned by the credential. PIV responses use 0x53 for ordinary objects and native 0x7E and 0x7F61 containers for Discovery and the BIT Group. Other application profiles may define other containers; whole-object selection does not substitute or remove them.

For nonzero Tag Length, the PD **SHALL** parse the outer container and select exactly one matching direct child, including that child's tag, length, and value. The enclosing object's tag and length are not included. The PD **SHALL NOT** search recursively, decompress the value, or re-encode the selected element. The PD **SHALL** inspect the complete child sequence before accepting a unique match. No match produces 0x1024 (data object or tag not found); multiple matches produce 0x1027 (ambiguous element selection); malformed container or child encoding produces 0x1028 (malformed credential-object encoding). Element extraction requires definite BER lengths, a single complete outer container, and child boundaries within that container. To select an inner tag, the PD **SHALL** search the elements directly inside the object's outer container. In BER-TLV encoding, *primitive* means the tag marks its value as data bytes; *constructed* means the tag marks its value as a sequence of TLV elements. For PIV objects returned under tag 0x53, the PD **SHALL** treat the contents as TLV elements even though the tag marks them as primitive. Child values are opaque regardless of their constructed bit; the PD **SHALL NOT** recursively validate them as ASN.1 structures. Whole-object selection does not require child parsing. An empty card response is valid whole-object data; a nonzero Tag Length against an empty response **SHALL** produce 0x1024 (data object or tag not found).

Offset and requested length The PD **SHALL** apply Offset and Requested Length once, before dividing the data into fragments. Offset zero identifies the first byte of the selected object's or element's tag. The PD **SHALL** start at Offset and return no more than Requested Length bytes. Requested Length zero requests all remaining bytes. If fewer bytes remain, the PD **SHALL** return those bytes. The resulting bytes are the *requested data*. Offset equal to the selected-data length produces an empty success; Offset beyond that length produces 0x0005 (invalid parameter). The requested data **SHALL NOT** exceed 65,535 bytes. An oversized through-end result produces 0x1029 (transfer length exceeded); a bounded range within a larger object is permitted. A nonzero Offset can skip tag and length bytes. The requested range may begin or end within a tag, length, or value. The PD **SHALL** return those bytes unchanged. The ACU is responsible for interpreting a partial result.

Figure 1 shows how selection and fragmentation relate.

Acceptance and completion The PD **SHALL** acknowledge an accepted operation with osdp_ACK. Acceptance does not indicate successful card access or parameter validation. The PD **SHALL** return the result on a

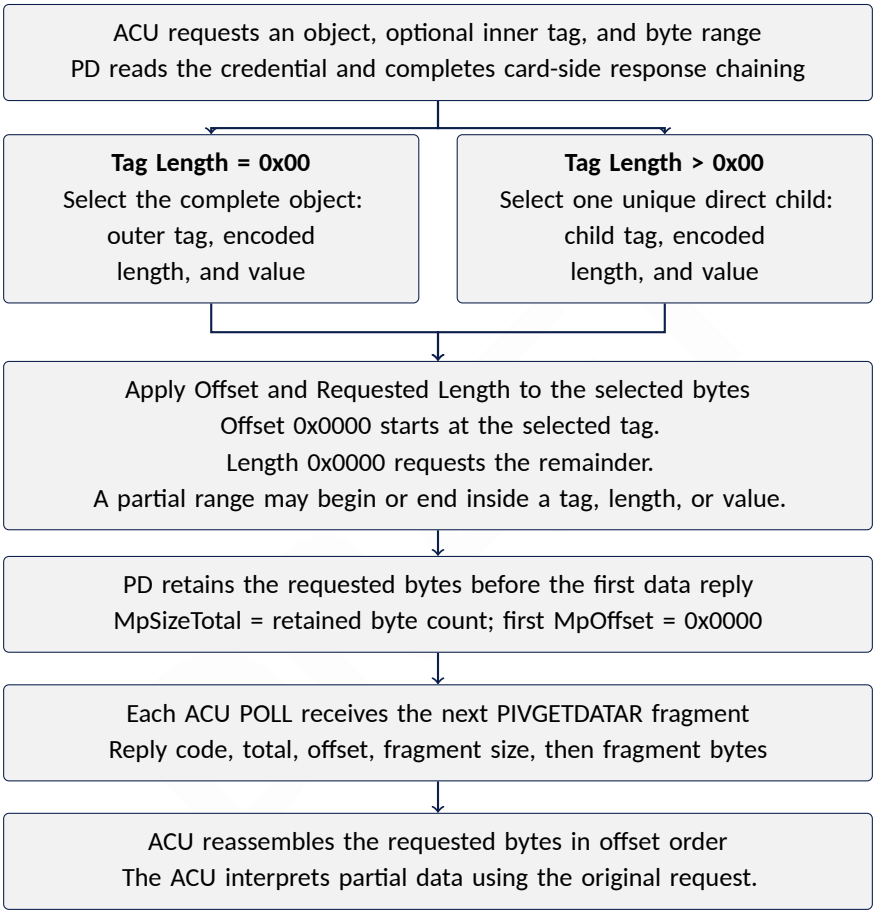


Figure 1: GET DATA selection precedes byte-range selection and OSDP fragmentation. ACK acknowledges acceptance. A failed read or selection ends with PIVERROR on a poll instead of data fragments. Child-selection errors are defined in the text.

subsequent osdp_POLL: osdp_PIVGETDATAR for success or osdp_PIVERROR for failure. While processing is incomplete, the PD **SHALL** answer polls with osdp_ACK. Unsupported commands and transport faults retain their OSDP osdp_NAK responses.

Only one operation may be outstanding at that PD during this read. Until the read completes or is aborted, the ACU **SHALL** send only osdp_POLL or osdp_ABORT to that PD. The PD **SHALL** reject a competing command with osdp_NAK 0x09 (unable to process command record) without replacing the active operation. Traffic to other PDs is unaffected. On osdp_ABORT, the PD **SHALL** discard pending data or errors, cancel further delivery, and return osdp_ACK. It **SHALL NOT** subsequently report completion for that operation.

The PD **SHALL** prepare and retain all requested data before sending the first data fragment. The data **SHALL** come from one credential operation. The PD **SHALL NOT** combine data from different credential presentations. Once delivery begins, the retained bytes remain unchanged. Card removal before all requested data is retained produces 0x1026 (credential not present or removed); removal afterward does not change the retained data. A card-operation timeout produces 0x0002 (smart-card timeout), insufficient PD resources produce 0x1021 (insufficient PD memory or storage), and unsatisfied credential security conditions produce 0x1023 (credential security status not satisfied). A failure produces one terminal osdp_PIVERROR, not an empty osdp_PIVGETDATAR followed by optional error detail. Error definitions appear in Subsection 2.2.

2.1.4 PIV Data Reply (osdp_PIVGETDATAR)

Table 4: PIV Data Reply (osdp_PIVGETDATAR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x80
0x02	MpSizeTotal	Total number of requested data bytes across all fragments. Least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Position of this fragment's first byte within the requested data. Least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Number of Card Data bytes in this reply. Least-significant byte first.	0x0000–0xFFFF
0x00-n	Card Data	The bytes returned in this reply, in their original order.	0x00–0xFF

The PD **SHALL** transfer the requested data using the OSDP 3.0 multi-part structure and usage rules. The ACU retrieves each fragment with osdp_POLL. Every reply contains the reply identifier, the three two-byte multi-part fields, and the next portion of the requested data. The multi-part fields use least-significant byte first, in the order shown in the table.

MpSizeTotal **SHALL** specify the number of requested data bytes, not the original object length. Its value **SHALL** remain unchanged throughout the transfer. *MpOffset* identifies the first byte of this fragment within the requested data. The first fragment **SHALL** have *MpOffset* zero, regardless of Offset in the command. Each

subsequent fragment **SHALL** start immediately after the preceding fragment. *MpFragmentSize* **SHALL** equal the number of data bytes in that reply.

Fragment boundaries may occur anywhere within the requested data, including within a tag or length field. The PD **SHALL NOT** add or repeat TLV headers for individual fragments. The ACU reconstructs the requested data by joining the fragments in order. Reply identifiers, multi-part fields, and OSDP packet headers and trailers are not part of that data and are not counted in *MpSizeTotal* or *MpFragmentSize*.

For example, a request for `0x0104` bytes starting at Offset `0x0100` produces *MpSizeTotal* `0x0104`. With `0x0072` data bytes per reply, the three fragments have *MpOffset* values `0x0000`, `0x0072`, and `0x00E4`, and sizes `0x0072`, `0x0072`, and `0x0020`.

For an empty success, the PD **SHALL** return one `osdp_PIVGETDATAR` with all three multi-part fields zero and no data bytes. The ACU **SHALL** interpret this first reply as successful completion with an empty result. For PIVGETDATAR, the early-termination indication applies only when *MpSizeTotal* is nonzero. For a nonempty result, delivery completes when the distinct accepted fragment lengths sum to *MpSizeTotal*. Normal OSDP retransmission rules apply: replaying a reply does not advance the data offset or execute the card operation again. The ACU **SHALL** discard partial data on cancellation or the OSDP 3.0 early-termination indication (zero fragment size and offset at or beyond total). That indication is not successful completion of a nonempty transfer.

2.2 PIV Error Codes

PD error codes are carried in `osdp_PIVERROR` using two-byte codes (least-significant byte first). The common PD codes remain available: `0x0001` (smart-card error), `0x0002` (smart-card timeout), `0x0004` (required TLV missing), `0x0005` (invalid parameter), `0x0006` (secure channel required), `0x0007` (invalid multi-part transfer), `0x0008` (multi-part fragment out of sequence), and `0x0009` (multi-part fragment out of bounds). The following PIV-specific codes occupy the `0x1000`-`0x10FF` range:

Table 5: Enhanced PIV Error Codes

Code	Description
`0x1020`	Function not supported by credential (PD capability gaps **SHALL** use `osdp_NAK` function-not-supported).
`0x1021`	Insufficient PD memory or storage to complete the requested operation.
`0x1022`	Insufficient credential memory or storage space.
`0x1023`	Credential security status not satisfied.
`0x1024`	Data object or tag not found.
`0x1025`	VCI not established.
`0x1026`	Credential not present or removed.
`0x1027`	Ambiguous element selection: more than one direct child matches the requested tag.
`0x1028`	Malformed credential-object encoding encountered during element extraction.

Code	Description
0x1029	Selected response exceeds the supported transfer length.

The assignments are tentative. Unlisted codes remain unassigned by this proposal.

2.2.1 PIV Error Reply (osdp_PIVERROR)

The PD **SHALL** use osdp_PIVERROR for errors in PIV operations that are not reported by a command-specific result. Each reply contains one two-byte error code, least-significant byte first. The originating command defines when the error is returned; PIVGETDATA errors are returned on a later poll. Transport faults continue to use osdp_NAK. This reply replaces the name osdp_PDERROR; its tentative byte value 0x8A and payload structure are unchanged.

Table 6: PIV Error Reply (osdp_PIVERROR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x8A (tentative)
0x02	Error Code	Two-byte error code, least-significant byte first.	0x0001–0x10FF
0x00 or 0x02	Status Word	SW1 followed by SW2 when a card command returned the failure being reported. Omitted for failures detected locally by the PD. See the status-word rules below.	–

When a credential command returns the failure being reported, the PD **SHALL** include that command's status word in SW1, SW2 order. For a valid protected response, the PD **SHALL** use the status word inside the SM response. The PD **SHALL** omit the status word for failures detected locally, including invalid OSDP parameters, missing trust anchors, timeouts, card removal, and failed verification of a protected response. A success status from an earlier credential command **SHALL NOT** be attached to a later failure.

2.3 Updated Function Codes

2.3.1 Function Code 12 – Smart Card Support

Function Code 12 advertises smart-card transport capabilities and profile support at the PD. The compliance byte uses the following bit assignments:

Table 7: Function Code 12 Compliance Flags

Mask	Description
0x01	PD supports transparent reader mode.
0x02	PD supports extended packet mode.
0x04	PD supports the OSDP 3.0 Enhanced PIV profile, including osdp_PIVGETDATA refinements and related commands.
0x08	PD supports OSDP 3.0 autonomous PIV operation (PIV Auto).

Bits not listed are reserved and **SHALL** be transmitted as zero.

2.3.2 Function Code 13 – Reader Interfaces

OSDP 2.2 defined Function Code 13 as a reader count indicator with a compliance byte fixed at 0x00. Enhanced PIV-capable PDs **SHALL** continue to populate the reader count field, but they MAY set bits within the compliance byte to advertise the downstream credential technologies they expose. A value of 0x00 remains valid and retains the original OSDP 2.2 meaning of "interface presence unspecified."

When a PD reports specific interfaces, it **SHALL** encode the compliance byte using the flags in Table 8. The reader count byte continues to report the number of attached credential interfaces.

Table 8: Function Code 13 Compliance Flags

Mask	Description
0x01	PD presents a contact card interface (ISO 7816 or equivalent).
0x02	PD presents a high-frequency contactless credential interface (13.56 MHz NFC).
0x04	PD presents a low-frequency credential interface (approximately 125 kHz prox).
0x08	PD supports credential Secure Messaging and Virtual Contact Interface (VCI) for applicable interfaces and modes.
0x10	PD presents a Bluetooth credential interface.
0x20	PD presents a barcode credential interface.
0x40	PD presents an ultra-high-frequency credential interface (UHF/RAIN RFID).

All other bits are reserved for future use and **SHALL** be transmitted as zero.

2.3.3 Function Code 14 – Biometric Support

Function Code 14 remains backward compatible with OSDP 2.2. A compliance value of 0x00, 0x01, 0x02, or 0x03 retains its original meaning and indicates that the PD does not support the OSDP 3.0 biometric command set. Devices that support the enhanced commands **SHALL** set Bit 7 (0x80). When Bit 7 is asserted,

the remaining bits of the compliance byte are interpreted as a capability bitfield. Bits not explicitly defined in Table 9 are reserved and **SHALL** be transmitted as zero.

Table 9: Function Code 14 Compliance Flags

Mask	Description
0x80	PD supports the OSDP 3.0 biometric command set.
0x01	PD performs off-card fingerprint comparison using FIPS 201 BIO ANSI INCITS 378 templates.
0x02	PD performs on-card fingerprint comparison using FIPS 201 OCC processes.
0x04	PD performs off-card iris comparison aligned with FIPS 201 BIO guidance.
0x08	PD performs off-card facial comparison using ANSI INCITS 385-2004 images stored on the credential.

DRAFT

3 New or Enhanced PIV Commands

This section introduces commands defined exclusively for the OSDP 3.0 Enhanced PIV profile.

Unless specified otherwise, each multi-part command or reply defined in this section places *MpSizeTotal*, *MpOffset*, and *MpFragmentSize*, in that order, immediately after *CMND*. Each field is a two-byte unsigned integer encoded least-significant byte first. The bytes following *MpFragmentSize* are a fragment of the logical message data: *MpSizeTotal* is the complete logical-data length, *MpOffset* is the zero-based position of the fragment, and *MpFragmentSize* is the exact number of fragment bytes in the current message. The receiver **SHALL** apply the OSDP multi-part sequencing and bounds rules and reassemble the complete logical data before interpreting command-specific fields. Fixed fields shown after the multi-part fields in a packet table occur at their stated position in the reassembled data and are not repeated in every fragment. The separate `osdp_FILETRANSFER` framing remains unchanged.

3.1 Credential preparation and Secure Messaging

OSDP Secure Channel protects communication between the ACU and PD. Credential Secure Messaging (SM) protects credential commands and replies between the PD and the credential. Credential SM can operate over contact or contactless. The Virtual Contact Interface (VCI) allows a contactless credential to perform operations that otherwise require contact, subject to the credential's access rules. When the credential requires pairing, the PD establishes SM before submitting the eight-digit pairing code. Pairing is the default, recommended configuration in SP 800-73-5 Part 1, Section 5.5. The pairing code and the cardholder's PIN serve separate purposes: pairing enables VCI; PIN verification authorizes PIN-protected operations.

Credential presentation The PD **SHALL** select the configured application on an interface allowed by `osdp_PIVMODE`. For a PIV application, the PD **SHALL** inspect the SELECT response's application property template (tag `0x61`). Its cryptographic algorithm template (tag `0xAC`) contains algorithm identifiers in tag `0x80`. The PD **SHALL** examine all such entries. An identifier of `0x27` or `0x2E` advertises credential SM; SP 800-73-5 Part 2, Section 3.1.1, requires this indication when the PIV application supports SM. The PD **SHALL** distinguish a malformed response from a valid response that advertises neither SM algorithm.

Discovery Object The Discovery Object (`0x7E`) describes the credential's PIN and VCI policy. Its PIN Usage Policy (tag `0x5F2F`) identifies the supported PIN options, the preferred PIN when both PIN types are enabled, and whether VCI requires pairing. The PD uses this information to prepare PIN and VCI operations. SP 800-73-5 Part 1, Section 3.3.2, defines the policy bytes.

The PD **SHALL** read the Discovery Object when preparing a credential-side PIN operation or attempting SM setup, unless the PD already holds the policy for the selected application on the current credential. Automatic reads **SHALL** be limited to those workflows. Collecting a PIN for later use requires no credential read. A status query reports the policy already read, or `0xFFFF` when unavailable. The ACU can also request the object explicitly with `osdp_PIVGETDATA`.

In the first policy byte, mask `0x08` indicates VCI support. When VCI is supported, mask `0x04` permits VCI without a pairing code; otherwise pairing is required. If the object is absent, VCI is unavailable. The SELECT response remains the source of SM capability information.

Automatic SM preparation After selecting a PIV application, the PD checks the requirements for SM setup. If PIVMODE flag `0x02` is set, the PD supports an SM algorithm advertised by the credential, and at least one trust anchor is loaded, the PD **SHALL** attempt SM setup. This applies to both contact and contactless credentials, in commanded mode and PIV Auto.

Selecting a different application starts this check again. Selecting the same PIV application again preserves the current SM session and PIN-verification status. Reselection alone therefore requires neither new SM setup nor repeated PIN verification. The credential's access rules still apply to each operation. These selection rules follow SP 800-73-5 Part 2, Section 3.1.1.

The PD's trust store contains the anchors loaded through `osdp_PIVVCILOADTA`. The PD checks its own store before starting SM. An empty store prevents setup. If anchors are present, the PD must still find one that can validate this credential's certificate, directly or through an intermediate certificate. Loading an anchor makes it available for the next attempt. To start a new attempt with a credential already present, the ACU reapplies ~~PIVMODE~~ordinary PIVMODE with a fresh Auto Series; the mode-change clearing rules apply. The Resume extension in Subsection 3.12 continues only an eligible pairing/PIN prerequisite failure and does not restart blocked SM.

The credential returns its SM card-verifiable certificate (CVC) during key establishment. The PD **SHALL** validate that certificate using a loaded anchor, directly or through an intermediate CVC, and verify the authentication cryptogram as specified in SP 800-73-5 Part 2, Section 4.1. The PD **SHALL** report SM as established only after those checks succeed.

Proceeding without SM When SM is disabled, unsupported, or unavailable because a trust anchor is missing, the PD **SHALL** check whether the operation requested by the ACU or configured in PIV Auto requires SM. If SM is required, the PD **SHALL** terminate the operation without sending the operation's command to the credential and return the defined failure reply. Otherwise, the PD **SHALL** perform the requested operation under the credential's access rules. The PD **SHALL NOT** substitute another key, algorithm, interface, or operation to avoid an SM requirement.

PIV key `0x9A` requires PIN verification or a supported biometric comparison within the credential, as specified in SP 800-73-5 Part 1, Section 5.1, Table 5. The PIN-based contactless PIV Auto workflow with key `0x9A` therefore requires SM, VCI, and successful PIN verification. The PD **SHALL** check those requirements in that order before sending the authentication command. Missing PIN verification returns Result `0x05`, Detail `0x02` (PIN verification required). If SM is unavailable, the PD reports `osdp_PIVSTATUS` Result `0x02` (SM failure) with the applicable detail code. PIV Auto with key `0x9E` can continue without SM and reports the credential-authentication result. A commanded PIN verification over contact can also proceed without SM.

The PD may start key establishment before discovering that the required trust anchor is missing. In that case, the PD **SHALL** discard any session keys derived during setup. The PD then handles the requested operation using the rules above.

In commanded mode, the ACU obtains the preparation result with `osdp_CARDSTATUS`. The PD reports the established SM and VCI state in `osdp_CARDSTATUSR`. A missing anchor leaves permitted data reads available. For example, the ACU can use `osdp_PIVGETDATA` to read the Secure Messaging Certificate Signer object (`0x5FC122`) and inspect the signer certificate and any intermediate CVC. The credential's own SM CVC

is returned during key establishment; that CVC is separate from the signer data object. The ACU can request `osdp_CARDSTATUS` Status Type `0x02` to obtain the setup outcome and the hash of the CVC received during that attempt, including when setup failed.

Failed security checks A missing or invalid credential CVC, invalid authentication cryptogram, or failure to verify an SM reply **SHALL** end the current operation with a failure result. The PD **SHALL** block further data access and authentication commands until the credential is removed, a different application is successfully selected, or PIVMODE is reapplied. PIVSELECT remains available to select a different application; selecting the same application preserves the blocked state. This prevents continued use of a session that failed its security checks. The rule also applies when automatic SM setup fails before an ACU command arrives. PIV-MODE, trust-store management, and SM diagnostic requests remain available. For a standard status request (Status Type `0x01`), the PD terminates the request without sending further commands or data to the credential and returns `osdp_PIVERROR` `0x1023` (credential security conditions not satisfied). This distinguishes a blocked credential from operation without SM.

For blocked operations, the PD reports `osdp_PIVSPER` `0x08` (security condition not met) for SPE, or `osdp_PIVERROR` `0x1023` (credential security conditions not satisfied) for commands that use PIVERROR.

If an operation requires SM, VCI, or PIN verification and that requirement is unmet, the PD **SHALL** report failure in the operation's completion reply. PIV Auto uses `osdp_PIVSTATUS`; SPE uses `osdp_PIVSPER`.

Figure 2 summarizes credential preparation and the decision to proceed with the requested operation.

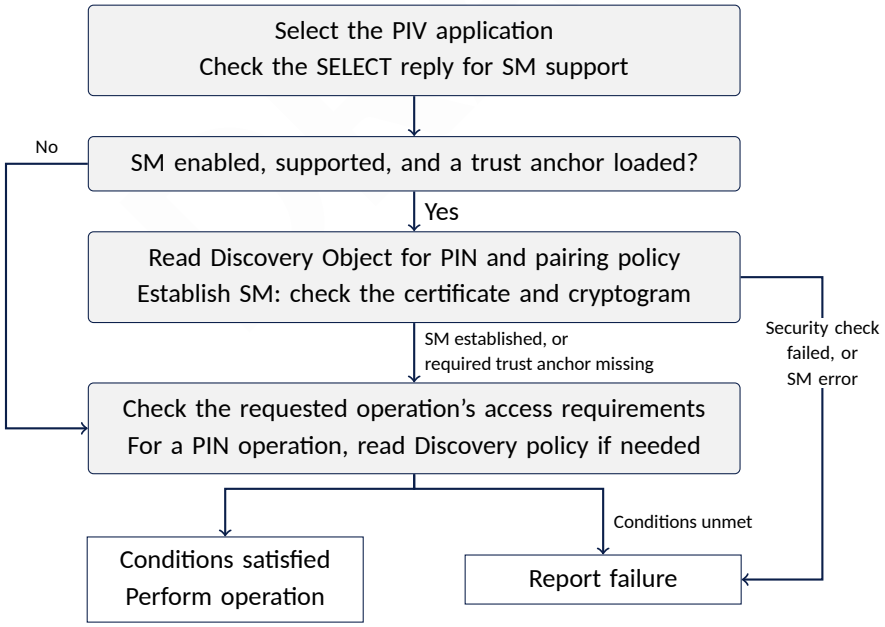


Figure 2: How the PD prepares a credential. The PD establishes SM when enabled and supported, using a loaded trust anchor. The credential's access rules determine which operations can proceed. Contactless PIN verification requires VCI.

Contactless operation When pairing is required, the ACU sends `osdp_PIVXMITPAIRING` after SM is established. The PD submits the pairing code to the credential through SM. A successful verification enables VCI. For a credential whose Discovery Object permits VCI without pairing, establishing SM also enables VCI. PIN verification over contactless requires VCI. Biometric comparison within the credential and pairing-code submission can use contactless SM without VCI, subject to the credential's access rules. SM and VCI preserve the credential's operation-specific restrictions, including the contact-interface restriction on ordinary PIV PUT DATA.

Protected credential commands When credential SM is established, the PD **SHALL** use that SM session to protect PIV commands that support SM. The PD **SHALL** verify and unwrap each protected reply before interpreting the credential data or status. These requirements apply to data access, authentication, PIN verification, and biometric operations. The PD **SHALL** follow SP 800-73-5 Part 2, Section 4.2, for encryption, MACs, counters, and credential-command and response chaining. The PD's GET RESPONSE exchanges follow that section's chaining rules.

The PD **SHALL** discard the SM session keys on credential reset, an SM error, or replacement by newly established session keys, as specified in SP 800-73-5 Part 2, Section 4.3. This requirement applies to the outer SW processing status defined in Section 4.2.7. The protected command status in tag `0x99` reports the result of the credential operation separately. When SM ends, the PD **SHALL** clear dependent VCI and PIN-verification status and apply the PIN-cache clearing rules.

A verified protected reply reporting a credential rejection **SHALL** end the requested operation with that failure while preserving SM. The PD **SHALL** update verification status according to the credential's response. An incorrect PIN (`0x63Cx`) clears PIN-verification status and the cached PIN; established pairing and VCI remain available. An incorrect pairing code (`0x6300`) clears pairing verification, making VCI unavailable when pairing is required. These credential rules are defined in SP 800-73-5 Part 2, Sections 3.2.1.1 and 3.2.1.3. The SM diagnostic outcome remains unchanged. A failed certificate, cryptogram, or protected-reply check follows the blocked-state rules above.

After an authenticated PIN rejection, the ACU can query remaining attempts with PIVSPE Mode `0x06`, acquire a corrected PIN, and request verification again. After a pairing rejection, the ACU can submit a corrected pairing code. Neither retry requires a new SM session.

After an SM failure, the ACU **SHALL** reapply PIVMODE to start a new SM attempt before retrying a protected operation. For a commanded retry, the ACU disables PIV Auto in that configuration, waits for SM establishment, supplies any required pairing code, and requests PIN acquisition and verification again. The PD **SHALL NOT** retry VERIFY automatically. Figures 3 and 4 show the ACU and PD exchanges after authenticated rejection. Figure 5 shows recovery after SM failure. The PD **SHALL** keep a completed operation's reply until it is delivered or canceled.

Commanded PIN verification For a PIN entered before credential presentation, the ACU first requests Mode `0x04` (acquire and cache) and waits for completion. After credential presentation, the PD prepares the credential. The ACU checks credential status and supplies any required pairing code before requesting cached-PIN verification. Figure 6 shows a successful contactless workflow with SM enabled, a suitable trust

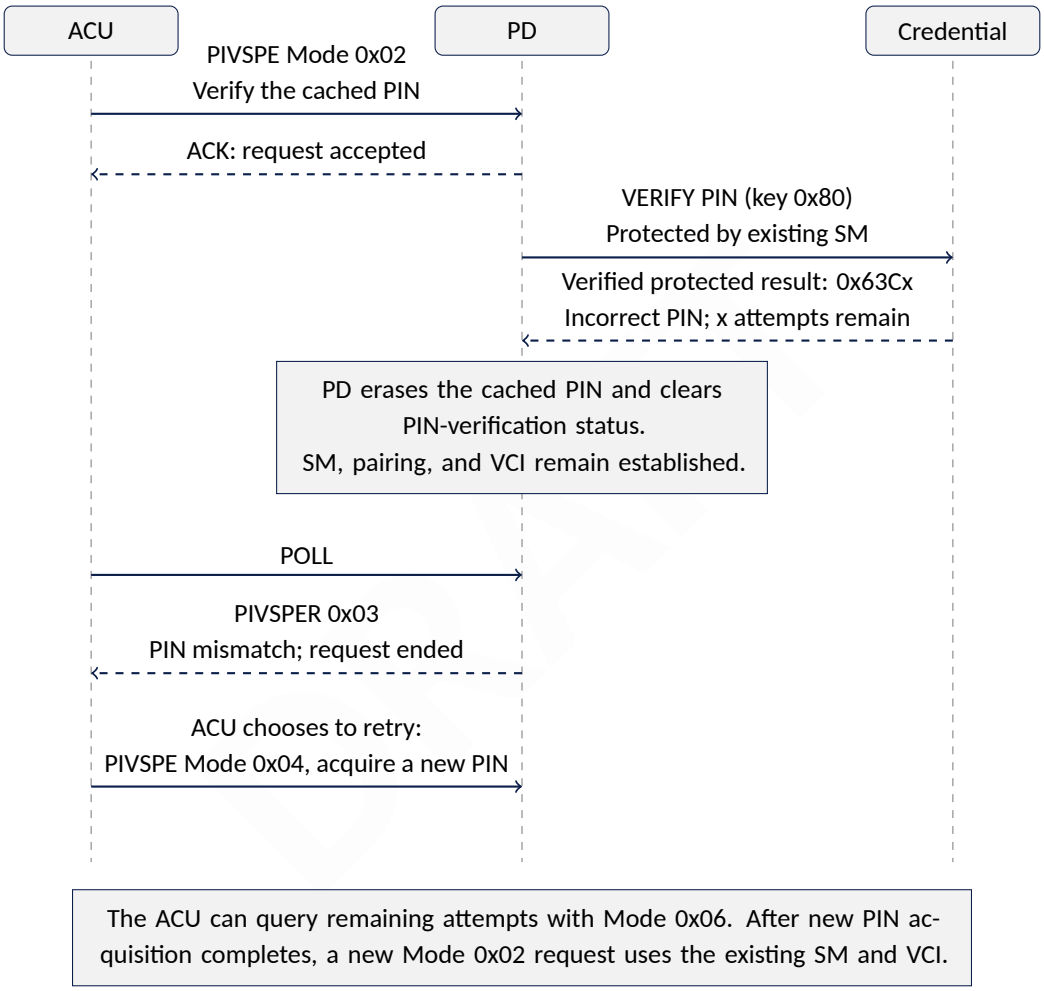


Figure 3: Incorrect PIN over established VCI. The PD verifies the protected reply before reporting the rejection. PIN entry occurs at the PD; the ACU never receives the PIN. Acquisition and its completion follow the normal ACK/POLL exchange.

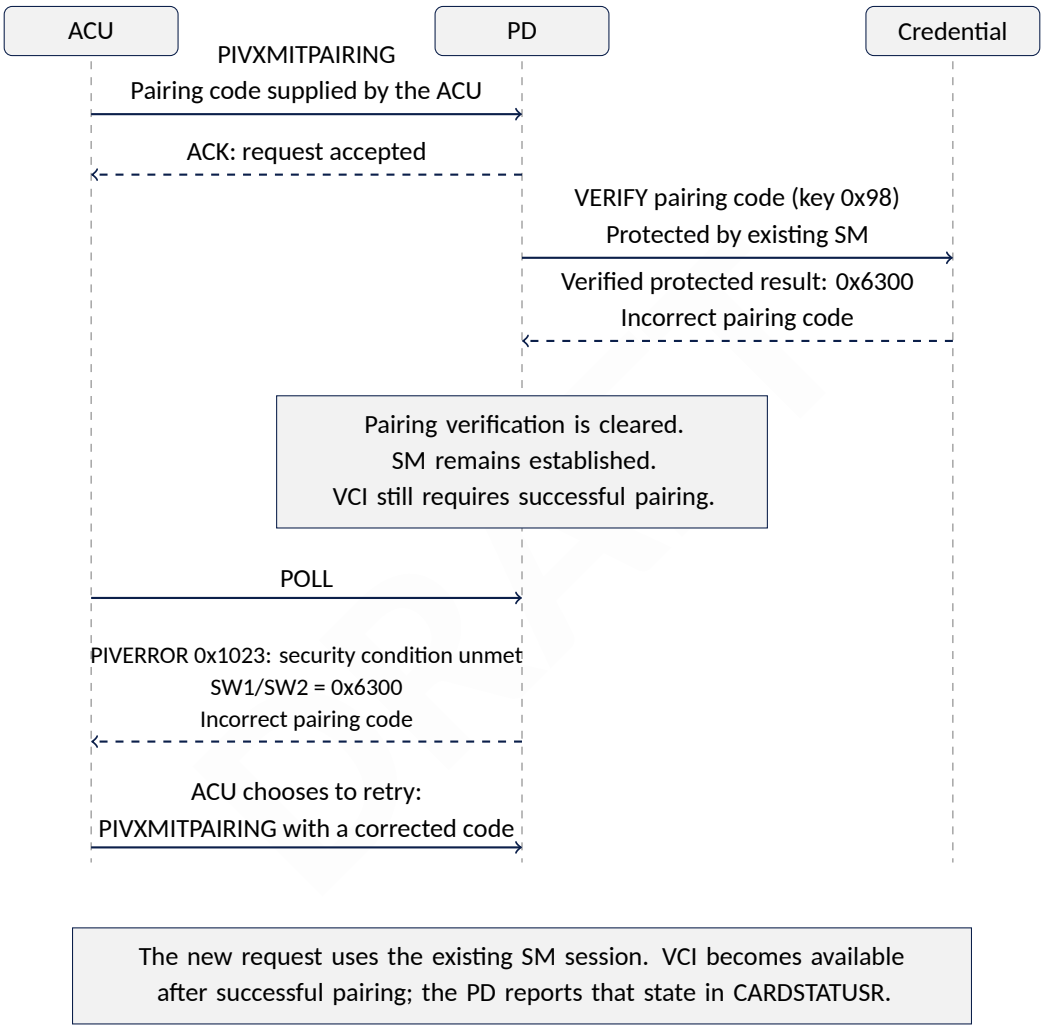


Figure 4: Incorrect pairing code on a contactless credential that requires pairing. The pairing code originates at the ACU. Each retry is a new command; the PD never repeats VERIFY automatically.

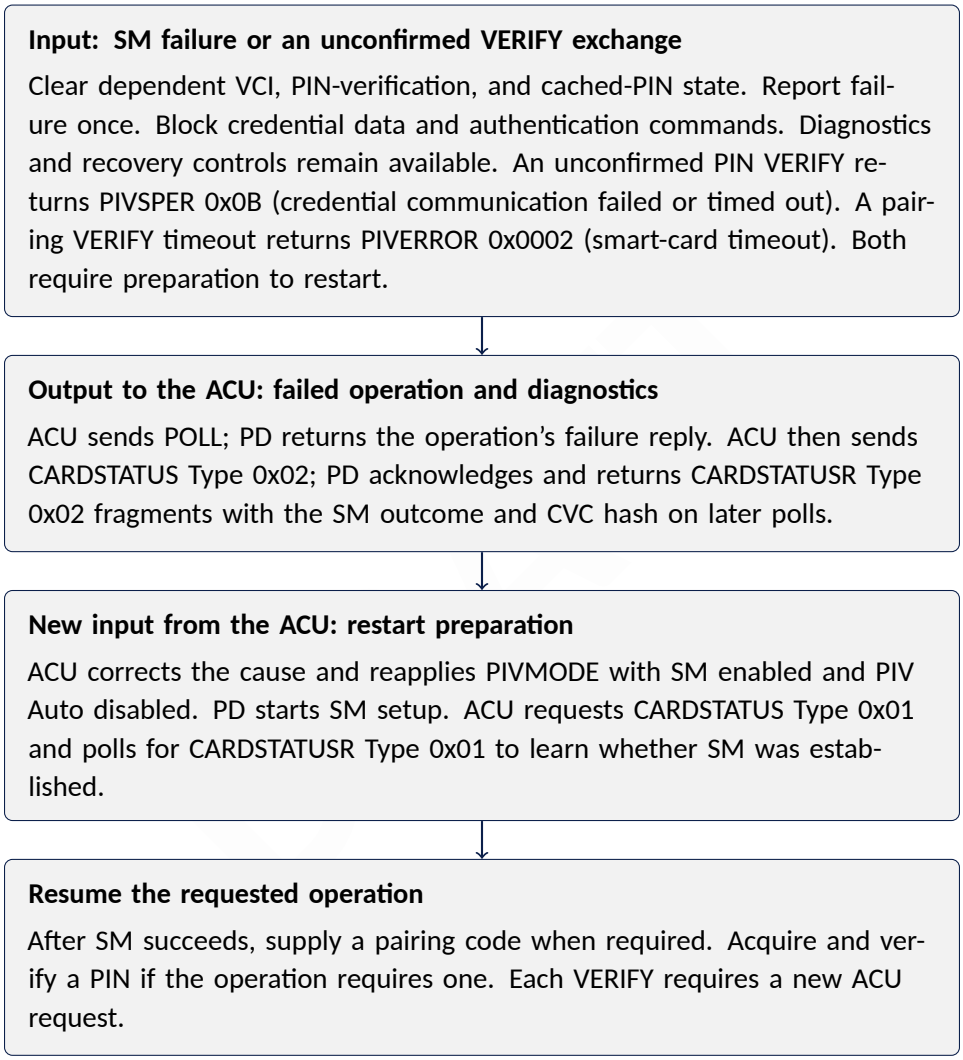


Figure 5: Recovery after SM failure or an unconfirmed VERIFY. The ACU receives the failed operation's result before requesting diagnostics and restarting preparation.

anchor loaded, and PIV Auto disabled. The ACU obtains the pairing code during enrollment or through another deployment-defined process. On contact, pairing is omitted and Mode 0x03 uses the selected contact interface, with SM when established.

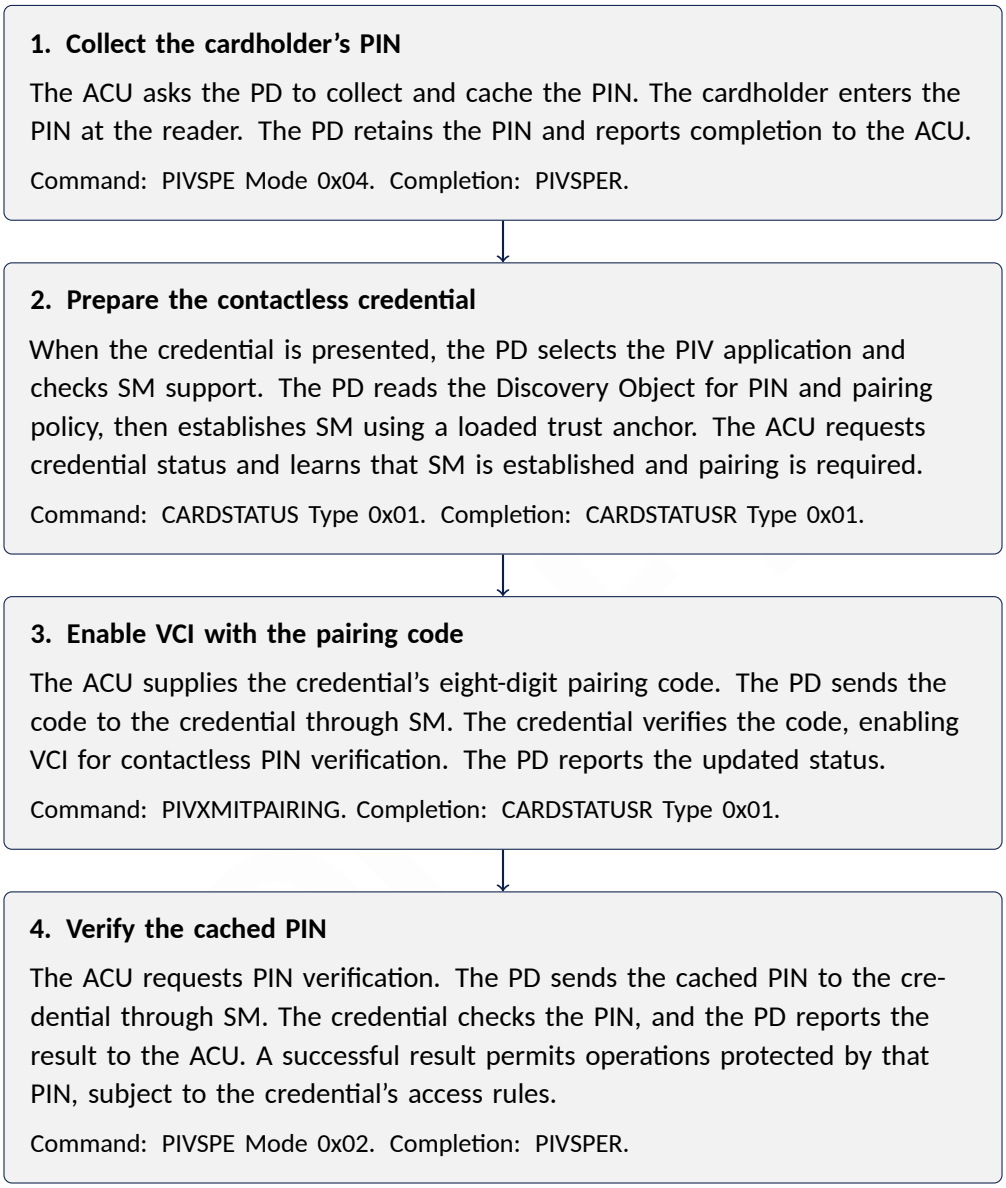


Figure 6: Contactless PIN verification with a pairing code. The ACU directs the reader (PD); the credential verifies the pairing code and PIN. Each OSDP command receives ACK when accepted and a completion reply on a later POLL. Appendix D shows the packet exchanges.

ACU recovery after a PIV Auto failure PIV Auto reports one result for the configured authentication attempt. The ACU can then request SM diagnostics, correct missing trust material, or choose another authentication workflow. Loading an anchor makes that anchor available to the PD. Reapplying ordinary PIVMODE starts a new automatic attempt and clears the previous PIN and SM state. The ACU **SHALL** finish trust recovery before pairing and PIN verification that depend on the new SM session.

After complete delivery of an eligible PIVSTATUS 0x05/0x01 or 0x05/0x02, the ACU may complete commanded pairing and PIN verification while Auto remains enabled, then send PIVMODE Resume with that status's pair and the next unused counters. Resume preserves completed SM, VCI, pairing, and PIN state instead of ordinary mode-change clearing. It performs one additional attempt with the same entry and key, without automatic VERIFY or authentication retry. Context loss, blocked SM, and counter exhaustion follow Subsection 3.12; they do not permit implicit fallback.

When SM is established but the configured authentication requires unsupported VCI, the PD **SHALL** terminate the attempt and report osdp_PIVSTATUS Result 0x05, Detail 0x03 (required VCI unsupported). The PD sends no authentication command for that attempt. Missing SM support is reported as Result 0x02, Detail 0x05 (required SM unsupported). A trust-anchor update cannot add either capability. The ACU must select a supported workflow or reject the transaction. Appendix D.12 shows trust recovery followed by commanded pairing, PIN verification, and authentication.

3.2 PIV Put Data (osdp_PIVPUTDATA)

The osdp_PIVPUTDATA command (command code 0xA8) instructs the PD to deliver an ISO 7816-4 PUT DATA operation to the credential. The ACU **SHALL** construct the payload as a complete BER-TLV object, including the desired outer tag (0x7E for the Discovery Object, 0x7F61 for a BIT Group Template, 0x5C/0x53 sequences for other data objects, and so on). The PD **SHALL** forward the payload to the credential using CLA = 0x00, INS = 0xDB, P1 = 0x3F, and P2 = 0xFF. If credential Secure Messaging (SM) is established, the PD **SHALL** use the established SM context and adjust the CLA accordingly. Chained APDUs **SHALL** be generated by the PD when the payload exceeds a single-card APDU.

Table 10: PIV Put Data (osdp_PIVPUTDATA)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA8 (tentative)
0x02	MpSizeTotal	Total size of the complete PUT DATA payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Number of payload bytes carried in this fragment, least-significant byte first.	0x0000–0xFFFF
0x00-n	Data	BER-TLV payload to be written to the credential. This field already includes the outer object tag selected by the ACU.	0x00–0xFF

Responses PIV PUT DATA requires the contact interface, including when SM is established. On contactless, the PD **SHALL** terminate the request without sending PUT DATA to the credential and return

osdp_PIVERROR 0x1023 (security conditions not satisfied). A successful write is acknowledged with osdp_ACK. If the PD does not implement the command, the PD **SHALL** return osdp_NAK function-not-supported. For a credential-side failure, the PD **SHALL** report osdp_PIVERROR, either immediately or on a subsequent osdp_POLL. The applicable codes are 0x1020 (function not supported by credential), 0x1021 (insufficient PD buffer space), 0x1022 (insufficient credential storage), 0x1023 (security conditions not satisfied), and 0x1026 (credential absent or removed). The status-word rules in Subsection 2.2.1 apply.

3.3 PIV Application Select (osdp_PIVSELECT)

The osdp_PIVSELECT command (command code 0xB0, tentative) instructs the PD to select the application identified by the supplied Application Identifier (AID) on the currently active credential. The command provides an OSDP-native selector for ISO/IEC 7816-4 SELECT by DF name; it is not a raw APDU tunnel and does not expose APDU header bytes to the ACU. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV) before controllers attempt to use this command.

Table 11: PIV Application Select (osdp_PIVSELECT)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xB0 (tentative)
0x01	Selection Flags	Selection controls. All bits are reserved for future use and **SHALL** be transmitted as zero.	0x00
0x01	Data Model	Data model for subsequent commands: 0x01 PIV, 0x02 TWIC, 0x03 CAC; 0x04-0x7F reserved; 0x80-0xFF private use. 0x00 is invalid.	0x01-0xFF
0x01	AID Length	Length of the Application Identifier.	0x05-0x10
0x05-0x10	AID	Application Identifier to select by ISO/IEC 7816 DF name.	0x00-0xFF

The ~~PD~~ *Data Model* field uses the same identifiers as a osdp_PIVMODE Configuration Entry: 0x01 PIV, 0x02 TWIC, and 0x03 CAC. Values 0x04-0x7F are reserved; 0x80-0xFF are private-use models supported by agreement between the ACU and PD. 0x00 is invalid. The ACU supplies the model and AID together; a matching osdp_PIVMODE Configuration Entry is not required. The command retains the current interface and mode-wide security settings. The model does not override the credential's access conditions. The payload is exactly four fixed bytes, including *CMND*, followed by *AID Length* bytes, for a total of 0x09-0x14 bytes. This layout replaces the earlier draft without a Data Model byte; PDs **SHALL NOT** interpret that earlier layout as this command.

The PD **SHALL** issue SELECT with CLA = 0x00, INS = 0xA4, P1 = 0x04, P2 = 0x00, Lc = AID Length, Data = AID, and Le = 0x00. The PD **SHALL** request the normal application property template over the interface selected by the current PIV mode. A selection within an established SM context follows the shared transport rules in Subsection 3.1. When ~~the credential selects a different application~~ a successful

selection changes the application or Data Model, the PD **SHALL** clear the previous application's SM session, VCI status, pairing code, cached PIN, PIN-verification status, and application information. This prevents the PD from carrying security state from one application into another. Selecting the PIV application when a credential is first presented preserves a PIN collected before presentation. Selecting the same PIV application again with the same Data Model preserves its SM session and PIN-verification status. If the credential rejects SELECT, the PD keeps the current application and its state.

The *AID Length* field **SHALL** be in the ISO/IEC 7816 application identifier range of 5 to 16 bytes. If the payload is malformed, its size is not exactly `0x04` plus AID Length, the AID length is outside that range, ~~or~~ the Data Model is zero, reserved, or unsupported by the PD, or any reserved selection flag is nonzero, the PD **SHALL** report `osdp_PIVERROR` `0x0005` (invalid parameter) without issuing SELECT to the credential. A successful selection is acknowledged with `osdp_ACK`~~;~~. Only after success, ~~subsequent PIV commands operate against the selected application until the credential is removed, a new `osdp_PIVSELECT` succeeds, or `osdp_PIVMODE` changes the operating~~ the PD **SHALL** commit the selected Data Model and AID as one operating context. Subsequent commands use that context until credential removal, another successful selection, or a mode change. A rejected command or credential SELECT **SHALL NOT** change the previous context. The PD **SHALL** update the selected AID reported by `osdp_CARDSTATUSR`; if the credential returns an FCI template containing tag `0x4F`, that value **SHALL** be reported, otherwise the PD **SHALL** report the requested AID. The ACK and CARDSTATUSR layouts are unchanged.

~~If the~~ On a TWIC credential, the PIV and TWIC applications have distinct AIDs. The ACU determines the intended application through system configuration, using an `osdp_PIVMODE` configuration entry or `osdp_PIVSELECT`. For `osdp_PIVSELECT`, the ACU supplies Data Model `0x01` with the PIV AID, or Data Model `0x02` with the TWIC AID; `0x02` covers both Legacy and NextGen TWIC applications. The PD **SHALL** apply the data model and access conditions of the selected application to subsequent commands. Selecting the PIV application invokes normal PIV processing, including PIN, SM, and VCI requirements where applicable. Biometric data access through the TWIC application does not require the PIV PIN; deciphering its encrypted biometric data requires the appropriate TPK.

Informative selection examples The following hexadecimal command payloads omit OSDP framing and Secure Channel for readability. The bytes are CMND, Selection Flags, Data Model, AID Length, then AID; neither example depends on a matching PIVMODE profile.

- Select the PIV application, including the PIV application on a TWIC:
  `B0 00 01 0B A0 00 00 03 08 00 00 10 00 01 00.`
- Select the TWIC application using its common nine-byte AID prefix:
  `B0 00 02 09 A0 00 00 03 67 20 00 00 01.`

After ACK, CARDSTATUSR reports the selected AID. A full TWIC AID ending in `01 01` identifies Legacy; `01 03` identifies NextGen. If only the requested prefix is available, it does not identify the generation; the ACU **SHALL NOT** infer NextGen facial support from that prefix alone. These AID distinctions are specified in *TWIC Specification, Part 2*, Section 4.1 and Appendix C.

If the PD does not implement the command, the PD **SHALL** return osdp_NAK function-not-supported. Credential-side failures **SHALL** be reported using osdp_PIVERROR: 0x1020 (application selection unsupported), 0x1023 (security conditions not satisfied), 0x1024 (requested AID absent), or 0x1026 (credential absent or removed). The status-word rules in Subsection 2.2.1 apply.

3.4 Card Status (osdp_CARDSTATUS)

The osdp_CARDSTATUS command (command code 0xA9, tentative) requests the current credential context from the PD. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV) before controllers attempt to use this command. The payload selects standard credential status or SM diagnostics.

Table 12: Card Status Request (osdp_CARDSTATUS)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA9 (tentative)
0x01	Status Type	0x01: standard PIV credential status. 0x02: SM diagnostic record. Values 0x00 and 0x03-0x7F are reserved and return osdp_PIVERROR 0x0005 (invalid parameter). Values 0x80-0xFF are reserved for private use.	0x00-0xFF

PDs **SHALL** return osdp_NAK function-not-supported when the command itself is not implemented. If the requested status type is not supported or the payload is malformed, PDs **SHALL** report the condition via osdp_PIVERROR (for example 0x0005 (invalid parameter)) on a subsequent osdp_POLL.

3.5 Card Status Reply (osdp_CARDSTATUSR)

Every osdp_CARDSTATUSR reply **SHALL** include Status Type immediately after the reply code. Type 0x01 selects the standard credential status fields below. Type 0x02 selects the diagnostic multi-part layout in Subsection 3.5. The ACU **SHALL** select the reply layout from this byte and verify that the type matches the pending request. An unknown or mismatched type **SHALL** be treated as an invalid reply; the ACU discards the reply and any partially reassembled record. The ACU can cancel the pending operation with ABORT and issue a new request.

The PD accepts osdp_CARDSTATUS Type 0x01 with osdp_ACK and returns osdp_CARDSTATUSR on a subsequent osdp_POLL after the current credential preparation completes. The PD **SHALL** use a finite, documented preparation timeout. If SM is established and the credential requires a pairing code, the PD reports that state so the ACU can supply the code. The ACU **SHALL** interpret the following combination as pairing required: Detected Credential Type 0x02, Credential Security Status 0x02, and an available PIN Usage Policy whose first byte has mask 0x08 set and mask 0x04 clear. Security Status 0x02 alone indicates SM without VCI; contact SM requires no pairing. While status is pending, the ACU sends only osdp_POLL or osdp_ABORT.

osdp_CARDSTATUSR also completes an accepted osdp_PIVXMITPAIRING command, using Status Type 0x01. Controllers continue to use osdp_RAW for ordinary card-data indications. Whenever credential SM is established, the status reply reports the CVC IIN and the loaded anchor IIN, including for contact SM. The values match for direct verification and differ when an Intermediate CVC is used. The *SM Trust Anchor ID* identifies the corresponding osdp_PIVVCITASTATUSR inventory entry, whose IIN matches *SM Anchor IIN*.

Table 13: Card Status Reply (osdp_CARDSTATUSR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x85 (tentative)
0x01	Status Type	Standard credential status, including successful pairing completion.	0x01
0x01	Detected Credential Type	Current card interface: 0x00 none, 0x01 ISO 7816 contact, 0x02 ISO 14443 contactless, 0x03 other ISO 7816 credential (for example DESFire, Seos), 0x04 low-frequency credential, 0x05 unknown type, 0x06-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
0x01	Credential Security Status	0x00 SM absent; 0x01 SM established and VCI available; 0x02 SM established without VCI. Contact SM uses 0x02. Values 0x03-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
0x01	SM Trust Anchor ID	Identifier assigned in osdp_PIVVCILOADTA for the anchor used to validate credential SM. The anchor's inventory IIN matches *SM Anchor IIN*. Set to 0x00 when SM is absent.	0x00–0xFF
0x08	SM CVC IIN	Issuer Identification Number from the credential SM CVC tag 0x42. Set to all zero when SM is absent.	0x00–0xFF
0x08	SM Anchor IIN	Issuer Identification Number of the loaded trust anchor used to validate credential SM. Set to all zero when SM is absent.	0x00–0xFF

Size (bytes)	Name	Meaning	Value
0x02	PIN Usage Policy	PIN Usage Policy bytes (tag 0x5F2F) returned verbatim when available. A value of 0xFFFF indicates the policy is not present or has not yet been read.	0x0000–0xFFFF
0x01	Selected AID Length	Length of the selected AID. 0x00 indicates no application is currently selected.	0x00–0x10
0x00-0x10	Selected AID	Application Identifier read from the credential (for example from the FCI template). Present only when the length is non-zero.	0x00–0xFF

Each accepted Type 0x01 status request **SHALL** complete with either osdp_CARDSTATUSR or osdp_PIVERROR. If a failed SM security check has blocked credential operations, the PD **SHALL** terminate a Status Type 0x01 request without sending further commands or data to the credential. The PD **SHALL** return osdp_PIVERROR 0x1023 (credential security conditions not satisfied) on the next poll. Otherwise, the standard status reply includes the selected AID when available and any previously read PIN Usage Policy bytes from Discovery Object tag 0x5F2F, unchanged. The status request itself requires no Discovery Object read. The PD reports 0xFFFF when the policy is unavailable. Security status reports the result of credential preparation. With no credential present, the PD reports credential type and security status zero, AID length zero, and PIN Usage Policy 0xFFFF. Removal during preparation returns osdp_PIVERROR 0x1026 (credential absent or removed). A preparation timeout returns osdp_PIVERROR 0x0002 (smart-card timeout). osdp_ABORT cancels the request with osdp_ACK and clears its pending completion reply.

SM diagnostics (Status Type 0x02) The ACU uses this status type to determine why SM setup failed and identify the CVC received by the PD. The PD **SHALL** accept this request while credential operations are blocked by a failed security check. The PD **SHALL** return the most recent setup outcome and the CVC hash for the current credential and selected application. This request reads retained PD information; the PD **SHALL NOT** send a command to the credential in response to the diagnostic request.

The PD acknowledges the request with osdp_ACK, then returns osdp_CARDSTATUSR on subsequent polls. Status Type 0x02 uses the OSDP 3.0 multi-part format in Table 14. Every diagnostic fragment **SHALL** contain Status Type 0x02 immediately after the reply code, followed by MpSizeTotal, MpOffset, and MpFragment-Size. Status Type identifies the layout before the ACU reads the multi-part fields. The repeated Status Type byte is outside the diagnostic record and **SHALL NOT** be counted in the multi-part lengths or offsets.

Table 14: SM Diagnostic Reply (CARDSTATUSR, Status Type 0x02)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x85 (tentative)
0x01	Status Type	SM diagnostics. Repeated in every fragment, outside the diagnostic record.	0x02
0x02	MpSizeTotal	Total diagnostic record length, least-significant byte first.	0x0023
0x02	MpOffset	Offset within the diagnostic record, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Diagnostic bytes in this reply, least-significant byte first.	0x0001–0xFFFF
0x01	Outcome	0x00: setup not attempted; 0x01: SM established; 0x02: SM unavailable; 0x03: security check or SM processing failed, credential operations blocked.	0x00–0x03
0x01	Detail	Reason from Table 15.	0x00–0x07
0x01	CVC Hash Present	0x01 when the PD received a complete CVC; otherwise 0x00.	0x00–0x01
0x20	CVC Hash	SHA-256 of the received CVC, including tag 0x7F21, its encoded length, and value, unchanged. All zero when CVC Hash Present is 0x00.	0x20 bytes

Outcome values 0x04-0xFF and Detail values 0x08-0xFF are reserved for future use.

Table 15: SM Diagnostic Detail Codes

Value	Description
0x00	No failure detail. Used with Outcome 0x00 or 0x01.
0x01	Trust store empty. Outcome 0x02; CVC Hash Present is zero because setup was skipped.
0x02	No matching trust path through the loaded anchors. Outcome 0x02. The ACU can inspect the signer object to determine whether an anchor or intermediate CVC is missing.
0x03	CVC missing, malformed, or invalid. Outcome 0x03. CVC Hash Present is zero when no complete CVC was received.
0x04	Authentication cryptogram invalid. Outcome 0x03.

Value	Description
0x05	No credential SM algorithm supported by both the PD and credential. Outcome 0x02.
0x06	Other SM processing failure. Outcome 0x03.
0x07	SM disabled by PIVMODE. Outcome 0x02.

The diagnostic record consists of Outcome, Detail, CVC Hash Present, and CVC Hash, in that order. The PD **SHALL** assemble the complete record before dividing the record into reply fragments. The multi-part fields describe that complete record. A fragment may end within any field. The ACU **SHALL** reassemble the record before interpreting the fields.

The PD **SHALL** compute and retain the hash of a complete received CVC even when certificate or cryptogram validation fails. The hash identifies the received bytes; the hash provides no evidence that the CVC is trusted. The diagnostic record requires only the outcome, detail, presence flag, and 32-byte hash. The PD has no obligation to retain the CVC or other credential objects for later retrieval. PIVGETDATA reads the requested object from the credential through the active interface. If no complete CVC was received, the PD **SHALL** set CVC Hash Present to zero, fill CVC Hash with zero bytes, and report the applicable reason.

The PD **SHALL** keep the diagnostic record until credential removal or reset, PD reset, successful selection of a different application, or reapplication of PIVMODE. A later SM failure updates Outcome and Detail while retaining the CVC hash from that session. Loading a trust anchor leaves the previous outcome unchanged. Before a setup attempt, the record has Outcome 0x00, Detail 0x00, CVC Hash Present zero, and an all-zero CVC Hash.

The ACU **SHALL** receive any pending PIV Auto result before requesting diagnostics. While a diagnostic reply is pending, the ACU sends only POLL or ABORT. A competing request receives osdp_NAK 0x09 (unable to process command). ABORT cancels the transfer with ACK. Credential removal during transfer terminates the transfer with osdp_PIVERROR 0x1026 (credential absent or removed); the ACU discards any partial record.

3.6 PIV Secure PIN Entry (osdp_PIVSPE)

The osdp_PIVSPE command (command code 0xAA, tentative) controls Secure PIN Entry for the Enhanced PIV profile. A PD **SHALL** advertise both Function Code 12 bit 0x04 (Enhanced PIV) and Function Code 15 Secure PIN Entry support before an ACU uses this command. The OSDP 2.2 osdp_XWR Mode-01 SPE command and osdp_PR01SPER reply remain unchanged.

The ACU is responsible for conveying a clear indication that PIN entry is in progress. While the PD is acquiring a PIN, it **SHALL** suppress osdp_KEYPAD reports so that no PIN digit is sent to the ACU.

Table 16: PIV Secure PIN Entry (osdp_PIVSPE)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAA (tentative)
0x01	SPE Mode	Secure PIN Entry operating mode. See Table 17.	0x00–0xFF
0x01	Start Timeout	Seconds allowed to begin key entry; 0x00 selects the documented PD default (a value from 0x01 through 0xFF seconds).	0x00–0xFF
0x01	Inter-key Timeout	Seconds allowed between keystrokes; 0x00 selects the documented PD default (a value from 0x01 through 0xFF seconds).	0x00–0xFF
0x01	Flags	Behaviour modifiers. See Table 18.	0x00–0xFF
0x02	Cache Validity	PIN cache validity in seconds (LSB first). Mode 0x04 requires 0x0001-0xFFFF; other modes transmit 0x0000.	0x0000–0xFFFF

Table 17: SPE Mode Values

Value	Description
0x00	Report the persistent SPE status without initiating PIN entry.
0x01	Acquire a PIN and verify it with the credential currently present.
0x02	Verify the cached PIN over established VCI.
0x03	Verify the cached PIN over the selected credential interface. Use established credential SM when present.
0x04	Acquire and cache a PIN.
0x05	Abort active PIN entry and clear the cached PIN.
0x06	Report attempts remaining without initiating PIN verification.
0x07–0x7F	Reserved for future use.
0x80–0xFF	Private use.

Table 18: SPE Flag Values

Mask	Description
0x01	Retain a cached PIN across credential removal until its validity expires or another clearing event occurs. Valid only with Mode 0x04.
0x02	Auto Execute When Credential Is Presented. Valid only with verification Modes 0x01-0x03.
0x04–0x80	Reserved flag bits; **SHALL** be transmitted as zero.

Modes are enumerated values. Modes 0x02 and 0x03 submit the cached PIN using PIV VERIFY. Mode 0x02 requires established VCI; Mode 0x03 uses the credential interface selected by the active PIV mode. Mode 0x01 acquires the PIN and uses the selected interface. The ACU selects Mode 0x02 to require VCI explicitly, or Mode 0x03 to permit verification on either the selected contact interface or established VCI. Both modes have the same behavior on contactless. Mode 0x02 returns 0x09 (VCI not established) on contact, even when contact SM is established; Mode 0x03 permits that workflow. When credential SM is established, the PD **SHALL** use that session for Modes 0x01-0x03 and 0x06. Contactless PIN verification requires VCI. Contact PIN verification may use the contact interface without SM. Mode 0x04 requires a *Cache Validity* from 0x0001 through 0xFFFF. All other modes **SHALL** transmit 0x0000 in that field. The PD **SHALL NOT** transmit an entered or cached PIN to the ACU.

A zero *Start Timeout* or *Inter-key Timeout* selects the PD's documented default. Each default **SHALL** be from 1 through 255 seconds. Nonzero values specify seconds and **SHALL** override the corresponding default. The ACU should supply explicit values when deployment or test requirements specify the timing. Fields that do not apply to a selected mode **SHALL** be transmitted as zero.

Only one osdp_PIVSPE operation may be outstanding. Modes 0x01-0x04 and 0x06 each initiate one operation. The PD returns osdp_ACK when it accepts the request, then returns one osdp_PIVSPER in response to a later osdp_POLL. While such an operation is pending, the ACU **SHALL** send only osdp_POLL, Mode 0x05, or osdp_ABORT. Returning osdp_PIVSPER ends the operation unconditionally, including after a PIN mismatch.

Mode 0x00 returns osdp_PIVSPER directly. It reports only persistent PIN state; it does not report transient active or waiting states. An ACU recovering from an interrupted session first uses Mode 0x05 or osdp_ABORT, then queries Mode 0x00. Mode 0x06 is accepted with osdp_ACK; the PD reads the present credential without performing PIN verification and returns the count in a subsequent osdp_PIVSPER. This query **SHALL NOT** decrement the credential retry counter. Auto Execute does not make the count available without a present credential. The PD **SHALL** return 0xFF (attempts unavailable) when the credential is absent or the active interface and security context do not permit the query. Contactless queries require VCI.

Mode 0x05 is idempotent. It aborts active or partial PIN entry, clears the cached PIN, and completes with osdp_ACK; the PD **SHALL NOT** send a later osdp_PIVSPER for the canceled operation. osdp_ABORT cancels active or partial PIN entry but preserves a cache that was completed before the active operation.

A successful verification **SHALL** preserve the cached PIN until expiration or a clearing event. A later explicit verification request may reuse that PIN. The PD **SHALL** clear the cache on Mode `0x05`, PIN mismatch, PD reset, loss of the OSDP Secure Channel, or a change to the PIV mode or a change from one selected application to another. Initial application selection and initial SM establishment **SHALL** preserve a PIN cached before credential presentation. Credential removal also clears it unless flag `0x01` was supplied with Mode `0x04`. Loss of credential SM while the credential remains present clears the cache. Removal follows the removal flag even when it also ends SM. PIN-verified status is bound to the current credential, selected application, interface, and credential security context. The PD **SHALL** clear that status on credential removal or reset, a change to the selected application, interface change, PIV mode change, or loss of credential SM or VCI, and **SHALL NOT** infer it only from an earlier successful reply.

PIN cache expiration The PIN cache holds a PIN in the PD for later verification by a credential. The PD sends the PIN to the credential in a VERIFY command. Successful verification allows operations that require the PIN, subject to the credential's access rules.

Cache Validity limits how long the PD stores the PIN for reuse. When the cache expires, the PD erases the stored PIN. A credential that has already accepted the PIN remains PIN verified until the credential's rules clear that state. Mode `0x00` reports both whether the PD has a usable cached PIN and whether the current credential has accepted PIN verification (Table 20).

Starting and expiring the cache. The PD **SHALL** start the cache timer when PIN acquisition completes successfully, using elapsed time unaffected by clock adjustments. Delivery of the completion reply, status queries, and verification leave the deadline unchanged. A new successful acquisition starts a new deadline. At expiration, the PD **SHALL** erase the cached PIN and mark the cache unavailable.

Expiration before VERIFY. Immediately before submitting VERIFY to the credential transport, the PD **SHALL** check the cache deadline. If the deadline has been reached, the PD **SHALL** send no VERIFY and return `osdp_PIVSPER 0x0A` (no cached PIN). The PD **SHALL** make the deadline check and submission decision as one indivisible step. At the deadline, the PIN is expired and the PD **SHALL NOT** submit VERIFY.

Expiration after VERIFY. Once VERIFY has been submitted, the PD **SHALL** finish the exchange even if the cache expires. The PD reports the credential's verification result, `0x0B` (credential communication failed or timed out), or `0x04` (credential removed), as applicable. The PD may retain a command copy needed to finish the exchange under the credential transport rules and **SHALL** erase that copy when the exchange ends. Each new VERIFY requires a new ACU request.

VERIFY without a confirmed result. If communication fails after VERIFY submission, the PD cannot determine whether the credential accepted the PIN. The PD **SHALL** clear its reported PIN-verification state and cached PIN, end any SM session, and require credential preparation to be restarted before another data or authentication operation. Restarting preparation restores a known security context after an exchange whose outcome is unknown. A verified credential reply determines the reported result even if delivery to the ACU is delayed.

Reporting expiration to the ACU. While idle, the PD erases an expired cached PIN without sending a reply. The ACU learns the current state from a status query or the result of a later verification request. If Auto Execute is waiting for credential presentation, expiration **SHALL** end that request with `0x0A` on a later poll.

An accepted verification request receives ACK followed by one terminal PIVSPER, unless the ACU cancels the request. Expiration after the operation finishes leaves its pending completion reply unchanged.

Figure 7 shows how expiration affects VERIFY. Appendix D.5 provides packet and timing examples.

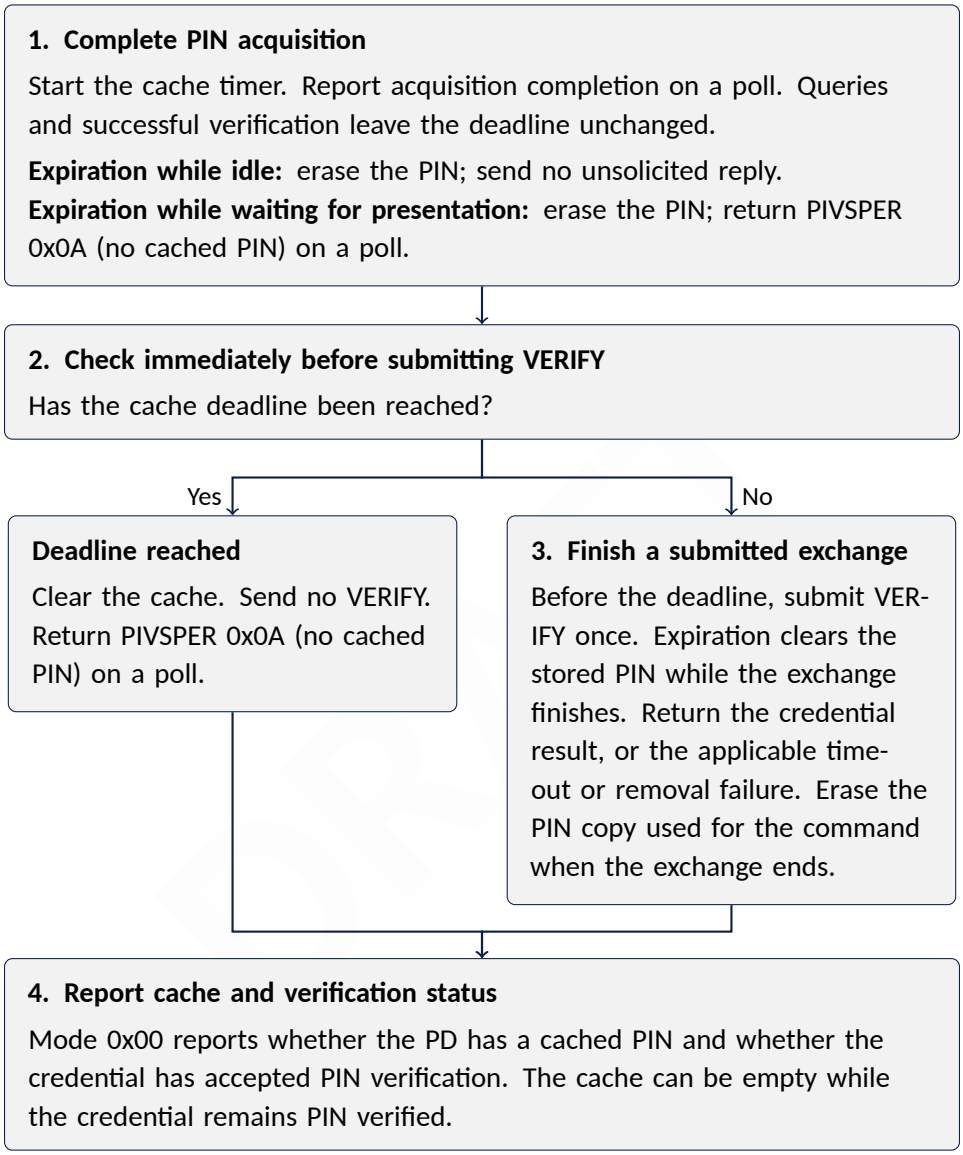


Figure 7: PIN cache lifetime and the VERIFY submission boundary. Idle expiration is silent. Expiration while waiting for presentation ends the pending request with PIVSPER 0x0A on a poll.

Flag 0x02 arms the verification selected by Mode 0x01, 0x02, or 0x03 to execute when a credential is presented. It **SHALL** be zero for Modes 0x00, 0x04, 0x05, and 0x06. A cached auto-execute workflow first acquires the PIN with Mode 0x04, then selects normal or VCI verification with Mode 0x03 or 0x02 and flag 0x02.

Without Auto Execute, verification begins after acceptance with the credential and required security context already available. An absent credential returns *Result* 0x04 (credential absent or removed). When Mode

0x02 is requested or verification uses contactless, missing VCI returns 0x09 (VCI not established). The ACU completes any required pairing before requesting verification.

With Auto Execute, the PD waits for presentation, completes credential preparation (Subsection 3.1), and attempts verification once. If preparation requires a pairing code, the PD completes the SPE operation with 0x09 (VCI not established). If the cached PIN expires while waiting, the PD completes with 0x0A (no cached PIN). A failed verification completes the operation without an automatic retry. The ACU may then perform setup and start a new SPE operation.

The ACU **SHALL** disable PIV Auto before arming SPE Auto Execute. A request to arm SPE Auto Execute while PIV Auto is enabled receives osdp_NAK 0x09 (unable to process command). This ensures that only one operation starts credential preparation and PIN verification when the credential is presented.

The PD **SHALL** reject an unsupported command with osdp_NAK 0x03, an incorrect command length with osdp_NAK 0x02, and an invalid mode, reserved bit, or field combination with osdp_NAK 0x09. Accepted credential-side outcomes are reported through osdp_PIVSPER, not osdp_PIVERROR.

Figure 8 summarizes reply timing and interpretation.

3.7 PIV Secure PIN Entry Reply (osdp_PIVSPER)

The osdp_PIVSPER reply (reply code 0x86, tentative) contains one mode-dependent *Result* byte.

Table 19: PIV Secure PIN Entry Reply (osdp_PIVSPER)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x86 (tentative)
0x01	Result	Mode-dependent SPE status, attempts count, or operation result.	0x00–0xFF

For Mode 0x00, *Result* is the status bitmap in Table 20. For Mode 0x06, it is the attempts value in Table 21. For Modes 0x01-0x04, it is the operation outcome in Table 22.

Table 20: PIVSPE Status Result (Mode 0x00)

Value	Description
0x00	No cached PIN is present and the current credential is not PIN verified.
0x01	A cached PIN is present.
0x02	The current credential and selected application are PIN verified.
0x03	A cached PIN is present and the current credential and selected application are PIN verified.
0x04–0x7F	Reserved for future use.
0x80–0xFF	Private use.

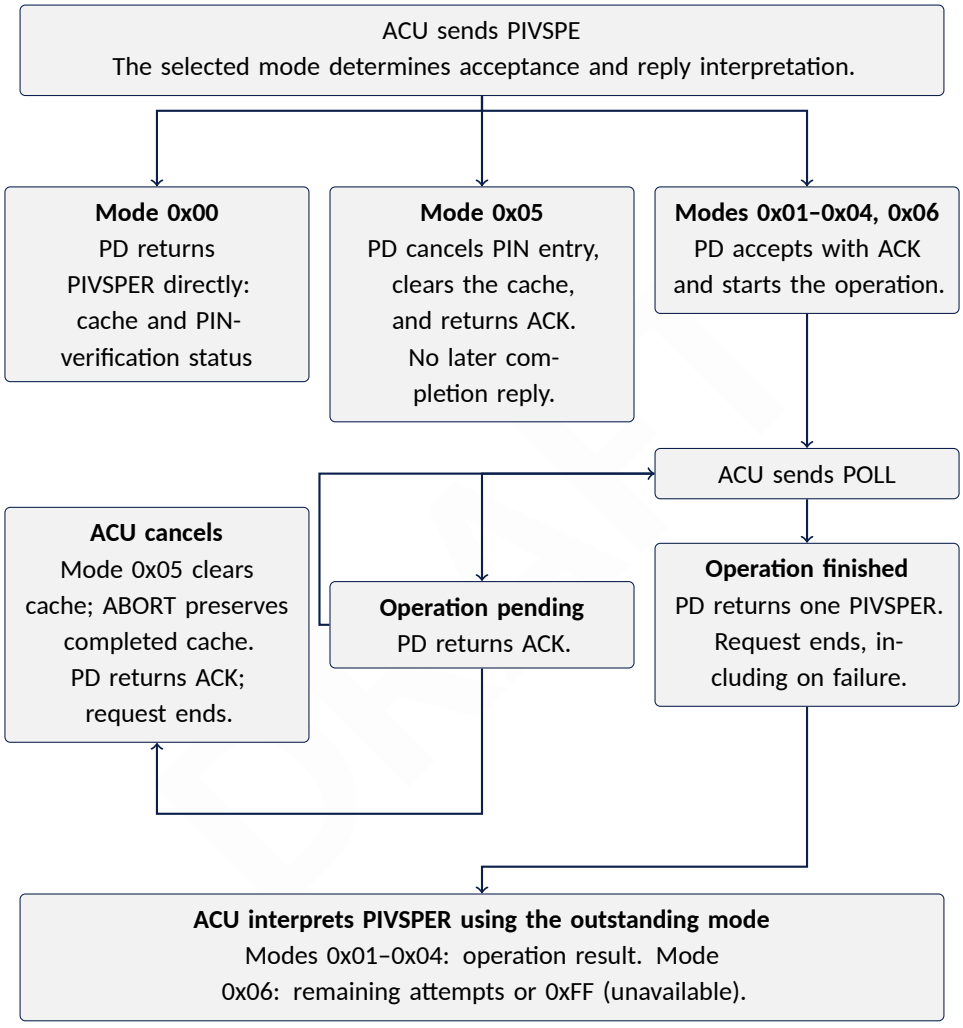


Figure 8: SPE request and reply lifecycle. While an operation is pending, the ACU sends only POLL, Mode 0x05, or ABORT. ABORT cancels the operation while preserving a previously completed cache. An unsupported or malformed request follows the NAK rules.

Table 21: PIVSPE Attempts Result (Mode 0x06)

Value	Description
0x00	The credential PIN is blocked.
0x01–0xFE	Number of PIN verification attempts remaining.
0xFF	Attempts remaining are unavailable, including when no credential is present.

Table 22: PIVSPE Result Codes

Value	Description
0x00	Requested operation completed.
0x01	Timeout – PIN entry did not begin within the allotted time.
0x02	Timeout – inter-digit interval exceeded.
0x03	PIN mismatch.
0x04	Credential absent or removed during operation.
0x05	Credential PIN blocked.
0x06	Wrong interface for the requested mode.
0x07	Incorrect parameters.
0x08	Security condition not met.
0x09	VCI not established for the requested mode.
0x0A	No cached PIN is available.
0x0B	Credential communication failed or timed out before a verified completion was received.
0x0C–0x7F	Reserved for future use.
0x80–0xFF	Private use.

3.8 PIV Biometric Match (osdp_PIVBIOMATCH)

The osdp_PIVBIOMATCH command (command code 0xAB, tentative) instructs the PD to acquire a live sample and compare it to the specified biometric template from the active application. Depending on the *Match Mode*, the comparison occurs either on the credential (on-card match) or within the PD (off-card match).

Table 23: PIV Biometric Match (osdp_PIVBIOMATCH)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAB (tentative)
0x01	Biometric Type	Selects the template type to compare. See Table 24.	0x00–0xFF
0x01	Match Mode	0x00 off-card match; 0x01 on-card match.	0x00–0x01

Table 24: Biometric Type Values

Value	Description
0x01	Cardholder fingerprint template.
0x02	Cardholder facial image.
0x04	Cardholder iris template.

For contactless on-card comparison, the PD **SHALL** use credential SM. This operation can proceed with SM established and VCI absent; pairing and VCI are required only when the credential's access conditions require them. Other biometric operations use established SM when present.

Controllers set *Match Mode* to select off-card or on-card comparison. Off-card flows require the PD to retrieve the appropriate template and, ~~where applicable (e.g., TWIC contactless)~~for encrypted TWIC biometric data on either contact or contactless, to have ~~a valid privacy key loaded via~~ the appropriate TPK available. The ACU can supply the key using osdp_PIVLOADTPK. For a biometric operation against the TWIC application, if a required TPK is missing or incorrect, the PD **SHALL** report osdp_PIVBIOMATCHR Result 0x07 (Invalid TPK). TPK failures **SHALL NOT** be reported as PIN-verification failures. On-card flows require the credential to support OCC. If the PD does not implement the command, it **SHALL** return osdp_NAK function-not-supported. Credential-side match outcomes (for example PIN verification required or credential removal during operation) **SHALL** be reported in the subsequent osdp_PIVBIOMATCHR result code per Table 26. Conditions that prevent the command from being attempted (for example unsupported credential function, missing VCI when required, or invalid command parameters) **MAY** be reported via osdp_PIVERROR on a subsequent osdp_POLL.

TWIC off-card workflows The following application-specific workflows follow *TWIC Specification, Part 2*, Sections 4.1, 4.2, 4.6.4, and 4.7.3, and *Part 3*, Sections 4.4.5–4.4.8 and 7. They use off-card comparison (*Match Mode* 0x00); the documented TWIC cards do not implement OCC or PIV credential SM/VCI. Selecting a PIV application on a TWIC therefore does not make contactless PIN-protected facial retrieval available. The ACU **SHALL** verify the PD's advertised modality support before requesting a fingerprint (*Biometric Type* 0x01) or facial (0x02) comparison.

1. **Legacy TWIC fingerprint.** Select the TWIC application with Data Model 0x02. Obtain the credential's TPK from its magnetic stripe, its contact-interface key container, or the PACS record for that credential.

If the ACU supplies the key, load it with `osdp_PIVLOADTPK` before requesting off-card fingerprint comparison. The PD reads object `DFC103` and deciphers its tag `0xBC` value using the TPK. No PIV PIN is required for this TWIC-application operation.

2. **Legacy TWIC facial.** The Legacy TWIC application contains no facial image object. For direct card retrieval, use the contact interface, select the PIV application with Data Model `0x01`, and verify the PIV application PIN using `osdp_PIVSPE` before requesting off-card facial comparison. The reference is PIV object `5FC108`; it is not encrypted with the TPK. If PIN verification has not been completed, the existing PIN-verification-required result applies, not Invalid TPK. Alternatively, the PACS can use a reference image collected during registration and indexed by the verified credential identifier. That backend workflow does not imply that PIVBIOMATCH accepts a backend image payload. The contact-only/PIN requirement is stated in Part 3, Section 4.4.7, footnote 37; registration is described in *Part 4*, Section 2.5.1.

3. **NextGen TWIC fingerprint.** Select the TWIC application with Data Model `0x02`. Obtain the TPK from the PDF417 barcode, the contact-interface key container, or the credential's PACS record. Load the key when supplied by the ACU, then request off-card fingerprint comparison using object `DFC103` and its encrypted tag `0xBC` value. This workflow requires no PIV PIN.

4. **NextGen TWIC facial.** Select the TWIC application with Data Model `0x02`, obtain the TPK as above, and request off-card facial comparison using object `DFC108`. The PD deciphers tag `0xBC` using the TPK; no PIV PIN is required. Automated comparison by a facial-capable PD uses PIVBIOMATCH/PIVBIOMATCHR. Operator picture verification instead retrieves, verifies, and displays the reference image under TWIC Part 3, Section 4.4.7; PIVBIOMATCH does not return that image.

The TPK is not readable from the credential over contactless. A Legacy magnetic stripe encodes the 16-byte key as 32 hexadecimal characters; the ACU decodes those characters before LOADTPK. A barcode or key-container read likewise supplies only the raw key value to LOADTPK. Loading a correctly sized key does not prove that it belongs to the presented credential. A cleared or expired cache without another valid key source is a missing TPK for a TWIC comparison. The existing cache timeout and clear-on-removal flag remain applicable.

Before accepting TWIC biometric authentication, the system **SHALL** verify the decrypted CBEFF record's signature and authorized document-signer trust, and bind its signed FASC-N to the identifier verified through the CHUID or active card authentication for the same credential, as required by the selected TWIC reader mode. These checks are separate from TPK decryption and live-sample comparison. A signature or identifier-binding failure **SHALL NOT** be accepted merely because decryption or matching succeeded. An integrity failure alone does not establish that the TPK was incorrect. The ACU retains responsibility for the access decision and the checks assigned to it by the deployment; BIOMATCH success alone does not assert completed TWIC authentication.

For modes requiring active card authentication, Legacy TWIC uses its PIV application's card-authentication key, whereas NextGen provides a key in the TWIC application. An application switch clears the previous application's security state, so any subsequent PIV PIN verification must apply to the newly selected PIV context.

3.9 PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

After `osdp_PIVBIOMATCH` completes, the PD reports the result as a poll response using reply code `0x87` (tentative).

Table 25: PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	`0x87 (tentative)`
0x01	Result	Outcome of the biometric match. See Table 26.	`0x00–0xFF`
0x00 or 0x20	Hash	0x20-byte SHA-256 hash of the biometric template tag `0xBC` data field. The PD **SHALL** include this field only when *Result* is `0x00` (off-card match success). The PD **SHALL** omit this field for on-card match success (`0x09`) and all failure result codes.	`0x00–0xFF`

When *Result* is `0x00` (off-card match success), the PD **SHALL** include the 32-byte SHA-256 hash of the biometric template tag `0xBC` data field in the reply. For on-card match success (*Result* `0x09`) and for all failure result codes, the PD **SHALL** omit the hash field; ACUs **SHALL NOT** rely on a hash received alongside any result other than `0x00`.

The hash input is the tag `0xBC` value as read from the credential, excluding its tag and length bytes. For the encrypted TWIC objects, this is the ciphertext, not the decrypted CBEFF record or a derived live-sample template. Decryption supplies the reference used for comparison; it does not change the reply's hash input.

Off-card biometric comparison demonstrates only that the live sample matched template data presented to the PD. Because off-card biometric objects can be copied or duplicated, an off-card biometric match does not by itself prove authenticity of the credential. On-card biometric comparison is a credential-controlled assertion; the biometric object may be attacker controlled and the PD does not receive the template data being matched. To obtain cryptographic proof associated with an on-card biometric success, the ACU **SHALL** perform an additional authentication using an appropriately verified credential key after the successful on-card comparison.

Table 26: PIVBIOMATCH Result Codes

Value	Description
0x00	Biometric match success (off-card match).
0x01	Biometric match timeout (no entry started).
0x02	Biometric match failed (no match).
0x03	PIN verification required.
0x04	Credential removed during operation.
0x05	PIN verification failed.
0x06	PIN verification timeout.
0x07	~~Reserved for future use~~Invalid TPK (the required TPK is missing or incorrect).
0x08	Reserved (PD/credential errors are reported via osdp_PIVERROR).
0x09	On-card match success.
0x0A–0x7F	Reserved for future use.
0x80–0xFF	Private use.

3.10 TWIC Privacy Key Load (osdp_PIVLOADTPK)

The osdp_PIVLOADTPK command (command code 0xAC, tentative) instructs the PD to cache a TWIC Privacy Key (TPK) for use with subsequent biometric operations against a TWIC credential. Function Code 14 is the biometric-support compliance byte; before using this command, controllers **SHOULD** verify that Function Code 14 advertises the OSDP 3.0 biometric command set and the relevant biometric modality bits for the intended TWIC operation.

Table 27: TWIC Privacy Key Load (osdp_PIVLOADTPK)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAC (tentative)
0x01	Cache Timeout	Number of seconds to retain the cached TPK. 0x00 clears any cached key immediately; 0x01-0xFF permit caching for up to 0xFF seconds.	0x00–0xFF
0x01	Control Flags	Bit 0 set requests automatic clearing when the credential is removed; all other bits **SHALL** be zero.	0x00–0xFF

Size (bytes)	Name	Meaning	Value
~~0x20~~ 0x10	TPK	~~Trusted Processing Key bytes~~ Raw AES-128 TWIC Privacy Key value to cache for the next TWIC biometric operation.	0x00–0xFF

The *TPK* field **SHALL** contain exactly 16 bytes of raw AES-128 key material, as specified in *TWIC Specification, Part 2*, Section 4.6.2. It carries only the key value, excluding TLV tags and lengths, the algorithm identifier, the key index, and any hexadecimal text encoding. The command payload, including *CMND*, *Cache Timeout*, and *Control Flags*, is exactly 0x13 bytes. The TWIC key container permits up to 32 bytes for future algorithms; that capacity does not change this command's 16-byte key-field length.

Controllers set *Cache Timeout* to define how long (in seconds) the PD may retain the key. A value of 0x00 clears any cached TPK immediately; the 16-byte *TPK* field remains present but is ignored. Values 0x01-0xFF allow caching for up to 255 seconds. Bit 0 of *Control Flags* directs the PD to clear the cached TPK when the credential is removed; all other bits are reserved and **SHALL** be transmitted as zero. If the PD does not support TWIC TPK caching it **SHALL** return osdp_NAK function-not-supported. Invalid payloads (~~for example~~ including an incorrect TPK field length or reserved bits set) **SHALL** elicit osdp_PIVERROR 0x0005 (invalid parameter).

3.11 PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING)

The osdp_PIVXMITPAIRING command (command code 0xAD, tentative) submits an eight-digit pairing code to the selected contactless PIV application over established credential SM. The ACU **SHALL** send it over OSDP Secure Channel 2.0. Credential SM, PIVMODE flag 0x02, and credential support for VCI are prerequisites.

Table 28: PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAD (tentative)
0x08	Pairing Code	Eight ASCII digits ('0'-'9') used by the PD during VCI establishment for the currently selected PIV application.	0x30–0x39

The pairing code contains eight ASCII digits ('0'-'9'). The PD accepts a valid request with osdp_ACK. When pairing is required, the PD submits the code using PIV VERIFY, reference 0x98, protected by the established SM context. Success completes with osdp_CARDSTATUSR on a later osdp_POLL, reporting security status 0x01 (SM and VCI). If VCI is already available, the PD completes with the same status reply without submitting the code again.

An accepted operation completes with osdp_PIVERROR on failure: 0x0005 (invalid parameter) for malformed digits or no selected PIV application, 0x1023 (security conditions not satisfied) for missing SM, disabled SM/VCI preparation, the wrong interface, or a rejected pairing code, 0x1020 (function not supported

by credential) when VCI is unsupported, 0x1026 (credential absent or removed), or 0x0002 (smart-card timeout). The PD includes the credential's failure status word as specified in Subsection 2.2.1. If a submitted pairing VERIFY times out without a confirmed result, the PD **SHALL** end SM, clear dependent VCI, PIN-verification, and cached-PIN state, and require an explicit preparation restart before further credential data or authentication commands. The PD **SHALL** return 0x0002 (smart-card timeout) without SW1/SW2 and **SHALL NOT** retry VERIFY automatically. While pairing is pending, the ACU sends only osdp_POLL or osdp_ABORT. Competing commands receive osdp_NAK 0x09 (unable to process command).

The PD **SHALL** clear the transient pairing code on completion, failure, cancellation, or loss of its credential context. osdp_ABORT completes with osdp_ACK and clears the pending completion. For an OSDP retransmission, the PD **SHALL** replay the previous reply without resubmitting the pairing code.

OSDP security preconditions Trust-anchor load/status, pairing-code transmit, credential SM establishment, and PIV operations that depend on SM or VCI **SHALL** require OSDP Secure Channel 2.0 between the ACU and PD with deployment-specific keys. OSDP Secure Channel protects the management commands and their results. Credential SM protects the PD-to-card exchange. A security-precondition failure returns osdp_PIVERROR 0x0006 (secure channel required) where supported; otherwise the command's existing failure reply applies.

3.12 Set PIV Mode (osdp_PIVMODE)

The osdp_PIVMODE command (command code 0xAE, tentative) defines the operating context for PIV, TWIC, and related credential transactions. OSDP 3.0 replaces the OSDP 2.2 "Application + AID" fields with a configuration payload that includes a required Global Interface Mask byte, PIV Auto challenge parameters when autonomous operation is enabled, and a TLV list of configuration descriptors so controllers can specify multiple application profiles, preferred interfaces, and optional Application Identifiers. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV).

Table 29: Set PIV Mode (osdp_PIVMODE)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAE (tentative)
0x02	MpSizeTotal	Total size of the reassembled configuration data beginning with PIV Mode Flags, least-significant byte first.	0x0008–0xFFFF
0x02	MpOffset	Offset of this fragment within the reassembled configuration data, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Number of configuration data bytes following the multi-part fields in this fragment, least-significant byte first.	0x0000–0xFFFF

Size (bytes)	Name	Meaning	Value
0x01	PIV Mode Flags	First byte of the reassembled configuration data; global behaviour flags (Section 3.12).	0x00–0xFF
0x01	Config Count	Number of ~~configuration entries~~ Configuration Entry TLVs (tag 0x01); excludes Resume.	0x01–0x20
0x01	Global Interface Mask	Bitmask limiting interfaces available to all configurations (bit 0 contact, bit 1 contactless, bit 3 Bluetooth; bit 2 reserved and set to zero).	0x00–0x0B
0x04	Series Counter	~~Conditional field present only when PIV Autois enabled.~~ Present only with Auto. Fresh ACU-assigned Series for ordinary configuration; installed Series for Resume. Least-significant byte first.	0x00000000 –0xFFFFFFFF
0x04	Sequence Counter	~~Conditional field present only when PIV Autois enabled.~~ Present only with Auto. Initial PD allocation value for ordinary configuration; exact next unused value for Resume. Least-significant byte first.	0x00000000 –0xFFFFFFFF
0x20	Supplemental Entropy	Conditional fixed-width field present only when PIV Auto is enabled. All zeroes are permitted but not recommended.	0x20 octets
0x00-n	TLV Data	~~Sequence of~~ Configuration Entry TLVs (~~Tag~~ tag 0x01)~~that follow the Global Interface Mask, or follow the conditional PIV Auto fields when PIV Auto is enabled~~, followed by optional Resume (0208 and previous Series/Sequence). Auto entries have a mandatory trailing Authentication Key Reference; see Section 3.12.	–

The *PIV Mode Flags* byte enables autonomous operation and credential SM/VCI preparation (Table 30). Bits not listed remain reserved and **SHALL** be transmitted as zero.

Table 30: PIV Mode Flags

Mask	Description
0x01	Enable autonomous PIV (PIV Auto). The PD prepares the credential, derives a KMAC256 challenge from the configured counters and entropy, and reports the authentication result using osdp_PIVSTATUS.
0x02	Enable credential Secure Messaging preparation over contact and contactless interfaces. On contactless, also enable VCI when the credential permits VCI without pairing. Applies to autonomous and commanded operation.
0x04	Enforce FICAM strict mode (PIV applications only).

Global and per-entry interface masks use the bit positions in Table 31; the Global Interface Mask bounds the interfaces a PD may use when processing all configuration entries.

Table 31: PIV Mode Interface Masks

Mask	Description
0x01	Permit ISO 7816 contact credential interface use.
0x02	Permit ISO 14443 contactless credential interface use.
0x04	Reserved and **SHALL** be transmitted as zero.
0x08	Permit Bluetooth credential interface use.
0x10–0x80	Reserved and **SHALL** be transmitted as zero.

The reassembled configuration data begins at *MpOffset* 0x0000 with *PIV Mode Flags*, followed by the configuration count (1-32) and the required Global Interface Mask byte. *MpSizeTotal* counts all configuration data from *PIV Mode Flags* through the final TLV byte. If and only if PIV Auto is enabled in *PIV Mode Flags*, the Global Interface Mask is followed by the Series Counter, Sequence Counter, and fixed 32-byte Supplemental Entropy field before the TLV list. When PIV Auto is not enabled, those three conditional fields are absent and the TLV list begins immediately after the Global Interface Mask. Each configuration appears as a TLV:

The configuration count **SHALL** equal the number of complete Configuration Entry TLVs in the reassembled payload. Consequently, *MpSizeTotal* **SHALL** be at least eight bytes when PIV Auto is disabled and at least ~~48~~ 49 bytes when it is enabled, and **SHALL** equal the exact encoded length of the required header, any conditional PIV Auto fields, ~~and~~ all configuration TLVs~~.~~, and any Resume TLV. Tags and lengths in this list are one byte each. The optional Resume TLV does not contribute to the configuration count. Truncated fields, unknown tags, trailing bytes, or a count that does not match the ~~TLV~~ Configuration Entry sequence **SHALL** be treated as malformed configuration data.

- **Global Interface Mask (required)** — One-byte mask (no tag) that limits the interfaces the PD may use for all configurations (see Table 31). A value of 0x00 indicates no global restriction; reserved bits **SHALL** be zero.

- **Configuration Entry (Tag 0x01)** — Contains the ~~data-model identifier, a per-entry interface mask, the AID length, and the optional AID~~following fields in order:
  - *Data Model* (0x01 PIV, 0x02 TWIC, 0x03 CAC, values 0x04-0x7F reserved, 0x80-0xFF private use). CAC entries use the same interface and AID encoding as PIV and TWIC entries.
  - *Interface Mask* — Uses the same bit positions as the global mask (Table 31) and **SHALL NOT** assert interfaces excluded by the Global Interface Mask. A value of 0x00 means "any interface permitted by the global mask."
  - *AID Length* (0-16) and *AID* bytes when provided.
  - *Authentication Key Reference* — One trailing byte present if and only if PIV Auto is enabled. For Data Model 0x01, the permitted values are 0x9A (PIV Authentication) and 0x9E (asymmetric Card Authentication). For Data Model 0x02, the permitted value is 0x9E (TWIC active card authentication). CAC and private-use Auto profiles are not defined here and **SHALL NOT** be accepted as this Auto profile. Non-Auto entries are unchanged.

The Configuration Entry value length **SHALL** be exactly 0x03 plus AID Length without Auto, or 0x04 plus AID Length with Auto. The key reference belongs to that entry; there is no configured algorithm byte and no implicit key default. The earlier Auto entry layout without the key reference **SHALL** be rejected, not guessed or interpreted as selecting 0x9E.

PDs process configuration entries in order, applying the global mask ~~(when present)~~ before honouring each entry~~'~~'s mask. Once an application has been selected for an Auto attempt, failure of discovery, a security prerequisite, or authentication **SHALL NOT** cause selection of another entry, key, or interface. Invalid masks, unknown data-model identifiers, unsupported Auto models, invalid key references, or malformed TLVs ~~**SHALL** result in `osdp_PIVERROR` (invalid parameter). Controllers **SHALL** send the command over a secure channel where required; the PD **SHALL**~~ return `osdp_PIVERROR` 0x0005; unmet channel requirements return 0x0006~~(secure channel required) if the necessary security conditions are not satisfied. Multi-part framing errors result in `osdp_PIVERROR` (invalid~~; multi-part ~~transfer). If the~~ framing errors return 0x0007; insufficient configuration storage returns 0x1021. The PD **SHALL** validate the complete reassembled command before replacing a configuration or clearing its credential context. Rejected configuration preserves the installed configuration and valid security state. The final fragment receives `osdp_ACK` only after complete validation and acceptance; credential work for the newly accepted Auto attempt begins after that ACK.

Per-credential algorithm discovery For each Auto attempt, the PD ~~cannot cache the requested configuration set,~~ **SHALL** retrieve the certificate corresponding to the configured key in the authentication application: object 5FC105 for PIV 0x9A, or 5FC101 for asymmetric 0x9E. TWIC DFC101 is a TPK container, not an authentication certificate. The PD **SHALL** extract certificate tag 0x70 and its single CertInfo byte in tag 0x71: 0x00 denotes uncompressed DER; 0x01 denotes GZIP-compressed DER. Missing, duplicated, or invalid required fields, failed decompression, and malformed DER certificate data produce PIVSTATUS Result 0x03. The PD **SHALL** discover the algorithm from the certificate's *subjectPublicKeyInfo*, following SP 800-73-5 Part 1, Appendix C.1, using the subject RSA modulus size or named EC curve, not the issuer's certificate-signature algorithm. It **SHALL NOT** reuse another credential's algorithm

or send trial authentication commands to probe algorithms. RSA profiles require `rsaEncryption` SubjectPublicKeyInfo; EC profiles require `id-ecPublicKey` with the named P-256 or P-384 curve. Other subject-key profiles, including RSA-PSS-restricted keys, **SHALL** produce Result 0x04, not a guessed algorithm or a different signature encoding.

Table 32: PIV Auto Subject Public-Key Profiles

Subject public key	`Algorithm`	Digest profile	`Digest bytes`
RSA-2048	0x07	SHA-256	0x20
RSA-3072	0x05	SHA-384	0x30
EC named curve P-256	0x11	SHA-256	0x20
EC named curve P-384	0x14	SHA-384	0x30

This table fixes this proposal's Auto digest choice, independently of the issuer's signature. A missing or malformed certificate produces Result 0x03; an unsupported subject-key algorithm, size, curve, or Auto profile produces Result 0x04. Neither permits another key or symmetric authentication as fallback. The PD reports the discovered PIV identifier when one can be determined, and 0x00 otherwise. No GENERAL AUTHENTICATE is sent until discovery succeeds. The PD verifies the returned key proof with this certificate's public key; the ACU retains certificate-path, validity, revocation, identity-binding, algorithm-acceptance, and access-decision responsibilities. PD success does not prove trust.

TWIC Auto application routing For a TWIC entry with AID Length zero, the PD **SHALL** select the common nine-byte prefix A00000036720000001; a supplied AID is honoured instead. Full selected AIDs A000000367200000010103 and A000000367200000010101 identify NextGen and Legacy, respectively. A prefix, absent certificate, or failed authentication **SHALL NOT** be treated as evidence of Legacy. The PD **SHALL** obtain positive full-AID generation evidence from the SELECT response; unavailable evidence produces Result 0x03, and an unsupported generation produces Result 0x04, without guessing. For NextGen, the PD **SHALL** read the certificate and perform 0x9E authentication in the TWIC application. Only for positively identified Legacy TWIC, it **SHALL** ~~respond with `osdp_PIVERROR` (insufficient PD memory or storage).~~ select the credential's PIV application and use its asymmetric 0x9E key and 5FC101 certificate. This is the configured TWIC route, not failure-driven fallback. Before that Legacy transition, the PD **SHALL** read the TWIC CHUID and retain its identity only for this attempt, then bind it to the PIV application's certificate below. The configured entry and TWIC authentication purpose remain unchanged; the actual selected application's access and state-clearing rules apply. CARDSTATUSR reports the actual selected AID. These routes follow TWIC Part 2, Sections 4.1, 4.4.2, 4.7.1, and 7.5.

For NextGen TWIC GET DATA, the PD **SHALL** use `Le = 0xFF`, not absent Le or 0x00, and retrieve the complete object using GET RESPONSE as specified in TWIC Part 2, Section 5.2 and Appendix E. Authentication command and response chaining follow Section 5.3 and Appendix E. A partial Dynamic Authentication Template is not success. The PD **SHALL** obtain FASC-N and UUID/GUID from CHUID object 5FC102, tags 0x30 and 0x34. PIV and NextGen TWIC success require a nonzero UUID. For a positively identified Legacy TWIC only, an actually retrieved 16-byte nil GUID is permitted when a well-formed, nonzero 25-byte FASC-N matches the FASC-N in the authentication certificate's subjectAltName. An absent or unreadable GUID

SHALL NOT be fabricated as nil. The PD and ACU **SHALL** enforce this Legacy binding; the ACU permits the exception only for its configured TWIC purpose. TWIC Part 2, Sections 4.3 and 6, define the relevant identity representations.

TWIC Auto success reports a certificate-associated key proof, not completed TSA reader authentication, CCL checking, biometric comparison, or TSA reader conformance. In particular, TWIC Part 3, Section 4.4.4, specifies a random or pseudorandom challenge of at least 127 bytes. The 32- or 48-byte KMAC digest and RSA encoding here are not claimed to establish that separate requirement; deployments requiring it need the corresponding TWIC authentication profile and qualification.

When PIV Auto is enabled, the PD prepares the presented credential as described in Subsection 3.1, derives the challenge, issues PIV `GENERAL AUTHENTICATE`, and reports completion through `osdp_PIVSTATUS`. The PD **SHALL** use the configured key, ~~algorithm, and~~ the algorithm discovered from that credential's role certificate, and the selected interface for that attempt. If SM is unavailable, the PD continues that same authentication only when the credential's access rules permit the authentication without SM, as for key `0x9E`. If the configured authentication requires SM, the PD reports the SM failure. A failed signature or cryptogram check ends the attempt even when the authentication could otherwise proceed without SM.

PIV Auto can establish VCI when the credential permits VCI without a pairing code. For an authentication that requires SM, the PD first checks whether SM is established. If SM is unavailable, the PD reports Result `0x02` (SM failure) with the applicable detail code. The PD then checks any VCI and PIN-verification requirements, in that order. If the selected authentication requires VCI and pairing is required, the PD completes with Result `0x05`, Detail `0x01` (pairing code required). The ACU can then begin commanded pairing. Required PIN verification that has not been completed returns Result `0x05`, Detail `0x02` (PIN verification required). PIV Auto performs one automatic attempt per presentation, subject to counter availability. Keeping the ~~same~~ credential presented does not restart the attempt. After a terminal prerequisite failure is delivered, the ACU may complete commanded pairing and PIVSPE PIN verification while Auto remains enabled, then explicitly resume as defined below. SPE Auto Execute remains prohibited while PIV Auto is enabled. Removal and a new presentation, or ~~successful reapplicationof `osdp_PIVMODE`~~ordinary PIVMODE reapplication, starts a new attempt ~~. Reapplication follows the~~ when counters are available. Ordinary reapplication follows mode-change clearing~~rules; loading a trust~~ ; loading an anchor alone does not restart ~~authentication~~Auto.

PIV Auto configuration **SHALL** be accepted only when OSDP Secure Channel 2.0 is active and the channel is not using a factory/default SCBK or other factory key material. A PD **SHALL** reject PIV Auto configuration or execution with `osdp_PIVERROR` `0x0006` (secure channel required) when these preconditions are not met.

~~When bit of *PIV Mode Flags* enables PIV Auto , the configuration **SHALL** specify a *Series Counter*, a *Sequence Counter*, and a~~

Counter ownership and exhaustion Ordinary Auto PIVMODE supplies a Series, an initial Sequence, and fixed-width 32-byte ~~*Supplemental Entropy* field. The supplemental entropy field is fixed width and **SHALL** always be present; all zeroes are permitted for compatibility but are~~ Supplemental Entropy. All-zero entropy is permitted but not recommended. The ACU **SHALL** ~~ensure that a series counter value does not repeat for~~ allocate a fresh Series for every ordinary Auto configuration and prevent its reuse for the same PD ~~and that no sequence counter value is issued more than oncewithin a series. One conforming implementation strategy is to maintain a monotonic series counter that increments at every ACU boot, similar to the SNMP engine~~

~~boots counter , and a monotonic sequence counter for challenges issued within that boot series~~across ACU or PD restart. Within that Series, the PD **SHALL** reserve the next unused Sequence before every attempt, including presentation preparation and accepted Resume. The supplied initial Sequence is used first; each reservation advances the allocator by one. Failure, cancellation, timeout, and removal do not release a reservation; removal does not reset the allocator. PIVSTATUS reports the reserved pair even if no challenge was generated. No new attempt starts while another operation or undelivered completion is pending.

Sequence 0xFFFFFFFF may be reserved once. That attempt may finish normally; afterward Auto is suspended until ordinary PIVMODE installs a fresh Series with normal clearing. The PD **SHALL NOT** wrap, start another Auto attempt, or issue further Auto credential commands or repeated exhaustion statuses. The terminal status carrying 0xFFFFFFFF informs the ACU to install a fresh Series before another attempt. Resume cannot change Series to avoid that clearing rule.

Explicit prerequisite continuation An optional Resume TLV is 0208, followed by the previous terminal PIVSTATUS Series and Sequence, four bytes each, least-significant byte first. It **SHALL** occur once, after all Configuration Entry TLVs, and only with Auto enabled. Its presence requests Resume; absence retains ordinary PIVMODE clearing. It is transient metadata, not installed configuration. Header Series **SHALL** equal the installed Series; header Sequence **SHALL** equal its exact next unused value. Supplemental Entropy may change; flags, global mask, count, entry order, models, entry masks, AIDs, and key references **SHALL** otherwise match the installed configuration exactly.

Resume requires complete delivery of a terminal PIVSTATUS 0x05/0x01 or 0x05/0x02 whose counter pair matches the TLV, with the same continuously present credential, selected application, interface, configured entry, and OSDP session. The PD **SHALL** recheck these conditions after reassembly and before acceptance. Removal, credential reset, context-changing selection, ordinary mode replacement, an active anchor change invalidating SM, blocked SM, or OSDP session replacement invalidates continuation. Delivering the final fragment of an old status after context loss **SHALL NOT** restore eligibility. Successful commanded pairing, PIN verification, permitted reads, and diagnostics preserve eligibility.

Acceptance **SHALL** atomically consume prior eligibility and reserve the supplied pair, then ACK before credential work. It preserves completed pairing, SM/VCI, PIN verification, and any otherwise valid completed PIN cache. The PD **SHALL NOT** select another entry, change interface or key, submit VERIFY automatically, or retry authentication automatically. The accepted attempt returns exactly one polled PIVSTATUS. A still-missing prerequisite returns 0x05/0x01 or 0x05/0x02 with the newly reserved pair. Removal after acceptance returns Result 0x01; SM failure follows existing blocked-state rules.

Before acceptance, malformed Resume, changed configuration, invalid counters, or exhaustion return PIVERROR 0x0005; unmet channel requirements return 0x0006; an absent credential returns 0x1026; stale or invalidated context returns 0x1023; another operation or undelivered completion returns NAK 0x09 (busy). Rejection starts no attempt and consumes no counters or valid security state. Normal OSDP retransmission of the same authenticated command bytes in the same session and OSDP sequence context replays the cached ACK or completion without another reservation or card operation. The same logical Resume sent with a new OSDP sequence is a new request and cannot reuse consumed eligibility.

The PIV Auto challenge **SHALL** be derived from the active Secure Channel 2.0 session key material using two KMAC256 operations. First, the PD derives a 32-byte PIV Auto KDK with key material formed by

concatenating S-ENC, S-MAC1, and S-MAC2; empty input data; output length 32 bytes; and customization string OSDP-PIV-AUTO-KDK-v1. Second, the PD derives the challenge digest using the PIV Auto KDK as the KMAC key, the fixed-width template as input data, output length L, and customization string OSDP-PIV-AUTO-CHALLENGE-v1. The ~~fixed-width~~ template **SHALL** ~~contain the series counter, sequence counter, credential FASC-N, credential UUID, challenged PIV key reference , PIV algorithm identifier, and the 32-byte supplemental entropy field. A successful PIV Auto transaction~~ be exactly 0x61 bytes in Table 33 order, with no implicit padding or length prefixes. Output length L and digest profile follow Table 32. UUID/GUID **SHALL** ~~include the credential UUID; if the UUID cannot be read~~have been read successfully; only the positive Legacy TWIC exception above permits nil. PIV Auto challenge derivation **SHALL NOT** use static factory keys. This serialization does not redefine the Secure Channel 2.0 key-binding requirement; its independent validation remains separate.

Table 33: Normative PIV Auto KMAC Challenge Template

Field, in order	Bytes	Encoding
Template label and version	0x0E	ASCII OSDP-PIV-AUTO, followed by byte 0x01.
Series Counter	0x04	Reserved Series, least-significant byte first.
Sequence Counter	0x04	Reserved Sequence, least-significant byte first.
FASC-N	0x19	Raw credential bytes.
UUID/GUID	0x10	Raw credential bytes, including an allowed actual Legacy nil GUID.
Authentication Key Reference	0x01	Configured key reference for this attempt.
Algorithm Identifier	0x01	Discovered subject-key identifier.
Supplemental Entropy	0x20	Exact configured or resumed 32-byte value.

For RSA, the PD **SHALL** ~~report PIV Auto failure instead of deriving a success challenge with an all-zero UUID. The output length is 32 bytes for~~ encode the KMAC digest with EMSA-PKCS1-v1_5, the profile's SHA-256 ~~profiles and 48 bytes for~~ or SHA-384 ~~profiles. PIV Auto challenge derivation~~ DigestInfo, and a block matching the discovered modulus length. The KMAC output is already the message hash and **SHALL NOT** ~~use static factory keys.~~ be hashed again. For EC, it **SHALL** submit the 32- or 48-byte digest directly, without rehashing. GENERAL AUTHENTICATE uses discovered P1, configured key reference P2, and tag 0x7C containing empty 0x82 and 0x81 carrying the encoded input. The returned 0x7C template is preserved whole, excluding transport status words.

3.13 PIV Auto Status Reply (osdp_PIVSTATUS)

When a PIV Auto attempt completes, including a failure during preparation, the PD reports the result as a poll response using reply code 0x89 (tentative). The first osdp_POLL response after completion **SHALL** contain osdp_PIVSTATUS; subsequent fragments, when required, use the same reply and advance *MpOffset* until the complete status payload is delivered. There is no corresponding osdp_PIVSTATUS request command.

Table 34: PIV Auto Status Reply (osdp_PIVSTATUS)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x89 (tentative)
0x02	MpSizeTotal	Total size of the complete PIV Auto status payload, least-significant byte first.	0x0000-0xFFFF
0x02	MpOffset	Offset of this fragment within the complete status payload, least-significant byte first.	0x0000-0xFFFF
0x02	MpFragmentSize	Length of the status payload bytes in this fragment, least-significant byte first.	0x0000-0xFFFF
0x01	Result	Result of the PIV Auto authentication attempt. See Table 35.	0x00-0xFF
0x01	Detail	Result-specific detail from Table 35. Set to 0x00 where no detail is defined.	0x00-0xFF
0x04	Series Counter	~~Series counter value used to derive the challenge, least-significant~~ Reserved Series, even on failure before challenge generation. Least-significant byte first.	0x00000000 -0xFFFFFFFF
0x04	Sequence Counter	~~Sequence counter value used to derive the challenge, least-significant~~ Reserved Sequence; also binds an eligible Resume. Least-significant byte first.	0x00000000 -0xFFFFFFFF
0x19	FASC-N	FASC-N read from the credential.	200-bit FASC-N
0x10	UUID	Credential UUID~~read from the credential. Success replies **SHALL** carry a nonzero UUID; failure replies may transmit all zeroes when the UUID is~~ /GUID. Success is nonzero except for the positively identified, FASC-N-bound Legacy TWIC exception. Failure may report zero when unavailable.	0x10 octets

Size (bytes)	Name	Meaning	Value
0x01	Key Reference	~~PIV key referencechallenged by the PD~~Configured authentication key reference, including failure before challenge generation.	~~For example 0x9A or~~ 0x9E
0x01	Algorithm Identifier	~~PIV GENERAL AUTHENTICATE algorithm identifier.~~ Discovered GENERAL AUTHENTICATE identifier; 0x00 on failure when none could be determined. Never configured by the ACU.	For example 0x07
0x02	Signed Response Length	Total length of the signed response, least-significant byte first.	0x0000–0xFFFF
0x00-n	Signed Response	Complete response bytes returned by the credential. This field is part of the reassembled status payload.	Variable
0x00 or 0x02	Status Word	SW1 followed by SW2 when the credential returned the reported failure. Omitted in the other cases specified below. Part of the reassembled payload.	–

Result and detail codes follow Table 35. When *Result* is 0x00, *Detail* is reserved and **SHALL** be 0x00. When *Result* is not 0x00, the PD **SHALL** set *Signed Response Length* to 0x0000 unless a credential response was received and is useful for ACU diagnostics.

The PD **SHALL** append SW1 followed by SW2 when a credential command returned the failure being reported. For a verified protected reply, the PD uses the protected status. The PD **SHALL** omit this field on success, local failure, timeout, credential removal without a failure reply, or failure to verify a protected reply. An earlier command's status **SHALL NOT** be used. These are the status-word rules of Subsection 2.2.1. The status word follows Signed Response in the reassembled payload and may cross a fragment boundary. After accounting for the fixed fields from Result through Signed Response Length in Table 34 (0x37 bytes) and *Signed Response Length*, the ACU **SHALL** accept exactly zero or two remaining bytes; any other length is malformed. Result 0x06 uses Detail 0x00; the complete status word carries the credential's error. The PD returns one terminal PIVSTATUS, with no follow-up PIVERROR for that attempt.

Table 35: PIV Auto Status Result Codes

Value	Description
0x00	~~Success~~Key-proof success. *Detail* is 0x00~~,~~; UUID is nonzero ~~, and~~ except for the defined, FASC-N-bound Legacy TWIC case. Signed Response carries the complete returned Dynamic Authentication Template. ACU trust and acceptance checks remain required.
0x01	Credential not present or removed before PIV Auto completed.
0x02	Credential SM unavailable or failed. Detail: 0x01 trust store empty; 0x02 no matching trust path; 0x03 missing or invalid CVC; 0x04 invalid authentication cryptogram; 0x05 required SM unsupported; 0x06 other SM processing failure; 0x07 required SM disabled. Detail assignments are tentative.
0x03	Required credential data object, FASC-N, UUID/GUID, key, ~~or certificateevidence is unavailable~~certificate, or generation evidence is missing, malformed, or cannot be obtained.
0x04	~~Requested PIV keyreference or algorithm~~ Configured key, credential generation, or discovered subject public-key algorithm/profile is not supported for ~~PIV~~ Auto.
0x05	Required credential security condition is absent. Detail: 0x01 pairing code required; 0x02 PIN verification required; 0x03 required VCI unsupported; 0x00 other unmet condition. Detail assignments are tentative.
0x06	Credential GENERAL AUTHENTICATE operation failed. Detail is 0x00; the Status Word field carries the credential failure status when available under the rules below.
0x07	Signed response verification failed.
0x08	Timeout or PD internal failure before a definitive credential result was available.
0x09–0x7F	Reserved for future standard result codes.
0x80–0xFF	Private use.

For Result 0x02, Details 0x03, 0x04, and 0x06 block further credential operations until the context is restarted, as specified in Subsection 3.1. Details 0x01, 0x02, 0x05, and 0x07 leave operations permitted without SM available.

For failure before challenge generation, the PD **SHALL** report the ~~configured~~ reserved series and sequence counters and ~~requested key referenceand algorithm~~ configured key reference. It reports the discovered algorithm identifier, or 0x00 if none could be determined. It **SHALL** set unavailable FASC-N and UUID fields to zero and *Signed Response Length* to zero. These identity fields describe data actually obtained from the credential. The PD **SHALL** set *Detail* to zero unless the result defines a detail value.

The reply reports the exact ~~series and sequence countersused for the challenge, the FASC-N and UUID read from the credential~~ reserved counters, ~~and the signed responsereturned by the credential~~credential identity, and returned signed response. The ACU ~~uses these fields to verify counter uniqueness, bind the response to the credential identity, and validate the signature~~ **SHALL** check the key reference against its

configured authentication purpose, independently discover the algorithm from the role certificate, and reject a mismatching reported algorithm. It verifies counter uniqueness, identity binding, the proof, and certificate path ~~using ACU~~ under its trust policy. ~~A success result~~ Success **SHALL** include a nonzero ~~credential UUID~~UUID, except for the bound Legacy TWIC case in Subsection 3.12. The PD **SHALL** preserve ~~the credential's signed response bytes as returned by the PIV operation~~returned proof bytes, excluding transport status words~~such as SW1/SW2.~~. PIVSTATUS's wire layout is unchanged by key discovery and Resume.

If the complete signed response does not fit in one OSDP response, the PD **SHALL** fragment `osdp_PIVSTATUS` using the OSDP 3.0 multi-part convention. *MpSizeTotal* is the total status payload size, *MpOffset* is the zero-based offset of the fragment, and *MpFragmentSize* is the number of payload bytes carried in that fragment. The logical status payload begins at offset zero with the fixed fields; each fragment carries the next contiguous bytes of that payload. A fragment boundary may occur within the fixed fields or within the signed response.

3.14 VCI Trust Anchor Load (osdp_PIVVCILOADTA)

The `osdp_PIVVCILOADTA` command (command code `0xAF`, tentative) loads, replaces, or deletes a credential SM trust anchor. The existing command name and record format apply to both contact and contactless SM. The ACU **SHALL** validate and authorize each anchor under its deployment certificate policy before loading it. The PD uses the loaded public key to validate the credential's SM CVC.

Table 36: VCI Trust Anchor Load (osdp_PIVVCILOADTA)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	`0xAF (tentative)`
0x02	MpSizeTotal	Total size of the reassembled operation data, including the 0x0C-byte metadata block and Anchor Data, least-significant byte first.	`0x000C–0x024C`
0x02	MpOffset	Offset of this fragment within the reassembled operation data beginning with Operation, least-significant byte first.	`0x0000–0xFFFF`
0x02	MpFragmentSize	Number of operation data bytes following the multi-part fields in this fragment, least-significant byte first.	`0x0000–0xFFFF`
0x01	Operation	`0x00` load/replace anchor; `0x01` delete anchor. Anchor inventory is requested with `osdp_PIVVCITASTATUS`.	`0x00–0x01`
0x01	Trust Anchor ID	Controller-assigned identifier for the anchor.	`0x01–0x80`

Size (bytes)	Name	Meaning	Value
0x08	Issuer Identifier Number	For load/replace, the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate and a match for tag 0x42 in the trust anchor record. For delete, this value **SHALL** match the stored anchor's IIN.	0x00–0xFF
0x02	Anchor Length	Total size of the variable-length VCI trust anchor record, least-significant byte first. Set to 0x0000 when deleting.	0x0000–0x0240
0x00-n	Anchor Data	Fragment bytes of the OSDP-defined VCI trust anchor record. Omitted for delete operations.	0x00–0xFF

The reassembled operation data begins at *MpOffset* 0x0000 with a 12-byte metadata block containing *Operation*, *Trust Anchor ID*, *Issuer Identifier Number*, and *Anchor Length*, followed immediately by *Anchor Data*. These fields may cross fragment boundaries and are interpreted only after the complete logical data has been reassembled. Controllers set *Operation* to 0x00 to load or replace the anchor identified by *Trust Anchor ID*. Set *Operation* to 0x01 with *Anchor Length* 0x0000 and *MpSizeTotal* 0x000C to delete the anchor. For a delete operation, the supplied IIN **SHALL** match the IIN of the stored anchor selected by *Trust Anchor ID*; a mismatch **SHALL** be rejected with osdp_PIVERROR 0x0005 (invalid parameter). Inventory is requested with osdp_PIVVCITASTATUS, not with osdp_PIVVCILOADTA. The *Issuer Identifier Number (IIN)* is an OSDP-defined anchor selector: it **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate associated with this anchor. For load/replace, this value **SHALL** match the IIN field (tag 0x42) in the VCI trust anchor record.

The *Anchor Length* declares the exact size of the variable-length VCI trust anchor record. For every operation, *MpSizeTotal* **SHALL** equal 0x000C plus *Anchor Length*. For an operation accepted for processing, the sum of all accepted *MpFragmentSize* values **SHALL** equal *MpSizeTotal*, and no fragment **SHALL** extend beyond *MpSizeTotal*. For load operations, the reassembled bytes following the metadata block **SHALL** contain exactly *Anchor Length* bytes. A conforming VCI trust anchor record **SHALL NOT** exceed 576 bytes (0x0240), making the maximum *MpSizeTotal* 0x024C. The PD **SHALL** accept any valid record length from the minimum BER-TLV encoding through 0x0240, subject to supported-key and storage-capacity checks.

PD storage accounting for VCI trust anchors is variable-length. A PD **SHALL NOT** treat every anchor as consuming the maximum 0x0240 bytes unless that is the PD's documented physical allocation model. When reporting capacity, loading, replacing, or deleting anchors, the PD **SHALL** account for stored anchors by their actual *Anchor Length* values and **SHALL** use the same accounting rule for both osdp_PIVVCITASTATUSR *Available Bytes* and osdp_PIVVCILOADTA storage acceptance. A replacement operation **SHALL** be evaluated after releasing the storage occupied by the existing anchor with the same *Trust Anchor ID*; if the replacement cannot be completed, the prior anchor **SHALL** remain available. A delete operation **SHALL** release the storage associated with the deleted anchor.

Table 37: VCI Trust Anchor Record Format

Tag	Length	Name	Value
0x7F50	Variable	VCI Trust Anchor Record	OSDP-defined record used to verify a secure messaging CVC or an Intermediate CVC.
0x80	0x01	Record Profile Identifier	0x01.
0x42	0x08	Issuer Identifier Number	Anchor selector. This value **SHALL** match the IIN in the osdp_PIVVCILOADTA metadata.
0x7F49	0x01–0x0226	SubjectPublicKeyInfo	DER-encoded X.509 SubjectPublicKeyInfo for the public key trusted by the ACU. Supported public-key algorithms are RSA public keys up to 4096 bits and EC public keys for P-256 or P-384.

The VCI trust anchor record is intentionally not a NIST SP 800-73 Intermediate CVC and does not carry X.509 path-validation state. The ACU is responsible for selecting, loading, unloading, and validating the VCI anchors it permits a PD to use before the PD relies on them. The PD uses the record only as a local public-key verifier selected by IIN. The maximum record size is derived from the largest supported key case: a 550-byte RSA-4096 SubjectPublicKeyInfo plus the fixed fields above and BER-TLV length overhead. The resulting worst case is 573 bytes; this specification rounds the maximum accepted record size to 576 bytes.

Use during credential SM establishment When the PD receives the secure messaging CVC from the credential, it **SHALL** first extract the Issuer Identification Number from tag 0x42. If a loaded VCI trust anchor has a matching IIN, the PD **SHALL** use that anchor's SubjectPublicKeyInfo to verify the secure messaging CVC signature. If verification succeeds, the PD may continue credential SM establishment using the public key in the secure messaging CVC.

If no loaded VCI trust anchor matches the secure messaging CVC IIN, the PD **SHALL** retrieve the Secure Messaging Certificate Signer data object (object identifier 0x5FC122, container identifier 0x1017) from the credential and inspect it for one Intermediate CVC (tag 0x7F21). If the object is absent or no Intermediate CVC is present, validation fails. If an Intermediate CVC is present, the PD **SHALL** verify all of the following before continuing:

- The secure messaging CVC IIN (tag 0x42) equals the Subject Identifier (tag 0x5F20) in the Intermediate CVC.
- The Intermediate CVC Role Identifier (tag 0x5F4C) is 0x12.
- A loaded VCI trust anchor exists whose IIN equals the Intermediate CVC IIN (tag 0x42).
- The Intermediate CVC signature verifies using the matching VCI trust anchor public key.
- The secure messaging CVC signature verifies using the public key carried in the Intermediate CVC.

The PD validates only the CVC chain needed for credential SM establishment: either the loaded trust anchor directly verifies the secure messaging CVC, or the loaded trust anchor verifies the Intermediate CVC that then verifies the secure messaging CVC. The PD **SHALL NOT** establish credential SM unless this CVC validation succeeds. The PD **SHALL NOT** perform full X.509 path validation, validity-period evaluation, certificate-policy evaluation, or revocation processing for VCI trust anchors; those checks are completed by the ACU before loading anchors and again as an acceptance policy after credential SM state is reported.

An absent signer object, an absent intermediate CVC, or the absence of a matching loaded anchor leaves the PD without a matching trust path. The PD reports SM Detail `0x02` for that condition. A malformed CVC, incorrect role or linkage, or failed signature check reports Detail `0x03` and blocks credential operations. These distinctions apply to both PIV Auto results and SM diagnostics.

ACU acceptance of credential SM After receiving `osdp_CARDSTATUSR` with credential SM established, the ACU **SHALL** validate the content-signing certificate corresponding to the reported *SM Anchor IIN* before treating the credential SM session as accepted for the deployment. This validation is an additional ACU acceptance check and does not replace the PD's required CVC validation during credential SM establishment. The ACU **SHALL** perform all of the following:

- Obtain the content-signing X.509 certificate. If the ACU does not already possess a cached copy whose subjectKeyIdentifier leftmost-eight-bytes equals the *SM Anchor IIN*, the ACU **SHALL** use `osdp_PIVGETDATA` to retrieve the Secure Messaging Certificate Signer data object (object identifier `0x5FC122`, container identifier `0x1017`) and extract the certificate from inner tag `0x70`.
- Verify that the leftmost eight bytes of the certificate's subjectKeyIdentifier equal the *SM Anchor IIN* reported in `osdp_CARDSTATUSR`.
- Verify that the certificate's SubjectPublicKeyInfo matches the SubjectPublicKeyInfo stored in the loaded VCI trust anchor record for that IIN.
- Verify that the certificate includes the PIV content-signing extended key usage OID (`2.16.840.1.101.3.6.7`) or the PIV-I equivalent (`2.16.840.1.101.3.8.7`).
- Verify that the current time falls within the certificate's validity period (`notBefore` $\leq$ now $\leq$ `notAfter`). If the certificate has expired or is not yet valid, the ACU **SHALL** reject the credential SM session and **SHALL** unload the corresponding trust anchor.
- Perform X.509 path validation and revocation checking in accordance with the deployment's certificate validation policy.

Table 38: VCI Trust Anchor Examples from NIST Vectors

Vector	Content Signer	Anchor IIN	SPKI	Record
card_2, card_5	Test PIV Content Signer 4 (EC P-384, direct)	B6103792270FBD08	0x78 bytes	0x8C bytes
card_3	Test PIV Content Signer 1 (RSA-2048, intermediate)	9BC5E7735F14EC7F	0x126 bytes	0x13D bytes
card_4	Test PIV Content Signer 3 (EC P-256, direct)	1905976BDDCA823D	0x5B bytes	0x6E bytes
card_16	Test PIV-I Content Signer 1 (RSA-2048, intermediate)	4CC9C7B3A4F1C83E	0x126 bytes	0x13D bytes

In the card_16 intermediate example, the secure messaging CVC IIN is 6F8FF48F42B3E22E; the Intermediate CVC Subject Identifier (tag 0x5F20) is also 6F8FF48F42B3E22E; and the Intermediate CVC IIN (tag 0x42) is 4CC9C7B3A4F1C83E, matching the loaded RSA trust anchor.

PDs **SHALL** reassemble the complete operation data before processing it. For a load or replacement, the PD **SHALL** verify that the reassembled *Anchor Data* portion is exactly *Anchor Length* bytes and is a single 0x7F50 record before committing it to storage. A successful operation returns osdp_ACK. Failures return:

- osdp_NAK function-not-supported when the PD does not support trust-anchor caching.
- osdp_PIVERROR 0x0005 (invalid parameter) for malformed metadata (unknown operation, reserved bits, invalid lengths, MpSizeTotal not equal to 0x000C plus *Anchor Length*, or a reassembled Anchor Data length not equal to *Anchor Length*).
- osdp_PIVERROR 0x0007 (invalid multi-part transfer) when multi-part sequencing fails.
- osdp_PIVERROR 0x1021 (insufficient PD memory or storage) when the PD lacks storage space for the anchor.

PDs are not required to maintain a real-time clock. PDs may retain either the supplied VCI trust anchor record or an equivalent internal representation so long as the public key and supplied IIN remain available for the verification flow above.

Trust anchors may be stored in volatile or non-volatile memory at the PD's discretion. Controllers should not assume retention beyond the device's documented behaviour.

Replacing or deleting the anchor used by an active SM session **SHALL** invalidate that session, its VCI state, and PIN-verified status. The PD **SHALL** clear security-dependent cached material and trust evidence for that session. A failed replacement preserves the existing anchor and session. Changes to unrelated anchors leave the active session unchanged. Trust-store updates take effect in the next explicitly started preparation attempt.

3.15 VCI Trust Anchor Status (osdp_PIVVCITASTATUS)

The osdp_PIVVCITASTATUS command (command code 0xA6, tentative) enumerates cached trust anchors and reports remaining storage capacity. The request carries no payload. The reply uses the standard multi-part fields (MpSizeTotal, MpOffset, MpFragmentSize) so PDs can stream the inventory across one or more fragments using reply code 0x88 (tentative).

Table 39: VCI Trust Anchor Status Request (osdp_PIVVCITASTATUS)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA6 (tentative)

Table 40: VCI Trust Anchor Status Reply (osdp_PIVVCITASTATUSR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x88 (tentative)
0x02	MpSizeTotal	Length of the complete status payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the status payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the data block in this fragment, least-significant byte first.	0x0000–0xFFFF
0x01	Total Anchor Count	Number of cached trust anchors.	0x00–0xFF
0x02	Available Bytes	Remaining storage capacity for additional variable-length trust anchor records, least-significant byte first.	0x0000–0xFFFF
0x00-n	Anchor Records	Zero or more 0x0C-byte inventory records. Each record contains: TrustAnchorID (0x01 byte for load/delete management), anchor selector IIN (0x08 bytes from the stored trust anchor record, not necessarily the secure CVC IIN), AnchorLength (0x02 bytes LSB first, maximum 0x0240), and Attributes (0x01 byte, reserved). Records that span fragments continue in subsequent replies.	–

The PD first reports the total number of anchors and the remaining free bytes. *Available Bytes* is the amount of trust-anchor storage that remains for additional variable-length VCI trust anchor record data after accounting for currently stored anchors by their actual *Anchor Length* values. Each record thereafter lists a trust anchor by management ID, anchor selector IIN, anchor length, and reserved attributes (transmitted as zero). The IIN in this inventory is the IIN from the stored VCI trust anchor record and is not necessarily the secure messaging CVC IIN reported in `osdp_CARDSTATUSR`. Error responses follow the standard pattern: `osdp_NAK` function-not-supported when the command is not implemented, and `osdp_PIVERROR` `0x0007` (invalid multi-part transfer) for multi-part errors.

Table 41: Trust Anchor Record Structure

Field	Size	Meaning	`Value Range`
Trust Anchor ID	0x01 byte	Identifier assigned by the ACU; reused to replace or delete an anchor.	`0x01–0x80`
Issuer Identifier Number (IIN)	0x08 bytes	Anchor selector supplied in the VCI trust anchor record.	`0x00–0xFF`
Anchor Length	0x02 bytes (LSB first)	Total size of the stored variable-length VCI trust anchor record supplied for this anchor.	`0x0000–0x0240`
Attributes	0x01 byte	Reserved for future use; transmitted as zero.	`0x00`

A References

A.1 Normative References

- Security Industry Association, *Open Supervised Device Protocol (OSDP) v2.2*, 2020.
- Transportation Security Administration, *TWIC Specification, Part 2 — TWIC Card Application*, December 2024, Sections 4.1–4.4, 4.6.2, 4.6.4, 4.7.1, 4.7.3, 4.8, 5.2–5.3, 6, 7.5, and Appendices C–E (application selection, authentication, identity, APDU chaining, biometric objects, and TWIC Privacy Key encoding).
- Transportation Security Administration, *TWIC Specification, Part 3 — TWIC Reader Requirements*, July 2024, Sections 4.4.4–4.4.8 and 7 (TWIC card authentication, biometric, and application-specific processing).
- National Institute of Standards and Technology, *NIST Special Publication 800-73-5, Interfaces for Personal Identity Verification (PIV) — Part 1: PIV Card Application Namespace, Data Model, and Representation*, 2024.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-4, Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, February 2015.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-5, Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, 2024.
- National Institute of Standards and Technology, *NIST Special Publication 800-78-5, Cryptographic Algorithms and Key Sizes for Personal Identity Verification*, 2024.
- National Institute of Standards and Technology, *Federal Information Processing Standards Publication 201-3, Personal Identity Verification (PIV) of Federal Employees and Contractors*, January 2022.
- National Institute of Standards and Technology, *NIST Special Publication 800-185, SHA-3 Derived Functions: cSHAKE, KMAC, TupleHash, and ParallelHash*, December 2016.
- International Organization for Standardization and International Electrotechnical Commission, *ISO/IEC 7816-4, Identification cards — Integrated circuit cards — Part 4: Organization, security and commands for interchange*.
- Internet Engineering Task Force, *RFC 5280, Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile*, May 2008.
- Internet Engineering Task Force, *RFC 1951, DEFLATE Compressed Data Format Specification version 1.3*; and *RFC 1952, GZIP file format specification version 4.3*, May 1996.
- Internet Engineering Task Force, *RFC 2119, Key words for use in RFCs to Indicate Requirement Levels*, March 1997; and *RFC 8174, Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*, May 2017.

A.2 Informative References

- Transportation Security Administration, *TWIC Specification, Part 4 — TWIC uses and registration by a PACS*, July 2024, Sections 2.5.1 and 2.5.2 (Legacy and NextGen registration guidance).
- National Institute of Standards and Technology, *NIST Internal Report 8347, Personal Identity Verification Cards, Version 2*, 2021.
- U.S. General Services Administration, *Federal Identity, Credential, and Access Management (FICAM) Architecture*, Revision 3, 2023.
- Open Physical, *High Speed Autonomous PIV: Concept Paper*, 2024.

DRAFT

B PIVGETDATA Worked Examples (Informative)

These examples show the enhanced request defined in Section 2.1.3, the OSDP 3.0 multi-part response defined in Section 2.1.4, and the error reply defined in Section 2.2.1.

The ACU and PD are simulated. Credential data comes from the identified existing card captures; constructed error cases are identified separately. These examples omit OSDP Secure Channel for clarity. CRC detects transmission errors. The exchanges provide neither authentication nor confidentiality. Some source captures used credential-side secure messaging; the examples use the data recovered after that processing.

The files under `worked-examples/pivdata/` contain the complete frame exchanges, expected results, source hashes, and validation instructions. Each example ID also appears in the data file, so the printed example can be matched to its complete exchange. Long responses are shortened in this appendix only.

B.1 Reading the exchanges

Field values use hexadecimal with a `0x` prefix. Packet dumps use space-separated hexadecimal byte pairs under *Packet bytes (hex)*. Two-byte field values are shown as numbers in the request table and in least-significant-byte-first order in the packet bytes.

An accepted PIVGETDATA command receives ACK. The ACU then sends POLL until the PD returns a data fragment or PIVERROR. The first data fragment has offset zero, even when Offset in the command is nonzero. Further polls retrieve the rest of the data in order, until the ACU has received the advertised total. Polls during processing receive ACK. The examples use a maximum frame size of `0x0080` bytes, including header and CRC; this leaves `0x0072` data bytes per PIVGETDATAR fragment when no security block is present.

A whole-object read includes the object's outer tag, length, and value. An inner-tag read includes the selected element's tag, length, and value, and excludes the enclosing object's tag and length. Offset and Requested Length then determine which bytes to return. Offset zero starts at the selected tag; skipping its tag and length returns only value bytes.

The PD divides the requested data across replies. Each TLV header occurs once in the reassembled data. Each reply has its own OSDP framing, reply identifier, and multi-part fields; the ACU joins only the Card Data bytes. Certificate and biometric values retain their original encoding. Selecting the FASC-N and UUID elements individually allows the ACU to read them regardless of their positions within the CHUID.

All exchanges use PD address 0x01 and CRC-16. Each request table gives the field values; the packet bytes below it show how those values are encoded.

Complete machine-readable exchanges include every frame. The fixture manifest records the inputs, expected results, source locations, and hashes. Hex includes the OSDP header and CRC, without an optional leading mark byte. Long replies show their first 0x18 and last 0x10 bytes; ellipsis text is not transmitted.

B.2 Source fixtures

S1: NIST SD33 card02, contact card capture.

S2: NIST SD33 card01, contact card capture.

The 19 examples below are a representative subset of the 37 checked exchanges in the linked JSON file.

B.3 Complete Discovery Object

Read the complete Discovery Object. Tag Length 0x00 keeps its outer tag 0x7E and its length, followed by all its contents.

Example ID: `discovery-whole`. Source: S1.

Table 42: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0014 returned bytes in 0x01 data reply.

Table 43: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 0F 00 05 A3 01 7E 00 00 00 00 00 4A 01	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 22 00 06 80 14 00 00 00 14 00 7E 12 4F 0B A0 00 00 03 08 00 00 10 00 01 00 5F 2F 02 68 20 DB F9	Total 0x0014 Offset 0x0000 Fragment 0x0014

B.4 Discovery PIN Usage Policy (0x5F2F)

Read the PIN Usage Policy from the Discovery Object. The reply contains tag 0x5F2F, length 0x02, and the two policy bytes.

Example ID: `discovery-pin-policy`. Source: S1.

Table 44: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x02	Bytes in Element Tag.
Element Tag	0x5F2F	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0005 returned bytes in 0x01 data reply.

Table 45: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 01 7E 02 5F 2F 00 00 00 00 03 F0	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 13 00 06 80 05 00 00 00 05 00 5F 2F 02 68 20 31 67	Total 0x0005 Offset 0x0000 Fragment 0x0005

B.5 Read the FASC-N element (0x30)

Read the FASC-N element from the CHUID without depending on its position. The returned element begins with tag 0x30 and length 0x19, followed by the FASC-N.

Example ID: `chuid-fascn`. Source: S1.

Table 46: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x001B returned bytes in 0x01 data reply.

Table 47: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 30 00 00 00 00 15 89	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 29 00 06 80 1B 00 00 00 1B 00 30 19 D6 50 18 58 55 E5 6D C8 12 79 45 A1 6C DB 90 6E 78 80 C0 82 86 50 18 43 F5 E1 F1	Total 0x001B Offset 0x0000 Fragment 0x001B

B.6 Read the FASC-N value

Read just the FASC-N bytes. Offset 0x0002 skips the selected element's 0x30 0x19 header; Requested Length 0x0019 limits the result to the FASC-N value.

Example ID: `chuid-fascn-value`. Source: S1.

Table 48: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0002	Bytes to skip from the selected tag.
Requested Length	0x0019	Maximum bytes to return.

Result: 0x0019 returned bytes in 0x01 data reply.

Table 49: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 30 02 00 19 00 96 DD	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 27 00 06 80 19 00 00 00 19 00 D6 50 18 58 55 E5 6D C8 12 79 45 A1 6C DB 90 6E 78 80 C0 82 86 50 18 43 F5 6C 73	Total 0x0019 Offset 0x0000 Fragment 0x0019

B.7 Read the UUID element (0x34)

Read the UUID (also called the GUID) element from the CHUID. The returned element begins with tag 0x34 and length 0x10, followed by the UUID.

Example ID: `chuid-guid`. Source: S1.

Table 50: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x34	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0012 returned bytes in 0x01 data reply.

Table 51: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 34 00 00 00 00 13 00	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 20 00 06 80 12 00 00 00 12 00 34 10 45 85 34 70 A4 03 48 44 AE 15 C6 49 59 59 FA D5 FA C7	Total 0x0012 Offset 0x0000 Fragment 0x0012

B.8 Read the UUID value

Read just the UUID bytes. Offset 0x0002 skips the selected element's 0x34 0x10 header; Requested Length 0x0010 limits the result to the UUID value.

Example ID: `chuid-guid-value`. Source: S1.

Table 52: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x34	Element to select.
Offset	0x0002	Bytes to skip from the selected tag.
Requested Length	0x0010	Maximum bytes to return.

Result: 0x0010 returned bytes in 0x01 data reply.

Table 53: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 34 02 00 10 00 08 EE	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 1E 00 06 80 10 00 00 00 10 00 45 85 34 70 A4 03 48 44 AE 15 C6 49 59 59 FA D5 75 CB	Total 0x0010 Offset 0x0000 Fragment 0x0010

B.9 Bounded CHUID prefix

Read the first 0x0010 bytes of the CHUID, starting with its outer tag and length. In this capture, the range ends partway through the FASC-N element.

Example ID: `chuid-prefix`. Source: S1.

Table 54: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0010	Maximum bytes to return.

Result: 0x0010 returned bytes in 0x01 data reply.

Table 55: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 02 00 00 00 10 00 A1 F4	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 1E 00 06 80 10 00 00 00 10 00 53 82 07 C8 EE 02 07 C4 30 19 D6 50 18 58 55 E5 5A 2E	Total 0x0010 Offset 0x0000 Fragment 0x0010

B.10 Complete certificate object

Read the complete certificate object, including its outer 0x53 header and all elements inside it. The result is larger than one reply, so the ACU polls for the remaining fragments.

Example ID: `certificate-whole`. Source: S1.

Table 56: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x066F returned bytes in 0x0F data replies.

Table 57: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 05 00 00 00 00 00 93 3F	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 6F 06 00 00 72 00 53 82 06 6B 70 82 06 62 30 82 06 5E ... 0x58 bytes omitted ... 65 73 74 20 43 41 31 33 30 31 06 03 55 04 4F 7E	Total 0x066F Offset 0x0000 Fragment 0x0072
POLL/reply	Omitted	0x0D exchanges; complete frames are in the JSON file.
ACU POLL	53 01 08 00 05 60 8B 33	Poll 0x0F.

Message	Packet bytes (hex)	Notes
PD PIVGETDATAR	53 81 41 00 05 80 6F 06 3C 06 33 00 49 D9 67 65 56 F7 94 17 43 D7 84 02 9F 61 BF D6 D3 99 B6 DC D1 DE 0D 87 F5 F8 1A A9 CD 4D B8 BB D0 E6 6E 4B F5 75 87 9D AB F1 EF 91 D2 1A 71 01 00 FE 00 F7 26	Total 0x066F Offset 0x063C Fragment 0x0033

B.11 Read the certificate element (0x70)

Read element 0x70, including its tag, encoded length, and certificate bytes. This omits the enclosing 0x53 header and the other elements in the object. The 0x70 header appears once in the returned data.

Example ID: `certificate-element`. Source: S1.

Table 58: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x70	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0666 returned bytes in 0x0F data replies.

Table 59: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 05 01 70 00 00 00 00 39 81	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 66 06 00 00 72 00 70 82 06 62 30 82 06 5E 30 82 04 46 ... 0x58 bytes omitted ... 43 41 31 33 30 31 06 03 55 04 03 13 2A 54 2D 25	Total 0x0666 Offset 0x0000 Fragment 0x0072

Message	Packet bytes (hex)	Notes
POLL/reply	Omitted	0x0D exchanges; complete frames are in the JSON file.
ACU POLL	53 01 08 00 05 60 8B 33	Poll 0x0F.
PD PIVGETDATAR	53 81 38 00 05 80 66 06 3C 06 2A 00 56 F7 94 17 43 D7 84 02 9F 61 BF D6 D3 99 B6 DC D1 DE 0D 87 F5 F8 1A A9 CD 4D B8 BB D0 E6 6E 4B F5 75 87 9D AB F1 EF 91 D2 1A 8B 7C	Total 0x0666 Offset 0x063C Fragment 0x002A

B.12 Read a compressed certificate element

Read element 0x70 from a card that stores a compressed certificate. The PD returns the original tag, length, and compressed bytes. The ACU can read the object's certificate-information element to determine how to interpret the value.

Example ID: `certificate-compressed-element`. Source: S2.

Table 60: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x70	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x05EA returned bytes in 0x0E data replies.

Table 61: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 05 01 70 00 00 00 00 39 81	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 EA 05 00 00 72 00 70 82 05 E6 1F 8B 08 00 00 00 00 00 ... 0x58 bytes omitted ... CE F4 47 FF F4 CF 5D BE EF 9C 7B CF 39 F7 79 A2	Total 0x05EA Offset 0x0000 Fragment 0x0072
POLL/reply	Omitted	0x0C exchanges; complete frames are in the JSON file.
ACU POLL	53 01 08 00 07 60 E9 55	Poll 0x0E.
PD PIVGETDATAR	53 81 2E 00 07 80 EA 05 CA 05 20 00 7F C4 30 BC D9 5C B8 05 0A 09 18 39 F8 6D A5 80 16 10 4D DC 8D 38 FC 05 BF 05 DC 1B AF 07 00 00 96 99	Total 0x05EA Offset 0x05CA Fragment 0x0020

B.13 Read a range within the certificate element

Read 0x0104 bytes starting at offset 0x0100 of the certificate element, counting from tag 0x70. The reply offsets are 0x0000, 0x0072, and 0x00E4; each identifies a position within the 0x0104 returned bytes.

Example ID: `certificate-offset256`. Source: S1.

Table 62: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x70	Element to select.
Offset	0x0100	Bytes to skip from the selected tag.
Requested Length	0x0104	Maximum bytes to return.

Result: 0x0104 returned bytes in 0x03 data replies.

Table 63: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 05 01 70 00 01 04 01 EC 6A	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 04 01 00 00 72 00 14 30 12 06 03 55 04 0B 13 0B 54 65 ... 0x58 bytes omitted ... 00 03 82 01 0F 00 30 82 01 0A 02 82 01 01 CE C4	Total 0x0104 Offset 0x0000 Fragment 0x0072
ACU POLL	53 01 08 00 07 60 E9 55	Poll 0x02.
PD PIVGETDATAR	53 81 80 00 07 80 04 01 72 00 72 00 00 C3 C3 53 BF 0F 03 48 37 48 A4 BA ... 0x58 bytes omitted ... 4A A1 F7 74 E7 A2 DE BC F3 87 AA 94 76 21 24 7E	Total 0x0104 Offset 0x0072 Fragment 0x0072
ACU POLL	53 01 08 00 05 60 8B 33	Poll 0x03.

Message	Packet bytes (hex)	Notes
PD PIVGETDATAR	53 81 2E 00 05 80 04 01 E4 00 20 00 FA 54 AA 2C DF 26 69 45 7E 6D CD 3A 48 EE 5E 4B 31 FD A9 4A D8 73 01 A7 5C C8 50 D2 0A 60 D1 CC 14 CD	Total 0x0104 Offset 0x00E4 Fragment 0x0020

B.14 Complete Key History object

Read the complete Key History object. Its outer 0x53 header and all elements inside it are returned unchanged.

Example ID: `key-history`. Source: S1.

Table 64: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC10C	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x000A returned bytes in 0x01 data reply.

Table 65: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 0C 00 00 00 00 00 71 77	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 18 00 06 80 0A 00 00 00 0A 00 53 08 C1 01 04 C2 01 00 FE 00 F3 05	Total 0x000A Offset 0x0000 Fragment 0x000A

B.15 BIT Group with no biometric templates

Read a Biometric Information Template (BIT) Group with no biometric templates. The reply keeps the outer 0x7F61 tag and the element that reports a count of zero.

Example ID: `bit-empty`. Source: constructed example.

Table 66: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x02	Bytes in Object Identifier.
Object Identifier	0x7F61	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0006 returned bytes in 0x01 data reply.

Table 67: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 02 7F 61 00 00 00 00 00 3E 14	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 14 00 06 80 06 00 00 00 06 00 7F 61 03 02 01 00 35 7C	Total 0x0006 Offset 0x0000 Fragment 0x0006

B.16 Empty 0x53 container

Read an object with no contents. The reply still contains two bytes, 0x53 0x00, because whole-object selection includes the outer tag and length.

Example ID: `empty-5300`. Source: constructed example.

Table 68: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0002 returned bytes in 0x01 data reply.

Table 69: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 05 00 00 00 00 93 3F	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 10 00 06 80 02 00 00 00 02 00 53 00 5D 38	Total 0x0002 Offset 0x0000 Fragment 0x0002

B.17 Zero-byte successful response

Handle a successful card response with zero data bytes. The PD sends one PIVGETDATAR with all three multi-part fields zero.

Example ID: `empty-success`. Source: constructed example.

Table 70: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0000 returned bytes in 0x01 data reply.

Table 71: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 0F 00 05 A3 01 7E 00 00 00 00 00 4A 01	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 0E 00 06 80 00 00 00 00 00 00 C0 05	Total 0x0000 Offset 0x0000 Fragment 0x0000

B.18 Existing child with an empty value

Read an element whose value is empty. The reply contains its tag and zero length, 0x30 0x00, for a total of two bytes.

Example ID: `zero-length-child`. Source: constructed example.

Table 72: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0002 returned bytes in 0x01 data reply.

Table 73: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 01 7E 01 30 00 00 00 00 DC 37	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 10 00 06 80 02 00 00 00 02 00 30 00 24 66	Total 0x0002 Offset 0x0000 Fragment 0x0002

B.19 Missing direct child

Request an element that is absent from the object. The PD accepts the command, then reports PIVERROR on the next poll.

Example ID: `missing-child`. Source: constructed example.

Table 74: Request fields

Field	Value	Notes
Command	`0xA3`	`osdp_PIVGETDATA`
Identifier Length	`0x01`	Bytes in Object Identifier.
Object Identifier	`0x7E`	Object to read.
Tag Length	`0x01`	Bytes in Element Tag.
Element Tag	`0x34`	Element to select.
Offset	`0x0000`	Start at the selected tag.
Requested Length	`0x0000`	All remaining bytes.

Result: `osdp_PIVERROR 0x1024` (data object or tag not found); see Section 2.2.1.

Table 75: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU `PIVGETDATA`	`53 01 10 00 05 A3 01 7E 01 34 00 00 00 00 DA BE`	Read request.
PD `ACK`	`53 81 08 00 05 40 39 35`	Command accepted.
ACU `POLL`	`53 01 08 00 06 60 D8 66`	Poll `0x01`.
PD `PIVERROR`	`53 81 0A 00 06 8A 24 10 A3 74`	`0x1024` (data object or tag not found)

B.20 Element request against an empty response

Request an element when the card returns no data. The PD reports that the tag was not found. A whole-object request for the same response would succeed with zero bytes.

Example ID: `empty-missing-child`. Source: constructed example.

Table 76: Request fields

Field	Value	Notes
Command	`0xA3`	`osdp_PIVGETDATA`
Identifier Length	`0x01`	Bytes in Object Identifier.
Object Identifier	`0x7E`	Object to read.
Tag Length	`0x01`	Bytes in Element Tag.
Element Tag	`0x34`	Element to select.
Offset	`0x0000`	Start at the selected tag.
Requested Length	`0x0000`	All remaining bytes.

Result: `osdp_PIVERROR 0x1024` (data object or tag not found); see Section 2.2.1.

Table 77: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU `PIVGETDATA`	`53 01 10 00 05 A3 01 7E 01 34 00 00 00 00 DA BE`	Read request.
PD `ACK`	`53 81 08 00 05 40 39 35`	Command accepted.
ACU `POLL`	`53 01 08 00 06 60 D8 66`	Poll `0x01`.
PD `PIVERROR`	`53 81 0A 00 06 8A 24 10 A3 74`	`0x1024` (data object or tag not found)

B.21 Ambiguous direct-child selection

Request a tag that occurs twice directly inside the outer container. Element selection requires a unique match, so the PD reports an error.

Example ID: `duplicate-child`. Source: constructed example.

Table 78: Request fields

Field	Value	Notes
Command	`0xA3`	`osdp_PIVGETDATA`
Identifier Length	`0x01`	Bytes in Object Identifier.
Object Identifier	`0x7E`	Object to read.
Tag Length	`0x01`	Bytes in Element Tag.
Element Tag	`0x30`	Element to select.
Offset	`0x0000`	Start at the selected tag.
Requested Length	`0x0000`	All remaining bytes.

Result: `osdp_PIVERROR 0x1027` (ambiguous element selection); see Section 2.2.1.

Table 79: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU `PIVGETDATA`	`53 01 10 00 05 A3 01 7E 01 30 00 00 00 00 DC 37`	Read request.
PD `ACK`	`53 81 08 00 05 40 39 35`	Command accepted.
ACU `POLL`	`53 01 08 00 06 60 D8 66`	Poll `0x01`.
PD `PIVERROR`	`53 81 0A 00 06 8A 27 10 F0 21`	`0x1027` (ambiguous element selection)

C High-Speed Autonomous PIV Data Templates (Informative)

This appendix ~~records the PIV Auto challenge, ISO/IEC 7816 GENERAL AUTHENTICATE templates, and fixture sources used by the demonstration~~ illustrates the per-entry authentication key selector, per-card certificate discovery, and explicit resumption of PIV Auto. The configuration, discovery, resume, and TWIC sequences added in this revision are illustrative; they have not been executed or used to regenerate cryptographic vectors. The ~~APDU examples are validated by the repository tests against the same helper code used by the simulator and replay tools.~~ normative rules are in Subsections 3.12, 3.12, and 3.12.

The existing APDU fixtures and generated signing examples are retained separately. Their prototype KDK input uses the SC1-style 48-byte session-key set and does not establish Secure Channel 2.0 interoperability. SC2 remains a required upstream dependency; its transport and final PIV Auto binding are outside these examples.

C.1 Vector Sources

Table 80: PIV Auto Vector Sources

Source	Use	Notes
NIST SD 33 / IR 8347	Live-card APDU fixtures	Cards 2, 3, 4, 5, and 16 provide VCI contactless enrolment, OPACITY, and key-challenge transcripts.
NIST SM/VCI captures	CVC and trust anchors	Provides validator-ready 7F21, 5FC122, and 7F50 objects.
GSA FICAM card-builder	Local keys and certificates	Used only as reusable local test material for source-backed simulation.
Generated YubiKey PIV	Missing algorithm profiles	Reusable RSA-1024, RSA-3072, and P-384 test keys are committed as public fixtures and loaded through `ykmanpiv`; RSA-1024 is informative legacy test material only.

C.2 Key Selection and Certificate Discovery

An Auto-enabled Configuration Entry ends with one Authentication Key Reference byte, after the AID bytes. There is no configured algorithm byte. With Auto disabled, the key-reference byte is absent and the previous entry encoding is unchanged. For example, the default PIV application with contact and contactless permitted is 01040103009E: tag `0x01`, length `0x04`, Data Model `0x01`, Interface Mask `0x03`, AID Length `0x00`, and key `0x9E`. Changing only the last byte to `0x9A` selects PIV Authentication without fixing the card's key type.

The logical configuration begins with three header bytes, two four-byte counters, and 32 entropy bytes. The shortest Auto-enabled configuration is therefore `0x31` bytes (49 decimal), including its one six-byte Configuration Entry. Configuration Count counts Configuration Entry TLVs only.

Table 81: Illustrative PIV Auto Configuration and Resume Bytes

Message	Field	Bytes
Card Authentication	Flags, Count, Global Mask, Series, Sequence	010103070000002A000000
Card Authentication	Supplemental Entropy	00000000000000000000000000000000000000000000000000000000000000000000000000000000
Card Authentication	Configuration Entry	01040103009E
Contactless PIV Authentication	Flags, Count, Global Mask, Series, Sequence	030102070000002A000000
Contactless PIV Authentication	Supplemental Entropy	00000000000000000000000000000000000000000000000000000000000000000000000000000000
Contactless PIV Authentication	Configuration Entry	01040102009A
Resume after pairing and PIN	Flags, Count, Global Mask, Series, Sequence	030102070000002B000000
Resume after pairing and PIN	Supplemental Entropy	00000000000000000000000000000000000000000000000000000000000000000000000000000000
Resume after pairing and PIN	Unchanged Configuration Entry	01040102009A
Resume after pairing and PIN	Final Resume TLV	0208070000002A000000

Concatenate the rows belonging to one message. The two initial logical payloads are 0x31 bytes; the resume payload is 0x3B bytes. For a single fragment, prefix the initial payload with AE310000003100, or the resume payload with AE3B0000003B00. These prefixes contain CMND, MpSizeTotal, MpOffset, and MpFragmentSize. OSDP packet headers and SC2 protection are not shown. All-zero entropy is permitted but not recommended for deployment.

Table 82: Illustrative Authentication Certificate Reads

Configured Key	Logical Card APDU	Certificate Object
PIV 0x9A	00CB3FFF055C035FC10500	0x5FC105; contactless requires VCI.
PIV 0x9E	00CB3FFF055C035FC10100	0x5FC101; no PIV PIN required.
NextGen TWIC 0x9E	00CB3FFF055C035FC101FF	0x5FC101 in the selected TWIC application.

These are logical APDUs; an established credential SM session protects applicable PIV commands. The PD reassembles the returned certificate object, extracts tag 0x70, and interprets its one-byte CertInfo in tag 0x71: 0x00 identifies uncompressed DER and 0x01 identifies GZIP-compressed DER. It decompresses when

indicated before parsing SubjectPublicKeyInfo. Missing or malformed certificate evidence ends the attempt with Result 0x03; a recognized but unsupported subject-key profile uses Result 0x04. Before an algorithm is determined, PIVSTATUS reports Algorithm Identifier 0x00; that failure value never selects a default card algorithm.

Table 83: Mixed PIV Subject Keys under the Same Configuration

Subject Public Key	Configured Key	Discovered Algorithm	Challenge Profile
RSA-2048	0x9A or 0x9E	0x07	32-byte digest; SHA-256 DigestInfo.
RSA-3072	0x9A or 0x9E	0x05	48-byte digest; SHA-384 DigestInfo.
Named curve P-256	0x9A or 0x9E	0x11	32-byte prehashed ECDSA input.
Named curve P-384	0x9A or 0x9E	0x14	48-byte prehashed ECDSA input.

The four subject-key types at each of the two references give eight PIV profiles. Within either reference, successive RSA and ECC cards use identical configuration bytes. The certificate changes the GENERAL AUTHENTICATE P1 byte, challenge encoding, and reported algorithm. For example, an RSA CA can sign a P-384 subject certificate using SHA-256: the PD still discovers 0x14 and uses the 48-byte P-384 profile. The issuer's signature algorithm does not select the card's authentication algorithm. No alternate key is tried if discovery fails.

C.3 Illustrative PIV Authentication Resume

In the contactless 0x9A example above, attempt 0x00000007/ 0x0000002A ends with Result 0x05, Detail 0x01 because pairing is required. The ACU receives the complete result, then sends osdp_PIVXMITPAIRING, for example AD3132333435363738, and waits for confirmed VCI establishment. It requests commanded PIN acquisition and verification with AA010000000000: Mode 0x01, default finite timeouts, Auto Execute clear, and Cache Validity zero. The ACU waits for the successful PIVSPER before sending the resume PIVMODE payload above.

The final 0208070000002A000000 identifies the completed prerequisite failure, not the counters of the new attempt. The header retains Series 0x00000007 and supplies its exact next unused Sequence 0x0000002B. Configuration Count remains one. The entropy may change; the flags, masks, ordered entries, AIDs, and key references remain unchanged. For this eligible resume, completed pairing and PIN verification remain bound to the same presentation, application, interface, and security contexts.

The PD reads certificate 0x5FC105, discovers its subject-key profile, and performs one new key proof. Pairing or PIN completion alone does not restart Auto. Credential removal, a context change, or an interrupted security session invalidates resumption; an ordinary mode replacement still clears state. Duplicate OSDP delivery does not repeat pairing, PIN verification, or the proof.

C.4 Illustrative TWIC Auto Key Proof

The TWIC Configuration Entry 010D020209A000000367200000019E selects Data Model 0x02, contactless interface, the nine-byte TWIC AID prefix, and asymmetric key 0x9E. It does not configure an algorithm or a TPK. The PD issues 00A4040009A000000367200000100 and examines the full AID returned in tag 0x4F:

- **Legacy:** A000000367200000010101. Read the TWIC CHUID with 00CB3FFF055C035FC102FF, retaining its identity only for this attempt. Then select the PIV application with 00A404000BA0000003080001000010000 and retrieve certificate 0x5FC101. Use that application's 0x9E key while retaining the configured TWIC acceptance policy. A present, correctly sized nil GUID remains zero in the challenge and PIVSTATUS only when the valid nonzero FASC-N matches the Card Authentication certificate. Missing identity data does not receive the nil-GUID exception.
- **NextGen:** A000000367200000010103. Remain in the TWIC application, read CHUID 0x5FC102 and certificate 0x5FC101 using Le 0xFF, and retain the populated GUID. Concatenate the initial data and subsequent GET RESPONSE data when status 0x61XX indicates more bytes. Le 0x00 can return 0x9000 with no data and is not a successful certificate read.

The documented NextGen profile uses RSA-2048: discover 0x07 from the certificate SPKI and use GENERAL AUTHENTICATE with P1 0x07 and P2 0x9E. Neither workflow submits a PIV PIN or uses the TPK for this proof. The Legacy application transition preserves the card-authentication purpose; it does not authorize fallback from PIV Authentication 0x9A to Card Authentication 0x9E. A TWIC-model 0x9A entry is invalid.

These new TWIC examples contain no captured card responses or newly computed signatures. Implementation fixtures must use public synthetic keys, certificates, and invented identifiers, with their provenance kept separate from physical-card captures. PIVSTATUS success reports the OSDP-defined key proof, not completion of a TSA reader mode or an access decision. Certificate trust, identity binding, expiration, and applicable cancellation checks remain required by deployment policy. TWIC Part 3, Section 4.4.4 requires at least 127 bytes of unique challenge data; these illustrative examples do not establish that fixed RSA padding and a shorter KMAC digest satisfy that requirement. TSA reader qualification remains a separate verification item.

C.5 Supported Profiles

Table 84: PIV Auto Supported Profiles

Profile	Alg	Key Ref	PIV Auto	Coverage
RSA-1024	06	9A, 9E	No	Informative legacy/non-current test-only material.
RSA-2048	07	9A, 9E	Yes	NIST SD 33 cards 2, 3, 16; generated-card; GSA-backed simulation.
RSA-3072	05	9A, 9E	Yes	Committed YubiKey material; requires YubiKey firmware with RSA-3072 support.
ECC P-256	11	9A, 9E	Yes	NIST SD 33 cards 4 and 5; generated-card coverage.
ECC P-384	14	9A, 9E	Yes	Committed YubiKey material. NIST card 5 also provides 9C/9D APDU examples.

C.6 Fixed-Width KMAC Challenge Template

Table 85: PIV Auto KMAC Challenge Template

Field	Size	Encoding	Notes
Template label	0x0E	ASCII plus version byte	OSDP-PIV-AUTO followed by 01.
Series Counter	0x04	LSB first	~~ACU-managed boot or epoch counter~~ ACU-provided series, for example 07000000.
Sequence Counter	0x04	LSB first	~~ACU-managed per-series counter~~ PD-allocated attempt counter within that series, for example 2A000000; see Subsection 3.12.
FASC-N	0x19	Raw bytes	Credential FASC-N from CHUID.
UUID	0x10	Raw bytes	Credential UUID~~. Successful PIV Auto examples use the credential UUID ; unavailable UUID is a failurecondition~~; the recognized Legacy TWIC nil-GUID exception follows Subsection 3.12. Unavailable UUID otherwise causes failure.
PIV Key Reference	0x01	Raw byte	Challenged key reference, normally 9A or 9E.

Field	Size	Encoding	Notes
PIV Algorithm	0x01	Raw byte	~~Algorithm~~ Discovered supported identifier: 05, ~~,~~07, 11, or 14; 06 remains informative legacy test material only.
Supplemental Entropy	0x20	Raw bytes	Fixed-width ACU entropy field; all zeroes are ~~valid for test vectors only~~permitted but not recommended.

The ~~challenge digest is produced in two steps. First, derive~~ fixed-width encoding and profile mapping are normative in Tables 33 and 32; this appendix illustrates them. The retained prototype vectors derive their KDK as `PIV-AUTO-KDK=KMAC256(key=S-ENC||S-MAC1||S-MAC2,data=empty,mac_len=32, custom="OSDP-PIV-AUTO-KDK-v1")`. ~~Second, derive the challenge as~~ This is SC1-style prototype input, not evidence of an SC2 key binding. The challenge stage is `KMAC256(key=PIV-AUTO-KDK, data=fixed-widthtemplate,mac_len=L,custom="OSDP-PIV-AUTO-CHALLENGE-v1")`. `L` is 32 bytes for SHA-256 profiles and 48 bytes for SHA-384 profiles. The supplemental entropy field is always present and fixed at 32 bytes; all zeroes are valid for test vectors but not recommended for deployments.

C.7 Dynamic Authentication Templates

C.7.1 RSA Key Challenge Input

Table 86: RSA Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Modulus length	Challenge block	EMSA-PKCS1-v1_5 encoded digest block with the same length as the RSA modulus.
0001FF...FF00	Variable	Padding prefix	EMSA-PKCS1-v1_5 block prefix before the digest information.
30...	Variable	DigestInfo	ASN.1 DigestInfo for SHA-256 or SHA-384 followed by the KMAC-derived digest; SHA-256 uses prefix 3031300D060960864801650304020105000420 and SHA-384 uses prefix 3041300D060960864801650304020205000430.

Table 87: RSA Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
RSA-1024	06	9A/9E	0x80 bytes	0x80 bytes	Informative legacy/non-current test-only APDU.
RSA-2048	07	9A/9E	0x100 bytes	0x100 bytes	0x112-byte extended APDU.
RSA-3072	05	9A/9E	0x180 bytes	0x180 bytes	0x192-byte extended APDU.

C.7.2 ECC Signature Input

Table 88: ECC Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Digest length	Challenge digest	KMAC-derived digest supplied directly to the ECC signing operation.
Digest bytes	0x20 or 0x30 bytes	Digest value	SHA-256-width digest for P-256; SHA-384-width digest for P-384.

Table 89: ECC Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
ECC P-256	11	9A/9E	0x20 bytes	DER ECDSA signature, about 0x46-0x48 bytes	0x2C-byte short APDU.
ECC P-384	14	9A/9E	0x30 bytes	DER ECDSA signature, about 0x66-0x68 bytes	0x3C-byte short APDU.

C.7.3 Worked Dynamic Authentication Examples

Worked examples are generated from `test-vectors/piv-auto-demo/generated/piv-auto-simulation-report.json`.

Table 90: PIV Auto Worked Example: PIV Authentication RSA-1024

Field	Value
Profile	9a-rsa1024
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-rsa1024. Informative legacy/non-current test-only coverage.
Algorithm ID	0x06
Key Reference	0x9A
KMAC Challenge Digest	5B2FC08860E3999EEF8F6CF5D2B0C9AE05CBB28A5B423BFE4F646F6F47FCA4F2
Tag 81 Challenge Input	0001FFFF

Field	Value
GENERAL AUTHENTICATE APDU	0087069E887C818582008181800001FFFF

Field	Value
Dynamic Authentication Response	7C8201048282010051535 6AB0012D988701AC373B029D86540C4DF473B031EDB5ADFDB84BC214A45497F6CEE5FFB15ECB0CD89B9F351EF5006AB20105EF289EE041F427802867BA48253B6B90F0BD6F77637B14DD36B32BE89F8BD12F2D9F0E5B408C6C4529 59F233EDB9FA0A96B2808711FB1B671A683460F5FD62847AC18E5717B7D008EDE5FB2AEDFFF44EDD90B2E2DB595456BA7D42644C57C2267C4B9961424839E352FE64EB5F344CBB12BC630FFCDE943055A80375CF6AEA11BAF6AEC2916D5A1B6D769CC5F935519634C2504A2CA4916368DAAA887DC61A514AC630DB093CD2AC3608CECD355A63DCA02E35FBF2CAEDA34ED51C7F60C69EFE7CCCFDA9115A9395001699099000
Response Tag 0x82 Length	0x0100 bytes

Table 93: PIV Auto Worked Example: Card Authentication RSA-2048

Field	Value
Profile	9e-rsa2048
Source	Committed RSA-2048 card-authentication key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x07
Key Reference	0x9E
KMAC Challenge Digest	0864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E
Tag 81 Challenge Input	0001FFFF

Field	Value
Dynamic Authentication Response	7C820104828201009BB1AA3875DB4E41107DD91CEAAC36FE4CE5F969E1F7B9B199DFAB38CFA1EE36FBF197CD2777C4A6CE082623C9F36A9DC2514FE54DD12934AFCAE16DFE14FFCFC0F58E0A0F46567444968F1DE78B496B4222DB913A87B850139B1E2A730124C4ED169919F1EB5FA3E15617E89BB9CED8FA996A63E126C239EB6D142A5746FBEC3DFA3B287F944EE45586BA5AEB64087BDA0661DF7254C42D9768141F7243382D61CC43A8CDBE4ADA849E05F013D947DF7FA608C38ABA3841B038AA69D77461B00053F59C4D34906BFBFD470A277D0722AAECD83117DE906BFCD36A16EFB00B3EACD4414F48528CCA4BE57826B4312424A65DABAD6FA1ECA3DAF55F6E6E1022C89000
Response Tag 0x82 Length	0x0100 bytes

Table 94: PIV Auto Worked Example: PIV Authentication RSA-3072

Field	Value
Profile	9a-rsa3072
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-rsa3072. Generated-card coverage.
Algorithm ID	0x05
Key Reference	0x9A
KMAC Challenge Digest	1503FD1E89AA91A5B37E6872AC3846F654B7B1D099B2F3AEB597472DFE38E71442816BC43454622219F61A3C6DAFDB4F
Tag 81 Challenge Input	0001FFFF

Field	Value
GENERAL AUTHENTICATE APDU	0087059A00018A7C8201868200818201800001FFFF

Field	Value
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFF

Table 96: PIV Auto Worked Example: PIV Authentication ECC P-256

Field	Value
Profile	9a-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9A
KMAC Challenge Digest	A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B45851
Tag 81 Challenge Input	A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B45851
GENERAL AUTHENTICATE APDU	0087119A267C2482008120A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B4585100
Dynamic Authentication Response	7C4A824830460221009D25A44E56572983819019DF26A2720E462A11D2CB3CD86A9142DEF4BCC1271B022100D084982BB9D3E2D3E6B0965956C71DBA0FE44FE328C0A2BD72951DFB4D9296FE9000
Response Tag 0x82 Length	0x0048 bytes

Table 97: PIV Auto Worked Example: Card Authentication ECC P-256

Field	Value
Profile	9e-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9E
KMAC Challenge Digest	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
Tag 81 Challenge Input	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
GENERAL AUTHENTICATE APDU	0087119E267C2482008120965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D00
Dynamic Authentication Response	7C4A824830460221008B204FA48FAA27A57D6E90F985199F284D2A8E0D0F2E051BBFBCACBA824619B9022100C6E04880D322DE7D10823EC78C8EA4CCB69FE394F694F1D12C797FA7BCAD769599000
Response Tag 0x82 Length	0x0048 bytes

Table 98: PIV Auto Worked Example: PIV Authentication ECC P-384

Field	Value
Profile	9a-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9A
KMAC Challenge Digest	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28
Tag 81 Challenge Input	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28
GENERAL AUTHENTICATE APDU	0087149A367C3482008130C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D2800
Dynamic Authentication Response	7C6A82683066023100D88ABABB9893D88341CFA2F1AB6E3D35E4AF02F4F8D085B5A3DE63574DCF694741D869066B8E6297429120614DCC9CE40231009A3F203BB5EA98839008B0D25F98C98DE8E87622F61279B2F2B29343912999A36311102B310F9985844BCC248350377B9000
Response Tag 0x82 Length	0x0068 bytes

Table 99: PIV Auto Worked Example: Card Authentication ECC P-384

Field	Value
Profile	9e-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9E
KMAC Challenge Digest	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689
Tag 81 Challenge Input	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689
GENERAL AUTHENTICATE APDU	0087149E367C3482008130F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC68900

Field	Value
Dynamic Authentication Response	7C698267306502303 4C6B2659FBE26A78D2615F8C0EBF2E9674292D9DE3CE8A31D9DA297B5AE236AC886E9E9C288D42474564BF0C46AE40C023100CCBD59A047D73F9FD58F60218FBCB4F99B18EBD7EB70EFE5CF64D5EB1A65F1676A867E1B1E5EB4CF9F34B44BE905D29C9000
Response Tag 0x82 Length	0x0067 bytes

C.8 APDUs

Table 100: PIV Auto APDU Reference: Card Preparation

Operation	Command APDU	Success Response	Purpose
Select PIV Application	00A404000BA0000003080000100 00100	9000	Selects the PIV application before reading objects or authenticating.
Get Discovery Object	00CB3FFF035C017E	7E...9000	Reads the Discovery Object so the PD can determine application and PIN/VCI policy.
Verify PIN (optional)	00200080083132333435 36FFFF	9000	Verifies example PIN 123456 padded with FF; failure responses include 63CX and 6983.

Table 101: PIV Auto APDU Reference: GENERAL AUTHENTICATE Command

Field	Encoding	Source	Notes
CLA	00	ISO/IEC 7816-4	Logical command before credential SM wrapping. Apply SP 800-73-5 Part 2, Section 4.2, when SM is established.
INS	87	ISO/IEC 7816-4	GENERAL AUTHENTICATE instruction.
P1	Algorithm ID	~~PIV profile~~ Certificate SPKI	05, 07, 11, or 14 for supported PIV Auto profiles; 06 appears only in informative RSA-1024 legacy examples.
P2	Key Reference	~~PIV profile~~ Configuration Entry	9A or 9E for PIV; 9E for TWIC Auto.

Field	Encoding	Source	Notes
Lc	Short or extended	Command length	Short APDUs use one length byte; extended APDUs use 00 followed by two length bytes.
Command data	7C...820081.....	Dynamic Authentication Template	Tag 82 is empty to request a response; tag 81 carries the exact challenge input.
Le	00	Response request	Requests the maximum response supported by the card interface.

Table 102: PIV Auto APDU Reference: GENERAL AUTHENTICATE Response

Case	Response APDU	PD Handling	OSDP Payload
Complete response	7C...82...9000	Strip only the ISO status word.	Carry the complete 7C Dynamic Authentication Template in osdp_CRAUTHR or osdp_PIVSTATUS.
Additional response bytes	61XX	Issue GET RESPONSE with Le XX.	Concatenate returned data before reporting the final 7C template.
GET RESPONSE command	00C00000XX	Retrieve remaining card response bytes.	Not exposed directly to the ACU.
Credential error	69.. or 6A..	Preserve command failure detail.	~~Report via osdp_PIVERROR when available; do not treat as a valid signed response~~Auto reports terminal PIVSTATUS Result 0x06 with the applicable status word; commanded CRAUTH uses PIVERROR.

Table 103: PIV Auto APDU Reference: OPACITY / VCI Context

Operation	Command Shape	Key Ref	Relationship to PIV Auto
OPACITY establish	0087[27/2E]04[Lc]7C...81...8200	04	Establishes credential SM after CVC and cryptogram verification. Contactless VCI also depends on Discovery policy and pairing when required.
PIV Auto challenge	0087[alg][9A/9E][Lc]7C...820081.....00	9A/9E	Performs the autonomous key challenge after card preparation and any required VCI setup.
Response handling	7C...82...9000	9A/9E	The PD reports the returned signed response bytes exactly, excluding only SW1/SW2.

DRAFT

D Credential SM and VCI Worked Examples (Informative)

These examples follow the preparation and PIN-verification flows in Subsection 3.1. A simulated ACU sends complete OSDP packets to a simulated PD. The PD replays existing card captures and reports the resulting credential state. These examples omit OSDP Secure Channel for clarity; the test harness assumes that the OSDP Secure Channel authorization requirements are satisfied.

The contact and contactless card01 captures supply the card commands and replies. The verifier checks the CVC signature against an existing trust anchor, recomputes SM key establishment, and checks protected command and response MACs and encryption. The replay retains intervening captured exchanges so the session counters and MAC chains remain intact. Those intervening exchanges preserve the original card session. The card captures include earlier provisioning and data-read commands; those commands impose no additional requirements on automatic card preparation. The examples replay recorded sessions; no connected card is used.

Table 104: Replay coverage

Case	Evidence	Purpose
Contact SM	Captured SM session and CVC validation	Report SM with VCI absent.
Incorrect PIN	Modeled authenticated rejection	Clear the failed PIN while preserving SM and VCI.
Commanded pairing and PIN	Captured protected VERIFY exchanges	Complete pairing before verifying the cached PIN.
PIV Auto failure	Modeled results and full status words	Report the cause across multipart replies.
PIN cache expiration	Modeled timing, with captured VERIFY for the submitted-command case	Separate cache availability from credential PIN verification.

Complete packets, APDUs, and source references are in the SM/VCI example file. The accompanying README gives the replay and independent-verification commands. Packet dumps use hexadecimal byte pairs, including the OSDP header and CRC. Each example starts at sequence `0x01` on an established link to PD address `0x01`. Standard status-query and pairing replies use the same `osdp_CARDSTATUSR` layout with Status Type `0x01`. Diagnostic replies carry Status Type `0x02` before the multi-part fields in every fragment. Timing examples identify the modeled elapsed time at each exchange; captured APDUs retain their original order.

D.1 Credential SM over contact

The PD prepares a contact credential and reports SM established without VCI. The reply identifies the CVC issuer and trust anchor used for that SM session.

Example ID: `contact-sm`. Credential exchanges come from the contact card01 capture. The complete replay retains its original APDU order.

Table 105: Reported security state

Field	Value	Notes
Status Type	0x01	Standard credential status.
Credential Security Status	0x02	SM established without VCI.
SM Trust Anchor ID	0x01	Loaded anchor used to validate SM.
SM CVC IIN	0xB6103792270FBD08	Issuer named by the credential's SM CVC.
SM Anchor IIN	0xB6103792270FBD08	Same issuer: direct trust path.

Table 106: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU CARDSTATUS	53 01 09 00 05 A9 01 8C 9B	Read current state after automatic credential preparation.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 06 85 01 01 02 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 98 79	Status 0x02: SM established without VCI.

D.2 PIN before credential presentation

The PD collects the PIN locally and reports completion before the credential is presented. Initial selection and SM establishment preserve the cache. The ACU then completes pairing and requests PIN verification over VCI.

Example ID: `cache-before-card`. Credential exchanges come from the contactless card01 capture. The complete replay retains its original APDU order.

Table 107: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 3C 00 FE 9F	Collect a PIN locally; cache for 0x003C seconds.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	Result 0x00: requested PIN operation completed.
ACU CARDSTATUS	53 01 09 00 07 A9 01 EC F5	Read current state after automatic credential preparation.
PD ACK	53 81 08 00 07 40 5B 53	Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 05 85 01 02 02 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 1E 84	Status 0x02: SM established without VCI.
ACU PIVXMITPAIRING	53 01 10 00 06 AD 30 30 30 30 30 30 30 32 EB 8B	Submit the captured pairing code.
PD ACK	53 81 08 00 06 40 6A 60	Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 07 85 01 02 01 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 56 3D	Status 0x01: SM and VCI established.
ACU PIVSPE	53 01 0E 00 05 AA 02 00 00 00 00 00 E7 52	Verify the cached PIN over established VCI.

Message	Packet bytes (hex)	Notes
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	Result 0x00: requested PIN operation completed.

D.3 PIV Auto needs commanded pairing

In this modeled case, the configured PIV Auto authentication requires VCI. SM is established, but pairing remains necessary. PIV Auto reports the unmet condition and ends the attempt. A later poll receives ACK; the ACU can begin commanded pairing.

Example ID: `auto-pairing-required`.

Table 108: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD PIVSTATUS	53 81 45 00 05 89 37 00 00 00 37 00 05 01 01 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 9A 07 00 00 FF 09	Result 0x05, Detail 0x01: pairing code required.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD ACK	53 81 08 00 06 40 6A 60	No completion pending.

D.4 SM failure diagnostics

In this modeled case, PIV Auto reports Result 0x02, Detail 0x04: invalid authentication cryptogram. The ACU then requests the SM diagnostic record. The PD returns the failure reason and the hash of the received CVC, with no further credential commands. Three fragments carry the 35-byte record. The CVC bytes used for the hash come from the existing capture.

Example ID: `diagnostic-cvc-hash`.

Table 109: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU CARDSTATUS	53 01 09 00 06 A9 02 BF F2	Request retained SM diagnostics; no credential command is sent.
PD ACK	53 81 08 00 06 40 6A 60	Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	Poll for completion.
PD CARDSTATUSR	53 81 1F 00 07 85 02 23 00 00 00 10 00 03 04 01 A5 50 00 73 BA 17 70 37 0A 1D 58 B1 5B 2F F2	Diagnostic: total 0x0023, offset 0x0000, fragment 0x0010.
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD CARDSTATUSR	53 81 1F 00 05 85 02 23 00 10 00 10 00 97 5D D1 50 7F 39 22 BB 44 20 69 9B 2D FB 42 83 52 B1	Diagnostic: total 0x0023, offset 0x0010, fragment 0x0010.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD CARDSTATUSR	53 81 12 00 06 85 02 23 00 20 00 03 00 C7 6C E9 B4 96	Diagnostic: total 0x0023, offset 0x0020, fragment 0x0003.

D.5 PIN cache expires while idle

The PD acquires a PIN at t=0 with a ten-second validity period. At t=10 the PD erases the PIN. A poll receives ACK because no operation is pending. A status query reports an empty cache; a subsequent verification request ends with no cached PIN. The PD sends no VERIFY. This example models the timing and credential state.

Example ID: `pin-expiration-idle`.

Table 110: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU POLL	53 01 08 00 07 60 E9 55	t=10 s. Poll for completion.
PD ACK	53 81 08 00 07 40 5B 53	t=10 s. No completion pending.
ACU PIVSPE	53 01 0E 00 05 AA 00 00 00 00 00 00 A7 D9	t=10 s. Report empty cache and no PIN verification.
PD PIVSPER	53 81 09 00 05 86 00 55 49	t=10 s. Cache empty; credential PIN verification absent.
ACU PIVSPE	53 01 0E 00 06 AA 03 00 00 00 00 00 32 DF	t=10 s. Request verification using the expired cache.
PD ACK	53 81 08 00 06 40 6A 60	t=10 s. Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 07 86 0A 7F 86	t=10 s. Result 0x0A: no cached PIN.

D.6 Expiration while waiting for a credential

The ACU arms cached-PIN verification with Auto Execute flag 0x02. The credential is absent when the cache expires at t=10. The PD ends the pending request with PIVSPER 0x0A (no cached PIN). This example models the waiting state.

Example ID: `pin-expiration-waiting`.

Table 111: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 07 AA 03 00 00 02 00 00 81 F6	t=0 s. Arm cached-PIN verification for credential presentation.
PD ACK	53 81 08 00 07 40 5B 53	t=0 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=0 s. Poll for completion.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. No completion pending.
ACU POLL	53 01 08 00 06 60 D8 66	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 0A 4F B1	t=10 s. Result 0x0A: no cached PIN.

D.7 Expiration before VERIFY submission

The PD accepts verification at t=9. Before the PD submits VERIFY, the deadline is reached at t=10. The PD returns PIVSPER 0x0A (no cached PIN) and sends no VERIFY. Acceptance alone does not reserve the PIN beyond its deadline. Timing is modeled.

Example ID: `pin-expiration-boundary`.

Table 112: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 07 AA 03 00 00 00 00 00 E1 98	t=9 s. Accept verification at t=9, before the cache deadline.
PD ACK	53 81 08 00 07 40 5B 53	t=9 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 05 86 0A 1F E8	t=10 s. Result 0x0A: no cached PIN.

D.8 Expiration after VERIFY submission

The PD submits VERIFY at t=9, before the deadline. The cache expires at t=10 while the exchange is completing. The PD reports successful verification, then reports an empty cache with the credential still PIN verified. Credential APDUs come from the contactless capture; the timing is modeled.

Example ID: `pin-expiration-submitted`.

Table 113: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVXMITPAIRING	53 01 10 00 07 AD 30 30 30 30 30 30 30 32 AE E4	t=0 s. Supply the captured pairing code after SM preparation.
PD ACK	53 81 08 00 07 40 5B 53	t=0 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=0 s. Poll for completion.
PD CARDSTATUSR	53 81 2A 00 05 85 01 02 01 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 D4 6F	t=0 s. Status 0x01: SM and VCI established.
ACU PIVSPE	53 01 0E 00 06 AA 02 00 00 00 00 00 92 9A	t=9 s. Accept verification at t=9, before the cache deadline.
PD ACK	53 81 08 00 06 40 6A 60	t=9 s. Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 07 86 00 35 27	t=10 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 05 AA 00 00 00 00 00 00 A7 D9	t=10 s. Empty cache; credential PIN verified.
PD PIVSPER	53 81 09 00 05 86 02 17 69	t=10 s. Cache empty; credential PIN verified.

D.9 PIN reuse preserves the original deadline

Two explicit contact verification requests use the same cached PIN at t=1 and t=2. Both succeed. At t=10, the cache expires and the credential remains PIN verified. The successes and timing are modeled; the example demonstrates the PD state rules.

Example ID: `pin-expiration-reuse`.

Table 114: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 07 AA 03 00 00 00 00 00 E1 98	t=1 s. Explicit contact verification using the same cached PIN.
PD ACK	53 81 08 00 07 40 5B 53	t=1 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=1 s. Poll for completion.
PD PIVSPER	53 81 09 00 05 86 00 55 49	t=1 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 06 AA 03 00 00 00 00 00 32 DF	t=2 s. Explicit contact verification using the same cached PIN.
PD ACK	53 81 08 00 06 40 6A 60	t=2 s. Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	t=2 s. Poll for completion.
PD PIVSPER	53 81 09 00 07 86 00 35 27	t=2 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 05 AA 00 00 00 00 00 00 A7 D9	t=10 s. Cache expired; credential PIN verification remains valid.
PD PIVSPER	53 81 09 00 05 86 02 17 69	t=10 s. Cache empty; credential PIN verified.

D.10 Credential failure status across two fragments

PIV Auto reports a modeled GENERAL AUTHENTICATE rejection: Result 0x06, Detail 0x00. The credential status is 0x6982 (security status not satisfied). The first fragment ends with SW1, 0x69; the second carries SW2, 0x82. The ACU reassembles the complete record before interpreting the status word.

Example ID: `auto-status-word`.

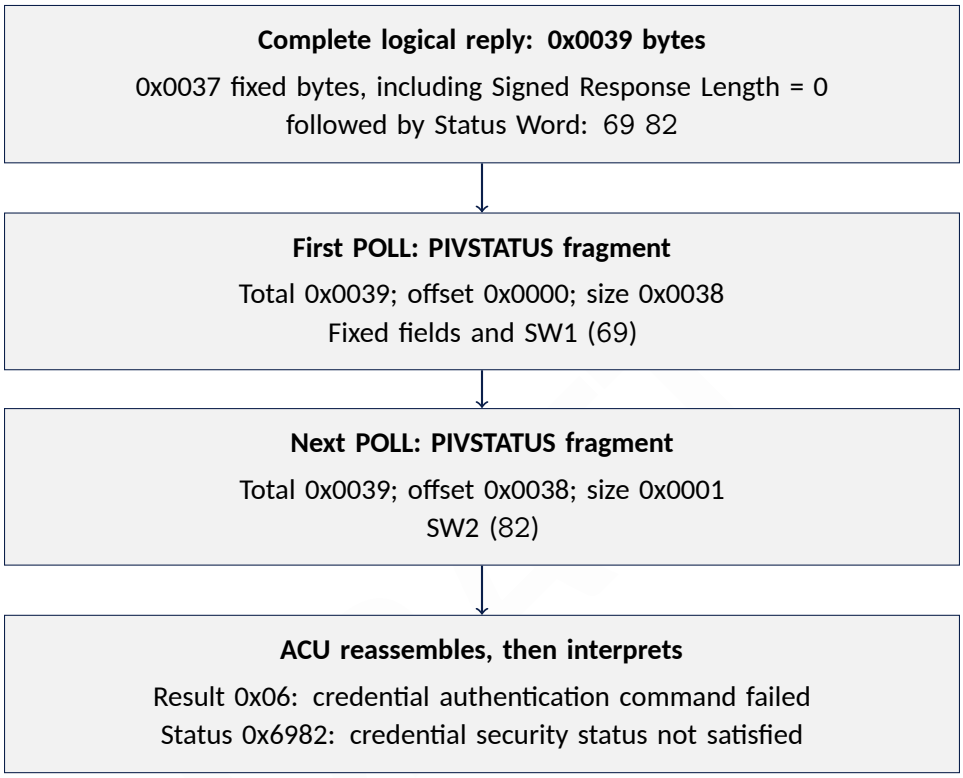


Figure 9: OSDP 3.0 multipart fields describe the entire logical reply. A fragment boundary can fall between SW1 and SW2.

Table 115: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD PIVSTATUS	53 81 46 00 05 89 39 00 00 00 38 00 06 00 01 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 9A 07 00 00 69 E4 69	PIV Auto credential rejection; SW1/SW2 cross the fragment boundary.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.

Message	Packet bytes (hex)	Notes
PD PIVSTATUS	53 81 0F 00 06 89 39 00 38 00 01 00 82 9D EB	PIV Auto credential rejection; SW1/SW2 cross the fragment boundary.

D.11 Incorrect PIN preserves SM and VCI

This modeled case starts with SM, VCI, and a cached PIN available. The credential returns an authenticated 0x63C2 (incorrect PIN, two attempts remain). The PD reports PIN mismatch and clears the cache and PIN-verification state. SM and VCI remain established. The SPE reply carries the mismatch result; a separate Mode 0x06 query retrieves remaining attempts. The ACU can acquire a corrected PIN and request verification again using the existing session.

Example ID: `sm-credential-rejection`.

Table 116: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 02 00 00 00 00 00 E7 52	Verify a cached PIN over an established VCI; initial state is modeled.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD PIVSPER	53 81 09 00 06 86 03 66 20	Result 0x03: PIN mismatch.
ACU PIVSPE	53 01 0E 00 07 AA 00 00 00 00 00 00 01 56	Cache erased and PIN-verification state cleared after PIN mismatch.
PD PIVSPER	53 81 09 00 07 86 00 35 27	Cache empty; credential PIN verification absent.
ACU CARDSTATUS	53 01 09 00 05 A9 01 8C 9B	Read credential status after ordinary rejection.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 06 85 01 02 01 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF FF 0B A0 00 00 03 08 00 00 10 00 01 00 A9 03	Status 0x01: SM and VCI established.

D.12 Recovering trust for contactless PIV Auto

This example uses a credential that supports SM and requires a pairing code for VCI. PIV Auto is configured for contactless authentication with key 0x9A, but the PD's trust store is empty. The ACU already holds the roots and validation policy authorized for the deployment. Figure 10 shows the recovery sequence.

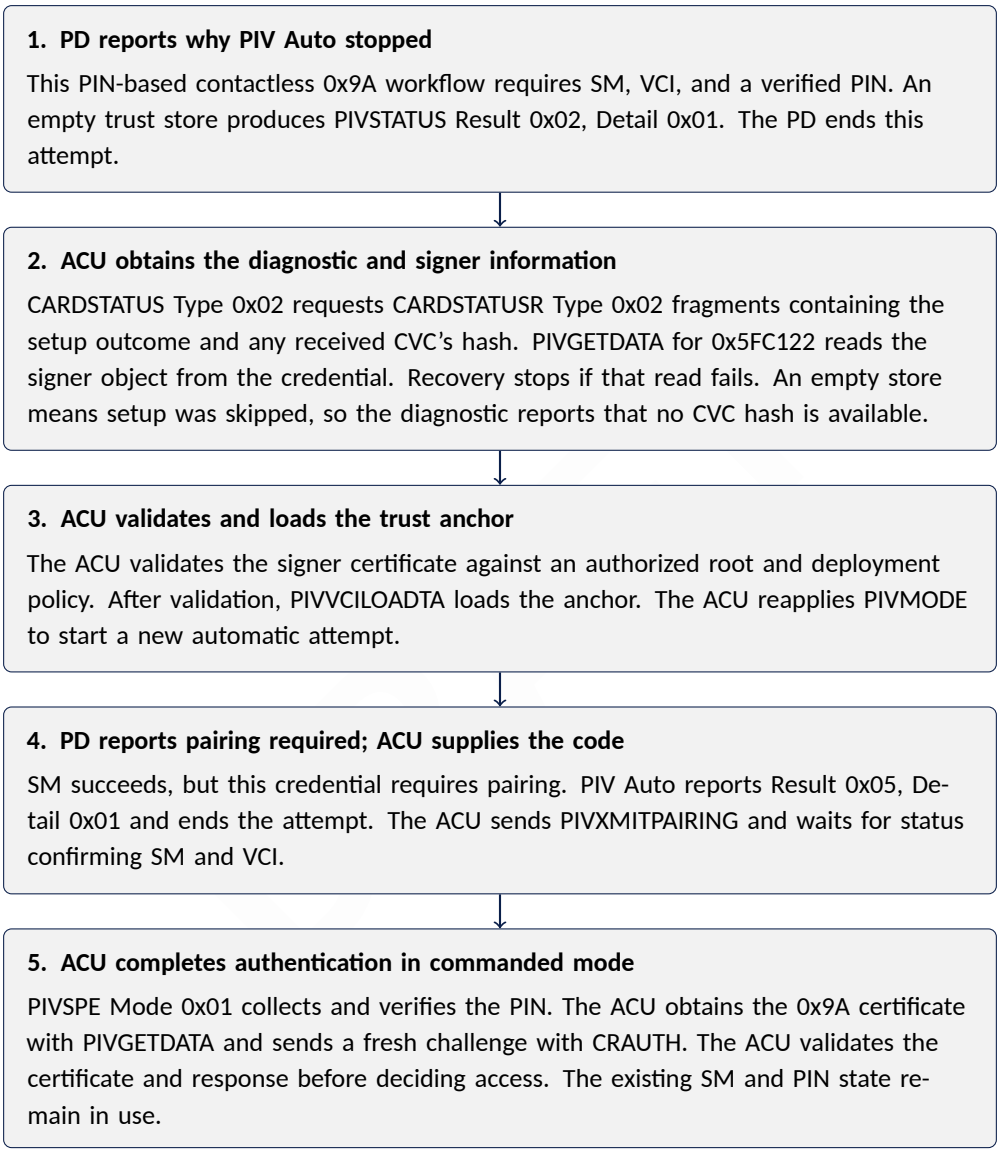


Figure 10: Recovery from an empty PD trust store for contactless PIV authentication with key 0x9A. Every step depends on a successful result from the preceding step. Unsupported SM or VCI requires a different ACU-selected workflow.

The first PIV Auto attempt ends with Result 0x02, Detail 0x01 (trust store empty). The ACU requests the SM diagnostic record, then reads the Secure Messaging Certificate Signer object with PIVGETDATA, Object Identifier 0x5FC122, Tag Length zero, Offset zero, and Requested Length zero. When the read succeeds, the reply contains the complete object, including signer certificate tag 0x70 and any intermediate CVC in

tag 0x7F21. The PD reads the credential through the contactless interface. If the object is absent or the credential denies access, the PD reports the corresponding PIVERROR. The ACU then selects another supported workflow or ends the transaction. Recovery relies on the credential returning the object.

The ACU validates the signer certificate's chain to an authorized root, validity period, revocation status, and PIV or PIV-I content-signing extended key usage. The ACU checks any intermediate CVC's signature and linkage as specified under PIVVCILOADTA. The anchor record contains the validated signer's public key and the IIN derived from that certificate's subjectKeyIdentifier. The ACU loads an anchor only after these checks succeed. An IIN match alone provides no authorization to trust a certificate.

After the load succeeds, the ACU reapplies the PIVMODE configuration. The PD starts a new attempt, establishes SM, and reports Result 0x05, Detail 0x01 (pairing code required). The ACU sends PIVXMITPAIRING and waits for CARDSTATUSR with security status 0x01 (SM and VCI established). The ACU then sends PIVSPE Mode 0x01, with Auto Execute clear, to collect and verify the PIN. After successful verification, the ACU reads the PIV Authentication certificate (Object Identifier 0x5FC105) and uses CRAUTH with key 0x9A and a fresh challenge. Certificate and challenge-response validation remain the ACU's responsibility.

The ACU waits for a successful result at each step before issuing the next command. An error ends this recovery sequence. These are commanded operations after the automatic attempt has ended. The PD returns each command's defined reply; the sequence generates no second PIV Auto result. PIVMODE remains configured for the next credential presentation. ~~Reapplying PIVMODE~~ An ordinary PIVMODE replacement after pairing or PIN verification ~~would clear~~ clears the state needed for the commanded authentication. Alternatively, after the completed pairing/PIN prerequisite failure, an eligible PIVMODE carrying the explicit Resume TLV can preserve that context and start one new Auto attempt with the exact next unused sequence counter. Subsection 3.12 defines this distinct operation; the illustrative resume bytes appear in the autonomous-template appendix.

If SM or VCI is unsupported, the ACU chooses a supported interface or authentication method explicitly, or rejects the transaction. If a signature or cryptogram check fails, the PD blocks credential operations and retains the diagnostic record. The ACU investigates that failure before requesting a new attempt. Trust recovery is appropriate for missing authorized trust material; failed cryptographic checks are security failures.

This is a control-flow example. The preceding captured SM/VCI packet examples cover pairing and PIN verification. The companion trust-recovery example records the command sequence and validation decisions.

E CRAUTH Worked Examples (Informative)

This example sends a captured P-256 challenge to key 0x9E and returns the captured signature template. The first command fragment contains only Format Marker 0xFF. The second contains Algorithm 0x11, Reference Identifier 0x9E, and the 0x20-byte challenge. The complete request record is 0x0023 bytes. The PD removes the marker and uses the algorithm, key reference, and challenge to construct GENERALAUTHENTICATE.

The credential data comes from the existing contactless card04 SM capture. Credential SM is assumed established as in that capture; the example shows the plaintext at the PD's SM boundary. The packets omit OSDP Secure Channel for clarity. The two failure cases are modeled to show terminal error handling. Full exchanges and source references are in the CRAUTH example file. The command and reply codes remain 0xA5 and 0x82.

Table 117: CRAUTH command and reply exchange

Message	Packet bytes (hex)	Notes
ACU CRAUTH	53 01 0F 00 05 A5 23 00 00 00 01 00 FF 6B 63	Total 0x0023; offset 0x0000; fragment 0x0001. Format marker only.
PD ACK	53 81 08 00 05 40 39 35	Fragment accepted.
ACU CRAUTH	53 01 30 00 06 A5 23 00 01 00 22 00 11 9E 9B C1 80 EF 9A 61 6F 9F 05 8B F4 60 44 C8 9A 30 1B BD 90 13 7A DF 12 F9 D9 A9 91 7B 09 4A EF A0 5C FF	Total 0x0023; offset 0x0001; fragment 0x0022.
PD ACK	53 81 08 00 06 40 6A 60	Fragment accepted.
ACU POLL	53 01 08 00 07 60 E9 55	Poll for the result.
PD CRAUTHR	53 81 36 00 07 82 4B 00 00 00 28 00 7C 49 82 47 30 45 02 21 00 83 6F 34 70 E4 CD 87 3C 6C A3 E3 B6 5A D3 6B 17 DF FC E2 7B 56 A8 25 EC 39 A9 DC E3 D4 CB B0 6F 70	Total 0x004B; offset 0x0000; fragment 0x0028.
ACU POLL	53 01 08 00 05 60 8B 33	Poll for the result.
PD CRAUTHR	53 81 31 00 05 82 4B 00 28 00 23 00 95 02 20 61 1E FE 89 BF A8 13 B1 34 BD 11 1D 64 00 01 8B 50 71 1E 6C 9C 5D C8 22 DC 75 24 0B FA 60 D6 EE D9 FA	Total 0x004B; offset 0x0028; fragment 0x0023.

Message	Packet bytes (hex)	Notes
ACU POLL	53 01 08 00 06 60 D8 66	Poll for the result.
PD ACK	53 81 08 00 06 40 6A 60	No completion pending.

Table 118: Modeled failure completions

Failure	Packet bytes (hex)	ACU action
Credential rejection	53 81 0C 00 07 8A 23 10 69 82 2A D5	PIVERROR 0x1023 (security conditions unmet), SW1/SW2 0x6982. End the request.
Failure after a data fragment	53 81 0A 00 05 8A 02 00 0E 51	PIVERROR 0x0002 (smart-card timeout). Discard the partial response.

Draft

F Deprecated `osdp_GENAUTH`

The `osdp_GENAUTH` command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use `osdp_CRAUTH` flows instead. Controllers and PDs **SHOULD** plan migrations away from `osdp_GENAUTH`.

DRAFT